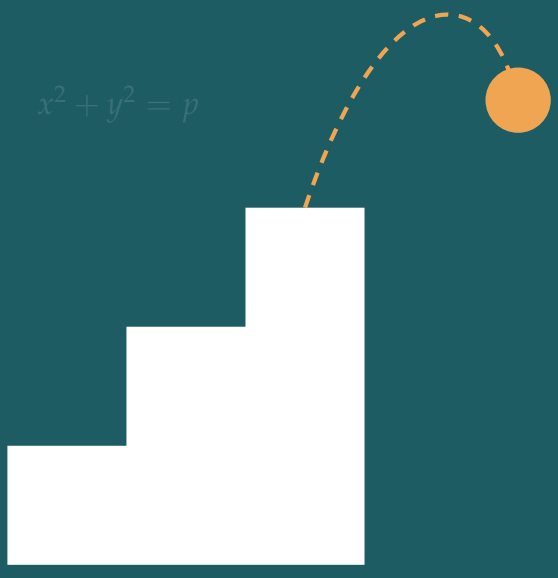
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1999 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1999 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1999 Worked Solutions

Worked solutions to all 42 questions of the 1999 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1999, Question 1

Pure • Counting Coprime Integers

> **STEP I 1999, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/99-s1-q1

The key idea

Divisibility by 2 and 5 depends only on the last digit, and divisibility by 3 and 7 depends only on the remainder modulo 21. So the integers we want repeat in a fixed pattern, block after block, and counting them is a matter of counting one block and multiplying. For the averages, the trick is symmetry: if n is in the set then so is its "partner" $N - n$, so the numbers pair off around the midpoint.

Solution

Part 1: not divisible by 2 or 5, below a million

"Not divisible by 2 or 5" means divisible by neither. An integer is divisible by 2 or by 5 exactly when its last digit is $0, 2, 4, 5, 6$ or 8. So we want the integers whose last digit is 1, 3, 7 or 9.

The integers $0, 1, \ldots, 999\,999$ split into $100\,000$ blocks of ten consecutive integers $10k, 10k+1, \ldots, 10k+9$, and each block contains exactly four of the integers we want. Hence the count is

$$4 \times 100\,000 = 400\,000.$$

(Inclusion–exclusion gives the same: $10^6 - \frac{1}{2}10^6 - \frac{1}{5}10^6 + \frac{1}{10}10^6 = 400\,000$.)

For the average, pair each such n with $10^6 - n$. If n ends in $1, 3, 7, 9$ then $10^6 - n$ ends in $9, 7, 3, 1$ respectively, and $0 < 10^6 - n < 10^6$ because $n \neq 0$. So $n \mapsto 10^6 - n$ maps the set to itself, and it is its own inverse. It has no fixed point, since $n = 500\,000$ is even. So the set splits into $200\,000$ pairs, each with sum 10^6. The total is $200\,000 \times 10^6$, and the average is

$$\frac{200\,000 \times 10^6}{400\,000} = 500\,000.$$

Part 2: not divisible by 3 or 7, below 4179

The number 4179 is chosen with care:

$$4179 = 3 \times 1393 = 3 \times 7 \times 199 = 21 \times 199.$$

Whether n is divisible by 3 or by 7 depends only on the remainder of n on division by 21. Among the remainders $0, 1, \ldots, 20$, the multiples of 3 are $0, 3, \ldots, 18$ (seven of them) and the multiples of 7 are $0, 7, 14$ (three), with 0 counted in both. So $21 - 7 - 3 + 1 = 12$ remainders are divisible by neither:

$$1,\ 2,\ 4,\ 5,\ 8,\ 10,\ 11,\ 13,\ 16,\ 17,\ 19,\ 20.$$

The integers $0, 1, \ldots, 4178$ make up exactly 199 complete blocks of 21, so the count is

$$12 \times 199 = 2388.$$

As a check, by inclusion–exclusion: $4179 - 1393 - 597 + 199 = 2388$.

> **Ask yourself** Why does the symmetry trick still work here, when the numbers have nothing to do with decimal digits?

The key is that 21 divides 4179. Then $4179 - n \equiv -n \pmod 3$ and $\pmod 7$, so $3 \mid 4179 - n$ if and only if $3 \mid n$, and the same for 7. Also $n = 0$ is excluded (it is divisible by everything), so every n in the set has $1 \le n \le 4178$, and then $1 \le 4179 - n \le 4178$ too. So $n \mapsto 4179 - n$ maps the set to itself. It has no fixed point because 4179 is odd. The 2388 integers therefore form 1194 pairs, each summing to 4179, and the average is

$$\frac{1194 \times 4179}{2388} = \frac{4179}{2} = 2089\tfrac{1}{2}.$$

Checks

A computer that runs through all the integers confirms the counts 400 000 and 2388 and the averages 500 000 and 2089.5 exactly. You can also do a small case by hand: below 21, the twelve remainders listed above add up to $126 = 12 \times \frac{21}{2}$, so their average is $10.5 = \frac{21}{2}$, exactly as the pairing argument predicts.

Common slips

- Reading "not divisible by 2 or 5" as "not divisible by both", i.e. not divisible by 10. That would give 900 000 integers.
- Forgetting to add back the multiples of 21 (or of 10) in inclusion–exclusion, since they have been subtracted twice.
- Assuming the average must be the midpoint of the range without checking that the set is symmetric. It works here only because 10 divides 10^6 and 21 divides 4179.

Beyond STEP

The count of integers in $1, \ldots, m$ coprime to m is Euler's totient function $\varphi(m)$. Here $\varphi(10) = 4$ and $\varphi(21) = 12$, and the counts are $\varphi(10^6) = 400\,000$ and $199\,\varphi(21)$. The pairing $n \leftrightarrow m - n$ shows that the integers coprime to m always add up to $\frac{1}{2} m\,\varphi(m)$ for $m > 2$, a fact that is a first step into the arithmetic of the integers modulo m. Davenport's *The Higher Arithmetic* is an excellent place to read more.

WORKED SOLUTION

STEP I 1999, Question 2

Pure • Locus and Centroid

STEP I 1999, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q2

The key idea

Write down the condition, expand, and you get x^2 and y^2 each with coefficient 3 and no xy term: that is the signature of a circle. Completing the square shows where its centre is, and the centroid appears naturally because $x_1 + x_2 + x_3$ is exactly what turns up in the linear terms. The constraint on a^2 then drops out of asking when the "radius squared" is allowed to be negative.

Solution

The locus

Let $P = (x, y)$ be a point of the locus, and write $A_i = (x_i, y_i)$ for $i = 1, 2, 3$. The condition is

$$\sum_{i=1}^{3} \left[(x - x_i)^2 + (y - y_i)^2 \right] = a^2.$$

Expanding, and using $x_1 + x_2 + x_3 = 3\bar{x}$ and $y_1 + y_2 + y_3 = 3\bar{y}$,

$$3x^2 - 6\bar{x}\,x + 3y^2 - 6\bar{y}\,y + \sum_i (x_i^2 + y_i^2) = a^2.$$

This is the equation of the locus. To see what it is, divide by 3 and complete the square:

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \frac{a^2 - \sum_i (x_i^2 + y_i^2) + 3\bar{x}^2 + 3\bar{y}^2}{3}.$$

The numerator tidies up nicely. Since $\sum_i x_i = 3\bar{x}$,

$$\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - 2\bar{x} \cdot 3\bar{x} + 3\bar{x}^2 = \sum_i x_i^2 - 3\bar{x}^2,$$

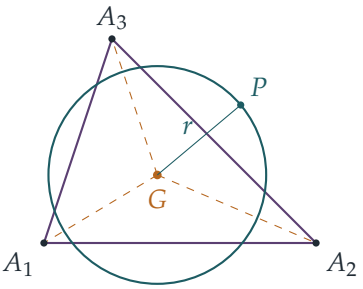
and similarly for y. So if we write

$$S = \sum_{i=1}^{3} \left[(x_i - \bar{x})^2 + (y_i - \bar{y})^2 \right]$$

for the sum of the squares of the distances of the three points from their centroid $G = (\bar{x}, \bar{y})$, the locus is

$$\boxed{(x-\bar{x})^2 + (y-\bar{y})^2 = \frac{a^2 - S}{3}}.$$

Interpretation. When $a^2 > S$ this is a circle with centre the centroid G of the three points and radius $\sqrt{(a^2 - S)/3}$. As a^2 varies we get a family of concentric circles about G. In the boundary case $a^2 = S$ the "circle" shrinks to the single point G.



Why a^2 cannot be less than S

Ask yourself What does it mean for the right-hand side of the boxed equation to be negative?

The left-hand side $(x-\bar{x})^2 + (y-\bar{y})^2$ is a sum of squares of real numbers, so it is never negative. If $a^2 < S$, the right-hand side is negative and no real point (x, y) satisfies the equation: the locus is empty. So for a point to move as described at all, we need $a^2 \geq S$.

It is illuminating to read the same working the other way round. What we actually proved is the identity, valid for every point P,

$$PA_1^2 + PA_2^2 + PA_3^2 = 3\,PG^2 + S.$$

Since $PG^2 \geq 0$, the sum $PA_1^2 + PA_2^2 + PA_3^2$ is always at least S, with equality only at $P = G$. In words: the centroid is the point that minimises the sum of the squared distances to the three points, and the minimum value is S. A constant sum a^2 smaller than this minimum is impossible.

Checks

- Take $A_1 = (0,0)$, $A_2 = (4,0)$, $A_3 = (1,3)$, so $G = (\frac{5}{3}, 1)$ and $S = \frac{34}{9} + \frac{58}{9} + \frac{40}{9} = \frac{44}{3}$. At $P = G$ the sum of squared distances is exactly $\frac{44}{3} \approx 14.67$; at $P = (0,0)$ it is $0 + 16 + 10 = 26$, and $3PG^2 + S = 3 \cdot \frac{34}{9} + \frac{44}{3} = 26$, as the identity predicts.
- Another useful form: $S = \frac{1}{3}(A_1A_2^2 + A_2A_3^2 + A_3A_1^2)$. In the example the squared sides are 16, 18 and 10, giving $\frac{44}{3}$ again.

- We checked the identity exactly, with fractions, on 2000 random triples of points, and confirmed numerically that points on the circle give the sum a^2 and that the minimum of the sum is S, attained at G.

Common slips

- Leaving the radius as $\sqrt{\left(a^2 - \sum(x_i^2 + y_i^2)\right)/3 + \bar{x}^2 + \bar{y}^2}$ without recognising it in terms of S; the final part then becomes very hard to see.
- Forgetting the degenerate case $a^2 = S$, where the locus is a single point, not a circle.

Beyond STEP

The identity $\sum PA_i^2 = 3PG^2 + \sum GA_i^2$ is a form of the parallel axis theorem for moments of inertia, and in statistics it is the decomposition of a sum of squares into "between" and "within" parts: the mean is the value that minimises the sum of squared deviations, which is the whole reason least squares fitting works. In linear algebra it is Pythagoras' theorem in disguise, because the vector from G to P is orthogonal (in the right sense) to the deviations $A_i - G$, which sum to zero. Strang's *Introduction to Linear Algebra* develops this projection viewpoint of least squares.

WORKED SOLUTION

STEP I 1999, Question 3

Pure • Cyclic Systems of Equations

STEP I 1999, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q3

The key idea

Each equation says $x_k = f(x_{k+1})$ with $f(x) = 1 + 1/x$, and the system closes up into a loop. Differences between neighbours obey a simple rule: each difference is the next one divided by a product of two numbers bigger than 1. So going round the loop the differences can only shrink. Since we come back to where we started, they must all be zero.

Throughout, read the indices cyclically: x_{n+1} means x_1 and x_{n+2} means x_2. Then every equation in the system has the single form

$$x_k = 1 + \frac{1}{x_{k+1}}, \qquad k = 1, 2, \ldots, n.$$

Solution

Part (i)

All the x_k are positive, so $1/x_{k+1} > 0$ for every k. Hence

$$x_k = 1 + \frac{1}{x_{k+1}} > 1 \qquad \text{for } k = 1, 2, \ldots, n,$$

and because of the cyclic indexing this covers every one of $x_1, \ldots, x_n$ (the last one is $x_n = 1 + 1/x_1$). So $x_1, x_2, \ldots, x_n > 1$.

Part (ii)

Subtract the second equation from the first:

$$x_1 - x_2 = \left(1 + \frac{1}{x_2}\right) - \left(1 + \frac{1}{x_3}\right) = \frac{1}{x_2} - \frac{1}{x_3} = \frac{x_3 - x_2}{x_2 x_3} = -\frac{x_2 - x_3}{x_2 x_3},$$

which is the required result. We used only $x_1 = 1 + 1/x_2$ and $x_2 = 1 + 1/x_3$ (and $n \geq 3$, so that x_3 is a different variable). Exactly the same calculation for any two consecutive

equations gives, for every k,

$$x_k - x_{k+1} = -\frac{x_{k+1} - x_{k+2}}{x_{k+1}x_{k+2}} \qquad \text{(indices cyclic)}.$$

Part (iii)

Write $d_k = x_k - x_{k+1}$ for the differences between neighbours, so $d_n = x_n - x_1$. The general form of (ii) reads $d_k = -d_{k+1}/(x_{k+1}x_{k+2})$. Taking absolute values, and using part (i), which gives $x_{k+1}x_{k+2} > 1$,

$$|d_k| = \frac{|d_{k+1}|}{x_{k+1}x_{k+2}} \leq |d_{k+1}|,$$

with *strict* inequality whenever $d_{k+1} \neq 0$. Chaining these round the loop,

$$|d_1| \leq |d_2| \leq |d_3| \leq \cdots \leq |d_n| \leq |d_1|.$$

The two ends are the same, so every inequality in the chain is an equality. In particular $|d_1| = |d_2|$. But if $d_2 \neq 0$ we would have $|d_1| < |d_2|$ strictly, so $d_2 = 0$, and then all the $|d_k|$, being equal to $|d_2|$, are zero.

Ask yourself Where exactly did we use that the system closes up, with $x_n = 1 + 1/x_1$?

Only in the last link of the chain, $|d_n| \leq |d_1|$, which comes from $x_n = 1 + 1/x_1$. Without it the differences could simply grow as k increases, and indeed an open chain $x_1, \ldots, x_n$ need not be constant. It is the return to the start that forces the shrinking to be no shrinking at all.

So $x_k = x_{k+1}$ for every k, that is, $x_1 = x_2 = \cdots = x_n$.

The value of x_1

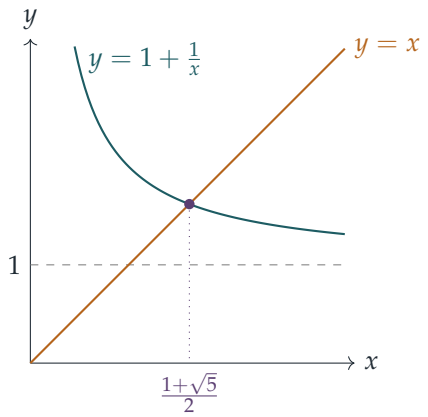
Call the common value x. Any one of the equations now says $x = 1 + 1/x$, so

$$x^2 - x - 1 = 0, \qquad x = \frac{1 \pm \sqrt{5}}{2}.$$

The negative root is ruled out because $x > 1$ by part (i), so

$$x_1 = \frac{1 + \sqrt{5}}{2}.$$

Conversely, setting every x_k equal to this value does satisfy all n equations, so the system really has this solution, and by (iii) it has no other. Geometrically, x is where the curve $y = 1 + 1/x$ meets the line $y = x$:



Checks

- With $x = \frac{1+\sqrt{5}}{2} \approx 1.6180$ we get $1 + 1/x \approx 1 + 0.6180 = 1.6180$, as required.
- The identity in (ii) can be tested on any open chain: take $x_3 = 2$, so $x_2 = \frac{3}{2}$ and $x_1 = \frac{5}{3}$. Then $x_1 - x_2 = \frac{1}{6}$ and $-(x_2 - x_3)/(x_2 x_3) = \frac{1/2}{3} = \frac{1}{6}$.
- Going all the way round the loop, $x_1 = f^n(x_1)$ with $f(x) = 1 + 1/x$. A numerical scan of $f^n(x) - x$ on $(0, 20)$ for $n = 3, \ldots, 11$ finds a single sign change, at $x \approx 1.618$, confirming that there is no other solution.

Common slips

- Proving (iii) from $|d_1| \leq |d_2|$ alone. One step only says the differences don't grow; you need to go all the way round the loop and come back to $|d_1|$.
- Assuming the x_k are equal and then finding x_1. Part (iii) is what justifies that, and it needs the strict inequality $x_{k+1}x_{k+2} > 1$ from part (i).
- Forgetting to reject $\frac{1-\sqrt{5}}{2}$, which is negative.

Beyond STEP

The heart of this question is that $f(x) = 1 + 1/x$ brings points closer together on $x > 1$, since $|f(a) - f(b)| = |a - b|/(ab)$. A map like this is a *contraction*, and the Banach fixed-point theorem says that a contraction of a complete space has exactly one fixed point, which iteration from any start converges to. Iterating here from $x = 1$ gives the ratios of consecutive Fibonacci numbers, $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$, and the continued fraction $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}$ for the golden ratio. For the analysis, read Abbott, *Understanding Analysis*. For continued fractions, read Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 1999, Question 4

Pure • Regions with Moduli

STEP I 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q4

The key idea

Modulus signs split the plane into regions where each bracket has a fixed sign, and in each region the condition becomes an ordinary linear (or, in the last part, hyperbolic) inequality. The other big time-saver is symmetry and translation: $|x|$ and $|y|$ make a set symmetric in both axes, so you only need to work in one quadrant and reflect; and replacing x by $x-1$ simply slides the picture one unit to the right.

Solution

Part (i)

The set is unchanged if we replace x by $-x$ or y by $-y$, so it is symmetric in both coordinate axes. In the first quadrant ($x \geq 0$, $y \geq 0$) the condition is just $x+y \leq 1$: the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$. Reflecting in both axes gives the four triangles that make up a square standing on one corner, with vertices

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,-1).$$

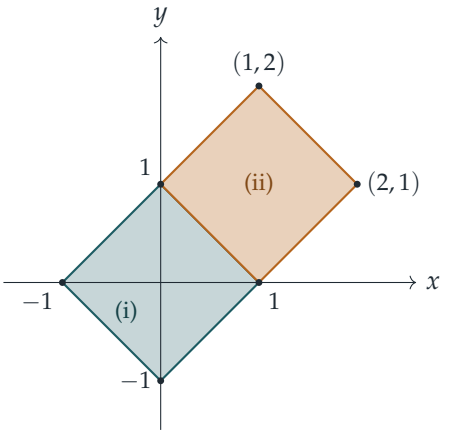
Equivalently, it is the intersection of the four half-planes $\pm x \pm y \leq 1$. The boundary is included (the inequality is $\leq$), and the area is 2.

Part (ii)

Put $X = x-1$, $Y = y-1$. Then the condition is $|X|+|Y| \leq 1$, which is part (i) in the (X,Y) coordinates. So the set is the square of part (i) translated by $(1,1)$: centre $(1,1)$, vertices

$$(2,1), \quad (1,2), \quad (0,1), \quad (1,0),$$

bounded by the lines $x+y=1$, $x+y=3$, $y=x+1$ and $y=x-1$. It touches both axes, at $(1,0)$ and $(0,1)$.



Part (iii)

Again shift: put $X = x - 1$, $Y = y + 1$, so the condition becomes

$$|X| \leq 1 + |Y|.$$

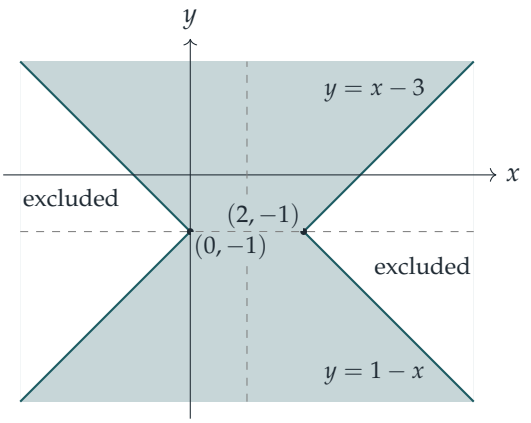
This is symmetric in both the X- and Y-axes, i.e. about the lines $x = 1$ and $y = -1$. In the quadrant $X \geq 0$, $Y \geq 0$ it reads $X \leq 1 + Y$: everything to the left of the line $X = 1 + Y$, which passes through $(1, 0)$ with gradient 1. Reflecting, the set consists of every point *except* two open wedges:

- on the right, $X > 1 + |Y|$: a right-angled wedge with apex $(X, Y) = (1, 0)$, opening to the right, bounded by $Y = \pm(X - 1)$;
- on the left, $X < -1 - |Y|$: its mirror image, with apex $(-1, 0)$, opening to the left.

Ask yourself Is this set bounded, as in (i) and (ii)?

No. The minus sign changes everything: making $|Y|$ large makes the condition *easier* to satisfy, so for each fixed Y the allowed X form the interval $-1 - |Y| \leq X \leq 1 + |Y|$, which widens as $|Y|$ grows. In particular the whole vertical line $X = 0$ (that is, $x = 1$) lies in the set.

Back in x, y coordinates the apexes are $(2, -1)$ and $(0, -1)$. The excluded right wedge is $x > 2 + |y + 1|$, bounded by the half-lines $y = x - 3$ and $y = 1 - x$ for $x \geq 2$; the excluded left wedge is $x < -|y + 1|$, bounded by $y = x - 1$ and $y = -1 - x$ for $x \leq 0$. The boundary lines themselves belong to the set. On the line $y = -1$ the set is just the segment $0 \leq x \leq 2$, its narrowest point.



The dashed lines $x = 1$ and $y = -1$ are the lines of symmetry.

Part (iv)

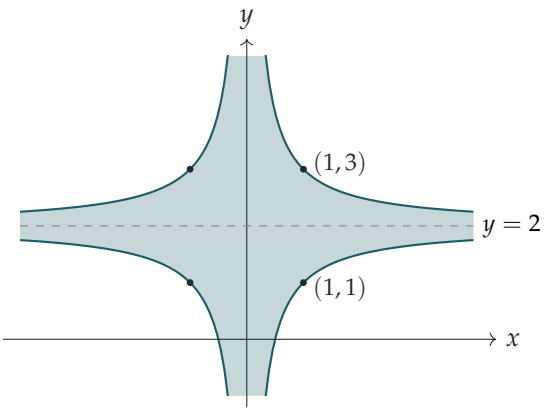
Put $Y = y - 2$, so the condition is $|x|\,|Y| \leq 1$, which is symmetric in both the x-axis and the line $Y = 0$ (that is, $y = 2$). In the quadrant $x \geq 0$, $Y \geq 0$:

- if $x = 0$ the condition holds for every Y, so the whole y-axis is in the set;
- if $x > 0$ we may divide by x, and the condition is $0 \leq Y \leq 1/x$: the region under the hyperbola $Y = 1/x$.

Reflecting in both lines of symmetry, the set is

$$x = 0, \quad \text{or} \quad 2 - \frac{1}{|x|} \leq y \leq 2 + \frac{1}{|x|} \quad (x \neq 0),$$

the region trapped between the four branches of the hyperbolas $(y-2)x = \pm 1$. It contains the whole y-axis and the whole line $y = 2$, and these are the asymptotes of the boundary curves. It looks like a fat plus sign centred at $(0, 2)$: the corners of the "square" $|x| \leq 1$, $|y - 2| \leq 1$ lie on the boundary at $(\pm 1, 1)$ and $(\pm 1, 3)$, the arms narrow towards $y = 2$ as $|x| \to \infty$, and the vertical arms narrow towards the y-axis as $|y| \to \infty$. The set is unbounded (but, curiously, has infinite area).



Checks

- Test a few points against the original inequalities. In (iii), $(1, 100)$ gives $0 - 101 \leq 1$, so it is in the set (confirming it is unbounded), while $(2.01, -1)$ gives $1.01 > 1$, just outside the right-hand apex. In (iv), $(2, 2.5)$ gives $2 \times 0.5 = 1$, on the boundary, and $(2, 2.51)$ is just outside.
- In (i), each of the four triangles has area $\frac{1}{2}$, so the square has area 2; a grid count gives 2.002.
- We tested 400,000 random points in $[-6, 6]^2$ for each part, comparing the original inequality with our description (half-planes, excluded wedges, or $|y - 2| \leq 1/|x|$); there were no disagreements.

Common slips

- In (ii) and (iii), shifting the wrong way: $|x - 1|$ moves the picture to $x = +1$, and $|y + 1|$ moves it to $y = -1$.
- In (iii), drawing a bounded diamond by analogy with (i). The minus sign reverses the role of $|y + 1|$, and the set is unbounded.
- In (iv), dividing by $|x|$ without treating $x = 0$ separately, and so missing the fact that the whole y-axis belongs to the set.

Beyond STEP

The set in (i) is the unit ball for the "taxicab" norm $\|(x, y)\|_1 = |x| + |y|$, and (ii) is the ball of radius 1 centred at $(1, 1)$. Different norms give differently shaped unit balls (a circle for the usual distance, a square with horizontal sides for $\max(|x|, |y|)$), and a first course in analysis or metric spaces shows that on $\mathbb{R}^2$ they all define the same notions of convergence and continuity. Bartle and Sherbert's *Introduction to Real Analysis* is a good place to meet these ideas.

WORKED SOLUTION

STEP I 1999, Question 5

Pure • Small Angle Approximations

STEP I 1999, Question 5

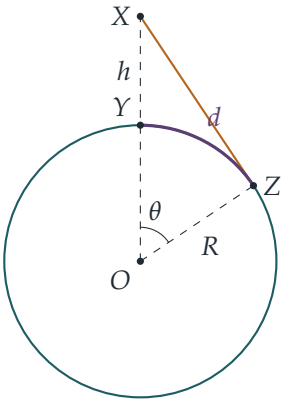
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q5

The key idea

"Just seen on the horizon" means the line of sight from Z to X grazes the Earth, so it is a tangent at Z and is perpendicular to the radius OZ. That gives a right-angled triangle with the centre O, and the arc YZ is R times the angle at O. The two approximations then come from looking at that angle when it is small, and when it is close to a right angle.

Solution



Part (i)

Let O be the centre of the Earth and let $\theta = \angle YOZ$. Since X is directly above Y, the points O, Y, X are collinear with $OX = R + h$. Since X is just visible from Z, the line ZX touches the Earth at Z, so it is perpendicular to the radius OZ. In the right-angled triangle OZX,

$$\cos\theta = \frac{OZ}{OX} = \frac{R}{R+h}.$$

The arc YZ of a circle of radius R subtending angle θ at the centre has length $R\theta$, so

$$d = R\arccos\left(\frac{R}{R+h}\right).$$

Here $0 < \theta < \frac{\pi}{2}$, since $0 < R/(R+h) < 1$.

Part (ii)

When h is small compared with R, $\cos\theta = R/(R+h)$ is close to 1, so θ is small and we may use $\cos\theta \approx 1 - \frac{1}{2}\theta^2$. Also

$$\frac{R}{R+h} = 1 - \frac{h}{R+h} \approx 1 - \frac{h}{R},$$

since $R + h \approx R$. Equating,

$$1 - \tfrac{1}{2}\theta^2 \approx 1 - \frac{h}{R} \quad\Longrightarrow\quad \theta^2 \approx \frac{2h}{R} \quad\Longrightarrow\quad \theta \approx \sqrt{\frac{2h}{R}},$$

taking the positive root as $\theta > 0$. Hence

$$d = R\theta \approx R\sqrt{\frac{2h}{R}} = \sqrt{2}\,(Rh)^{1/2},$$

which is the required form with $k = \sqrt{2}$.

Part (iii)

Now $R/(R+h)$ is small, so $\cos\theta$ is near 0 and θ is close to $\frac{\pi}{2}$, not small. The given approximations only work for small angles, so we measure the angle that *is* small.

Ask yourself Which angle is small here, and which approximation applies to it?

Write $\theta = \frac{\pi}{2} - \phi$, where $\phi = \angle OXZ$ is the angle at the satellite, which is small when the satellite is far away. Then $\cos\theta = \sin\phi$, so

$$\sin\phi = \frac{R}{R+h}.$$

We need this to second order in R/h, because the answer has an R^2/h term. By the binomial expansion, valid as $R/h < 1$,

$$\frac{R}{R+h} = \frac{R}{h}\left(1 + \frac{R}{h}\right)^{-1} = \frac{R}{h} - \frac{R^2}{h^2} + \cdots.$$

With $\sin\phi \approx \phi$,

$$\phi \approx \frac{R}{h} - \frac{R^2}{h^2}.$$

Therefore

$$\begin{aligned} d = R\theta &= R\left(\frac{\pi}{2} - \phi\right) \\ &\approx \frac{\pi}{2}R - \frac{R^2}{h} + \frac{R^3}{h^2}. \end{aligned}$$

The last term is smaller than R^2/h by a further factor of R/h, so we drop it, giving

$$d \approx \frac{\pi}{2}R - \frac{R^2}{h},$$

which is the required form with $a = \frac{\pi}{2}$ and $b = -1$.

Using $\sin\phi \approx \phi$ is safe to this order: the error in it is about $\frac{1}{6}\phi^3$, which is of order R^3/h^3, so it changes d only at order R^4/h^3.

Checks

- Limits make sense. As $h \to 0$, $d \to 0$. As $h \to \infty$, $d \to \frac{\pi}{2}R$, a quarter of the way round the Earth: a very distant satellite sees almost a full hemisphere. The term $-R^2/h$ is negative, so d approaches $\frac{\pi}{2}R$ from below, as it must since $\theta < \frac{\pi}{2}$.
- Numerically, with $R = 1$ and $h = 10^{-4}$, the exact d divided by $\sqrt{2Rh}$ is 0.99996. With $h = 1$ and $R = 0.01$, the exact d and $\frac{\pi}{2}R - R^2/h$ differ by 9.9×10^{-7}, which is about R^3/h^2, as predicted.
- For a satellite 300 km above an Earth of radius 6370 km, the exact d is about 1918 km and $\sqrt{2Rh}$ gives 1955 km.

Common slips

- Using $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ in part (iii), where θ is near $\frac{\pi}{2}$ and not small.
- Expanding $R/(R+h)$ only to first order in part (iii), which loses the R^2/h term and gives the wrong b.

Beyond STEP

Both parts are asymptotic expansions: the exact d is expanded in powers of a small parameter, h/R or R/h, and we keep terms up to a stated order while bounding the error. Making "$\approx$" precise with big-O notation and Taylor's theorem with remainder is a first-year analysis topic; see Abbott, *Understanding Analysis*. Siklos, *Advanced Problems in Mathematics*, has more STEP practice in this kind of approximation.

WORKED SOLUTION

STEP I 1999, Question 6

Pure • Extrema of Quadratics

STEP I 1999, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q6

The key idea

A straight line on a closed interval has its extreme values at the ends, and which end gives which depends only on the sign of the gradient. For a quadratic with $c > 0$ the graph is a $\cup$-shaped parabola: the greatest value is still at an end, but the least value is at the vertex *if the vertex lies in the interval*, and at the nearer end otherwise. So the whole question comes down to asking where the vertex $x = -b/(2c)$ sits relative to $[-10, 10]$.

Solution

Part (i)

Write $f(x) = bx + a$, so $f(10) = a + 10b$ and $f(-10) = a - 10b$.

- If $b > 0$, f is strictly increasing, so the least value is $f(-10) = a - 10b$ (at $x = -10$) and the greatest is $f(10) = a + 10b$ (at $x = 10$).
- If $b = 0$, $f(x) = a$ for every x, so the greatest and least values are both a.
- If $b < 0$, f is strictly decreasing, so the greatest value is $f(-10) = a - 10b$ (at $x = -10$) and the least is $f(10) = a + 10b$ (at $x = 10$).

All three cases are summed up by

$$\text{greatest} = a + 10|b|, \qquad \text{least} = a - 10|b|.$$

Part (ii)

Let $g(x) = cx^2 + bx + a$ with $c \geq 0$. If $c = 0$ this is part (i), so the answer is $a \pm 10|b|$ as above. From now on take $c > 0$ and complete the square:

$$g(x) = c\left(x + \frac{b}{2c}\right)^2 + a - \frac{b^2}{4c}.$$

The graph is a ∪-shaped parabola with vertex at $x_0 = -\dfrac{b}{2c}$, where g takes its smallest value over *all* real x. Also g decreases for $x < x_0$ and increases for $x > x_0$. The end values are

$$g(10) = a + 100c + 10b, \qquad g(-10) = a + 100c - 10b.$$

Greatest value.

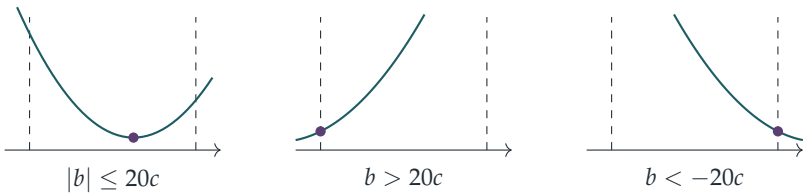
Ask yourself Why must the greatest value of g on the interval be at one of the ends, wherever the vertex is?

Because $c > 0$, $g(x)$ depends on x only through the distance $|x - x_0|$, and it increases as that distance increases. On $[-10, 10]$ the distance $|x - x_0|$ is largest at one of the ends (whichever is further from x_0). So the greatest value is the larger of $g(10)$ and $g(-10)$, and these differ by $g(10) - g(-10) = 20b$. Hence:

- if $b > 0$, the greatest value is $a + 100c + 10b$, at $x = 10$;
- if $b < 0$, the greatest value is $a + 100c - 10b$, at $x = -10$;
- if $b = 0$, the greatest value is $a + 100c$, at both ends.

In every case, greatest $= a + 100c + 10|b|$. This also holds when $c = 0$.

Least value. Now it matters where x_0 is. The vertex lies in the interval exactly when $-10 \le -\frac{b}{2c} \le 10$, that is (multiplying by $2c > 0$) when $-20c \le b \le 20c$, or $|b| \le 20c$.



The dashed lines are $x = \pm 10$, and the dot marks the least value on the interval.

- $|b| \le 20c$ **(vertex in the interval).** The least value is the global minimum,

$$a - \frac{b^2}{4c}, \quad \text{at } x = -\frac{b}{2c}.$$

- $b > 20c$ **(vertex to the left, $x_0 < -10$).** On $[-10, 10]$ we have $x > x_0$, so g is increasing there and the least value is $g(-10) = a + 100c - 10b$, at $x = -10$.
- $b < -20c$ **(vertex to the right, $x_0 > 10$).** Now g is decreasing on the whole interval, so the least value is $g(10) = a + 100c + 10b$, at $x = 10$.

The two outside cases combine to give least $= a + 100c - 10|b|$ when $|b| > 20c$.

Ask yourself What happens on the boundary $|b| = 20c$? Do the two formulas clash there?

No. If $b = 20c$ then $a - \frac{b^2}{4c} = a - 100c$ and $a + 100c - 10b = a - 100c$ as well, because the vertex is then exactly at $x = -10$. The case $b = -20c$ works the same way. So the answer changes continuously as b crosses $\pm 20c$, and it doesn't matter which case includes the boundary.

Summary. For $-10 \le x \le 10$ and $c \ge 0$:

$$\text{greatest} = a + 100c + 10|b|, \qquad \text{least} = \begin{cases} a - \dfrac{b^2}{4c}, & c > 0 \text{ and } |b| \le 20c, \\ a + 100c - 10|b|, & |b| > 20c, \\ a, & b = c = 0. \end{cases}$$

The case $c = 0$, $b \neq 0$ falls under $|b| > 20c = 0$ and gives back part (i).

Checks

- Take $g(x) = x^2 + 4x + 1$: here $|b| = 4 \le 20$, so the least value is $1 - 4 = -3$ at $x = -2$, and the greatest is $1 + 100 + 40 = 141$ at $x = 10$. Direct substitution agrees.
- Take $g(x) = x^2 + 30x$: here $b = 30 > 20$, so the vertex $x = -15$ is outside the interval and the least value is $g(-10) = 100 - 300 = -200$. Indeed $g(-10) = -200$ and g is increasing on $[-10, 10]$.
- A short computer check compared the formulas with a brute-force search over 401 points in $[-10, 10]$ (exact rational arithmetic) for 1000 random (a, b, c), including $c = 0$ and the boundary $|b| = 20c$. All cases agree.

Common slips

- Quoting $a - \frac{b^2}{4c}$ as the least value without checking that the vertex lies in $[-10, 10]$.
- Forgetting the cases $c = 0$ (no vertex, and you can't divide by c) and $b = c = 0$.
- Mixing up which end gives the least value when the vertex is outside: it's the end *nearer* the vertex.

Beyond STEP

The fact behind this question is the Extreme Value Theorem: a continuous function on a closed, bounded interval attains a greatest and a least value, and these occur either at an end or at a stationary point inside. The observation that a convex function

such as $cx^2 + bx + a$ with $c \geq 0$ attains its maximum at an end is the first step in convex optimisation, which is behind much of modern applied mathematics. Abbott's *Understanding Analysis* proves the Extreme Value Theorem carefully.

WORKED SOLUTION

STEP I 1999, Question 7

Pure • Second-Order Differential Equations

STEP I 1999, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q7

The key idea

The function $\sin(k\sin^{-1}x)$ looks awkward to differentiate twice, but it becomes tame once we write $x = \sin\theta$, so that $y = \sin k\theta$. Multiplying the first derivative by $\sqrt{1-x^2}$ before differentiating again keeps the algebra short. For the last part, the point is that a second-order differential equation with a given value and gradient at one point has only one solution, so the polynomial we find *must* be $\sin(3\sin^{-1}x)$.

Solution

The differential equation

Let $y = \sin(k\sin^{-1}x)$ for $-1 < x < 1$. By the chain rule, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\sin^{-1}x = \dfrac{1}{\sqrt{1-x^2}}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k\cos(k\sin^{-1}x)}{\sqrt{1-x^2}}, \qquad \text{so} \qquad \sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = k\cos(k\sin^{-1}x).$$

Differentiating this second form with respect to x (product rule on the left, chain rule on the right):

$$\sqrt{1-x^2}\,\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{x}{\sqrt{1-x^2}}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{k^2\sin(k\sin^{-1}x)}{\sqrt{1-x^2}} = -\frac{k^2y}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$, which is non-zero for $-1 < x < 1$, gives

$$(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} + k^2y = 0,$$

which is $(*)$, as required.

The cubic solution when $k = 3$

Put $y = Ax^3 + Bx^2 + Cx + D$. The condition $y = 0$ at $x = 0$ gives $D = 0$, and $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$ at $x = 0$ gives $C = 3$. Now

$$y' = 3Ax^2 + 2Bx + C, \qquad y'' = 6Ax + 2B,$$

and substituting into $(*)$ with $k^2 = 9$:

$$(1 - x^2)(6Ax + 2B) - x(3Ax^2 + 2Bx + C) + 9(Ax^3 + Bx^2 + Cx + D) = 0.$$

This must hold for all x, so we compare coefficients:

$$\begin{aligned} x^3: &\quad -6A - 3A + 9A = 0 \quad \text{(true for every } A\text{)},\\ x^2: &\quad -2B - 2B + 9B = 5B = 0,\\ x^1: &\quad 6A - C + 9C = 6A + 8C = 0,\\ x^0: &\quad 2B + 9D = 0. \end{aligned}$$

So $B = 0$ (consistent with $D = 0$), and $6A = -8C = -24$, giving $A = -4$. The required solution is

$$y = 3x - 4x^3.$$

Expressing $\sin 3\theta$

Ask yourself We have two solutions of the same equation: $\sin(3\sin^{-1} x)$ and $3x - 4x^3$. Why must they be the same function?

Both satisfy $(*)$ with $k = 3$, and both satisfy the same conditions at $x = 0$. For the polynomial we built this in. For $y = \sin(3\sin^{-1} x)$ we have $y(0) = \sin 0 = 0$ and, from the first derivative above, $y'(0) = 3\cos 0/1 = 3$.

On $-1 < x < 1$ we may divide $(*)$ by $1 - x^2 \neq 0$. This gives $y'' = \dfrac{xy' - 9y}{1 - x^2}$, a linear second-order equation with continuous coefficients, and such an equation has exactly one solution with a given value of y and y' at $x = 0$. (Think of it as stepping out from $x = 0$: the equation tells you y'' from y and y', so the starting values fix everything.) Hence

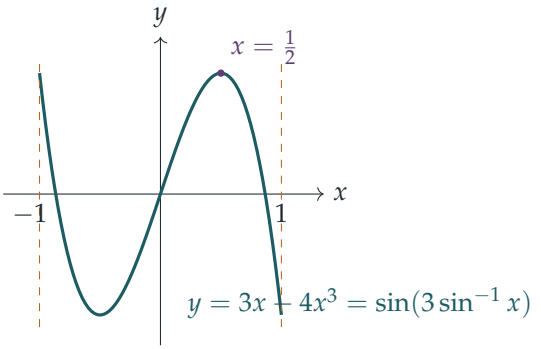
$$\sin(3\sin^{-1} x) = 3x - 4x^3 \qquad \text{for } -1 < x < 1,$$

and by continuity also at $x = \pm 1$.

Now put $x = \sin\theta$ with $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$, so that $\sin^{-1} x = \theta$:

$$\boxed{\sin 3\theta = 3\sin\theta - 4\sin^3\theta.}$$

This holds for every θ, not just $|\theta| \le \frac{\pi}{2}$. Both sides have period 2π. If $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$, put $\varphi = \pi - \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $\sin\varphi = \sin\theta$ and $\sin 3\varphi = \sin(3\pi - 3\theta) = \sin 3\theta$, so the identity for φ gives it for θ.



The graph is the familiar curve $\sin 3\theta$ seen through $x = \sin\theta$: as θ runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, 3θ sweeps out one and a half oscillations, hence the two turning points at $x = \pm\frac{1}{2}$ (where $3\theta = \pm\frac{\pi}{2}$).

Checks

- $\theta = \frac{\pi}{2}$: $\sin\frac{3\pi}{2} = -1$ and $3 - 4 = -1$. $\quad\theta = \frac{\pi}{6}$: $\sin\frac{\pi}{2} = 1$ and $\frac{3}{2} - \frac{1}{2} = 1$.
- Numerically, at $\theta = 0.7$: $\sin 2.1 = 0.86321$ and $3(0.64422) - 4(0.64422)^3 = 0.86321$.
- The same identity follows from $\sin(2\theta + \theta)$ and the double-angle formulae, which is a good independent check.

Common slips

- Differentiating $\dfrac{k\cos(k\sin^{-1}x)}{\sqrt{1-x^2}}$ directly with the quotient rule: correct, but messy. Clearing the square root first is much cleaner.
- Forgetting to say *why* the cubic equals $\sin(3\sin^{-1}x)$. Matching the equation alone is not enough: both solutions must also share $y(0)$ and $y'(0)$.

Beyond STEP

Equation $(*)$ is Chebyshev's equation. When $k = n$ is a whole number it has a polynomial solution, and $\cos(n\cos^{-1}x)$ is the Chebyshev polynomial $T_n(x)$, central in numerical approximation. The uniqueness theorem we used (the Picard–Lindelöf theorem) is proved in first-year differential equations courses. Power-series methods for equations like this are treated in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP I 1999, Question 8

Pure • Harmonic Series Bounds

STEP I 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q8

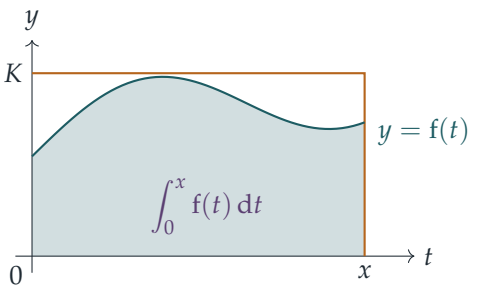
The key idea

An integral is an area, so trapping a curve inside a rectangle traps its area between 0 and the rectangle's area. The suggested integrand splits by partial fractions into something whose integral is exactly $\ln\left(\frac{n}{n-1}\right) - \frac{1}{n}$, and the bounds on the integrand give the inequality. After that, everything telescopes: adding up over n turns a chain of logarithms into $\ln N$ and a chain of differences of reciprocals into almost nothing.

Solution

Bounding an integral

Draw the graph of $y = \mathrm{f}(t)$ for $0 \le t \le x$. Because $0 \le \mathrm{f}(t) \le K$, the whole curve lies inside the rectangle with corners $(0, 0)$ and (x, K), on or above the t-axis.



The integral is the shaded area under the curve. That area can't be negative, since the curve never dips below the axis, and it can't exceed the area Kx of the rectangle, since the region sits inside it. So

$$0 \le \int_0^x \mathrm{f}(t)\,\mathrm{d}t \le Kx.$$

The inequality for $\ln\left(\frac{n}{n-1}\right)$

Take $n > 1$. For $0 \le t \le 1$ we have $n - t \ge n - 1 > 0$, so the integrand is defined. Splitting it into partial fractions,

$$\frac{t}{n(n-t)} = \frac{1}{n} \cdot \frac{n-(n-t)}{n-t} = \frac{1}{n-t} - \frac{1}{n},$$

so

$$\int_0^1 \frac{t}{n(n-t)}\,\mathrm{d}t = \Big[-\ln(n-t) - \frac{t}{n}\Big]_0^1 = \ln n - \ln(n-1) - \frac{1}{n} = \ln\Big(\frac{n}{n-1}\Big) - \frac{1}{n}.$$

Now we bound the integrand on $[0,1]$. The numerator t is non-negative and the denominator $n(n-t)$ is positive, so the integrand is at least 0. As t increases from 0 to 1 the numerator grows and the positive denominator shrinks, so the integrand is increasing and its largest value is at $t = 1$:

$$0 \le \frac{t}{n(n-t)} \le \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Using the first result with $x = 1$ and $K = \frac{1}{n-1} - \frac{1}{n}$,

$$0 \le \ln\Big(\frac{n}{n-1}\Big) - \frac{1}{n} \le \frac{1}{n-1} - \frac{1}{n},$$

as required.

Summing from $n = 2$ to N

Add the inequality for $n = 2, 3, \ldots, N$ (where $N \ge 2$). The logarithms telescope, because

$$\sum_{n=2}^{N} \ln\Big(\frac{n}{n-1}\Big) = \ln\Big(\frac{2}{1} \cdot \frac{3}{2} \cdots \frac{N}{N-1}\Big) = \ln N,$$

and so do the right-hand sides:

$$\sum_{n=2}^{N}\Big(\frac{1}{n-1} - \frac{1}{n}\Big) = \Big(1 - \frac{1}{2}\Big) + \Big(\frac{1}{2} - \frac{1}{3}\Big) + \cdots + \Big(\frac{1}{N-1} - \frac{1}{N}\Big) = 1 - \frac{1}{N}.$$

Therefore

$$0 \le \ln N - \sum_{n=2}^{N} \frac{1}{n} \le 1 - \frac{1}{N} \le 1.$$

(For $N = 1$ the sum is empty and $\ln 1 = 0$, so the statement holds then too.)

> **Ask yourself** Which half of this double inequality tells us the harmonic series diverges, and which half gives an upper bound?

The left half says the sum is *at most* $\ln N$; the right half says it is *at least* $\ln N - 1$. For divergence we need a lower bound, so we use the right half.

Divergence

From $\ln N - \sum_{n=2}^{N} \frac{1}{n} \leq 1$ we get $\sum_{n=2}^{N} \frac{1}{n} \geq \ln N - 1$, and adding the $n = 1$ term,

$$\sum_{n=1}^{N} \frac{1}{n} \geq \ln N.$$

Since $\ln N \to \infty$ as $N \to \infty$, the sum also tends to infinity.

The bound for $N < 10^{30}$

Now we use the left half: $\sum_{n=2}^{N} \frac{1}{n} \leq \ln N$, so

$$\sum_{n=1}^{N} \frac{1}{n} \leq 1 + \ln N.$$

Since $2^{10} = 1024 > 1000 = 10^3$, raising to the tenth power gives $10^{30} < 2^{100}$. So if $N < 10^{30}$ then $N < 2^{100}$ and, as ln is increasing,

$$\ln N < 100 \ln 2 < 100,$$

using $\ln 2 < 1$, which holds because $2 < \mathrm{e}$. Hence

$$\sum_{n=1}^{N} \frac{1}{n} \leq 1 + \ln N < 101.$$

Checks

- Simpson's rule on $\int_0^1 \frac{t}{n(n-t)}\,\mathrm{d}t$ agrees with $\ln\left(\frac{n}{n-1}\right) - \frac{1}{n}$ for many values of $n > 1$. For example, with $n = 2$ both give $\ln 2 - \frac{1}{2} \approx 0.193$, which lies between 0 and $\frac{1}{2}$.
- For every N up to 2×10^5, $\ln N - \sum_{n=2}^{N} \frac{1}{n}$ lies in $[0, 1]$. At $N = 2 \times 10^5$ it is about 0.4228, close to $1 - \gamma \approx 0.4228$, where γ is Euler's constant.
- The bound is generous: $1 + 30 \ln 10 \approx 70.1$, and in fact $\sum_{n=1}^{10^{30}} \frac{1}{n} \approx 69.65$. The harmonic series diverges, but extremely slowly.

Common slips

- Using the wrong half of the double inequality: divergence needs the *lower* bound on the sum, and the final part needs the *upper* bound.
- Forgetting the $n = 1$ term: the inequalities are about $\sum_{n=2}^{N}$, while the conclusions are about $\sum_{n=1}^{N}$.
- Not justifying that the integrand's maximum is at $t = 1$. Saying it is increasing (or simply that $t \leq 1$ and $n - t \geq n - 1$) is all it takes.

Beyond STEP

The quantity $\sum_{n=1}^{N} \frac{1}{n} - \ln N$ decreases and is bounded below, so it converges. Its limit is Euler's constant $\gamma \approx 0.5772$, and nobody yet knows whether γ is rational. Comparing a sum with an integral, as we did here, is the *integral test*, one of the first tools for series in a first-year analysis course. Abbott's *Understanding Analysis* is a good place to see it made rigorous.

WORKED SOLUTION

STEP I 1999, Question 9

Mechanics • Kinematics at Constant Speed

STEP I 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

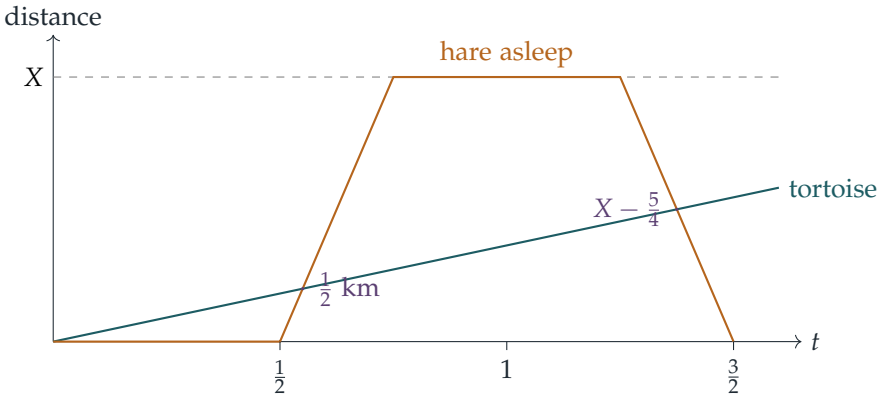
step.maths.org/questions/99-s1-q9

The key idea

Everything happens at constant speeds, so every journey time is just distance over speed. Each time the animals meet they are at the same place at the same moment, so we write down the time on the clock for each of them and set the two equal. Two meetings give two equations in v and V. Working with $1/v$ and $1/V$ (hours per kilometre) keeps the equations linear.

Solution

We model both animals as moving at constant speed along the same straight course, ignoring any time to turn round. We measure time t in hours from the start. A distance–time sketch (drawn here for the value of X found at the end) shows the two crossings we use.



The first meeting

The tortoise reaches the point $\frac{1}{2}$ km from the start at time $\dfrac{1/2}{v}$. The hare starts at $t = \frac{1}{2}$ and reaches the same point at time $\frac{1}{2} + \dfrac{1/2}{V}$. They are there together, so

$$\frac{1}{2v} = \frac{1}{2} + \frac{1}{2V}, \qquad \text{that is} \qquad \frac{1}{v} = 1 + \frac{1}{V}. \tag{1}$$

The second meeting

The hare reaches the vegetable patch at time $\frac{1}{2} + \dfrac{X}{V}$, sleeps for half an hour, and leaves at $1 + \dfrac{X}{V}$. She then runs $\frac{5}{4}$ km back, so she meets the tortoise at time

$$1 + \frac{X}{V} + \frac{5}{4V} = 1 + \frac{X + \frac{5}{4}}{V}.$$

The tortoise is still on the way out, $X - \frac{5}{4}$ km from the start, so she has been walking for $\dfrac{X - \frac{5}{4}}{v}$ hours. Setting the two times equal:

$$\frac{X - \frac{5}{4}}{v} = 1 + \frac{X + \frac{5}{4}}{V}. \tag{2}$$

Ask yourself We have two equations and two unknowns, but (2) has $1/v$ multiplied by X. How do we get V on its own?

We substitute (1) into (2). The X/V terms then cancel, which is what makes V come out so neatly:

$$\begin{aligned}
\left(X - \tfrac{5}{4}\right)\left(1 + \frac{1}{V}\right) &= 1 + \frac{X + \frac{5}{4}}{V} \\
X - \tfrac{5}{4} + \frac{X - \frac{5}{4}}{V} &= 1 + \frac{X + \frac{5}{4}}{V} \\
X - \tfrac{9}{4} &= \frac{5/2}{V}.
\end{aligned}$$

Hence

$$V = \frac{5/2}{X - \frac{9}{4}} = \frac{10}{4X - 9},$$

as required. (For this to be a positive speed we need $X > \frac{9}{4}$, which we keep in mind for the last part.)

Then from (1),

$$\frac{1}{v} = 1 + \frac{4X - 9}{10} = \frac{4X + 1}{10}, \qquad \text{so} \qquad v = \frac{10}{4X + 1}.$$

Finding X

The hare leaves the patch at $1 + \dfrac{X}{V}$ and takes another $\dfrac{X}{V}$ hours to get home, so she finishes at $1 + \dfrac{2X}{V}$. Setting this equal to $\frac{3}{2}$ gives $\dfrac{2X}{V} = \frac{1}{2}$, so $V = 4X$. Therefore

$$\begin{aligned} 4X &= \frac{10}{4X - 9} \\ 16X^2 - 36X - 10 &= 0 \\ 8X^2 - 18X - 5 &= 0 \\ (2X - 5)(4X + 1) &= 0. \end{aligned}$$

The root $X = -\frac{1}{4}$ is impossible for a distance (and anyway we need $X > \frac{9}{4}$), so

$$X = \tfrac{5}{2} \text{ km}.$$

With this value the hare's speed is $V = 10\,\text{km/h}$ and the tortoise's is $v = \frac{10}{11}\,\text{km/h}$.

Checks

Put $X = \frac{5}{2}$, $V = 10$, $v = \frac{10}{11}$ back into the story. The tortoise reaches $\frac{1}{2}$ km at $t = 0.55$, and the hare, starting at $t = 0.5$, needs 0.05 h to get there: they meet. The hare reaches the patch at $t = 0.75$ and leaves at $t = 1.25$. She reaches 1.25 km from the patch (which is 1.25 km from the start) at $t = 1.375$, and the tortoise reaches 1.25 km at $1.25 \times 1.1 = 1.375$: they meet again. The hare gets home at $1.25 + 0.25 = 1.5$. Substituting random values of $X > \frac{9}{4}$ into the formulas for V and v and replaying the race also gives meetings at exactly the right places.

Common slips

- Forgetting one of the hare's half-hour sleeps, especially the one at the patch, when writing her clock time.
- Taking the second meeting point as $\frac{5}{4}$ km from the *start* rather than from the vegetable patch.
- Keeping both roots of the quadratic: a negative distance must be rejected, with a reason.

Beyond STEP

A distance–time graph like the one above turns a word problem into a question about where straight lines cross, which is the starting point for kinematics in any mechanics course. Working with reciprocals of speeds (*pace*) made the equations linear here; choosing variables that linearise a problem is a trick that recurs throughout applied

mathematics. For a first-year treatment of motion that grows from these ideas, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1999, Question 10

Mechanics • Simple Harmonic Motion

STEP I 1999, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q10

The key idea

The particle splits the spring into two springs, AP and PB, each with the same modulus λ as the whole spring (modulus is a property of the material) but with natural lengths a and b. Write Hooke's law for each piece in terms of a single coordinate, the distance of the particle from the upper end, and the equation of motion turns out to be linear in that coordinate. The coefficient of the coordinate gives the period, and we just look at what it does and doesn't depend on.

Solution

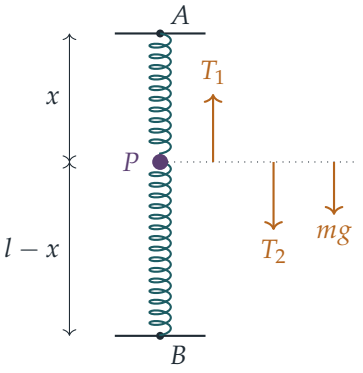
The model

Let the particle have mass m. The spring is light, so each piece has the same tension all along it, and it obeys Hooke's law in both extension and compression (it is a spring, not a string, so it can push as well as pull). Suppose first that A is the upper end. Let x be the distance of the particle below A at time t, so $AP = x$ and $PB = l - x$.

Since AP has natural length a and PB has natural length b, Hooke's law gives

$$T_1 = \frac{\lambda(x-a)}{a}, \qquad T_2 = \frac{\lambda(l-x-b)}{b},$$

where T_1 is the tension in AP and T_2 is the tension in PB. A positive tension pulls the particle towards the fixed end of that piece; a negative value means that piece is compressed and pushes the particle away. Either way the formulas give the correct signed force, so we never need to know in advance which pieces are stretched.



Simple harmonic motion

Resolving downwards for the particle, the forces are its weight mg, the tension T_1 upwards and the tension T_2 downwards:

$$\begin{aligned} m\ddot{x} &= mg - \frac{\lambda(x-a)}{a} + \frac{\lambda(l-x-b)}{b} \\ &= mg + \lambda - \frac{\lambda x}{a} + \frac{\lambda l}{b} - \frac{\lambda x}{b} - \lambda \\ &= mg + \frac{\lambda l}{b} - \lambda\left(\frac{1}{a} + \frac{1}{b}\right)x. \end{aligned}$$

The constants $\pm\lambda$ cancel, and the right-hand side is a linear function of x with negative gradient. Write

$$\omega^2 = \frac{\lambda}{m}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{\lambda(a+b)}{mab}, \qquad x_0 = \frac{mg + \lambda l/b}{\lambda(a+b)/(ab)} = \frac{a(mgb + \lambda l)}{\lambda(a+b)}.$$

Then the equation of motion becomes

$$\ddot{x} = -\omega^2(x - x_0),$$

so, with $X = x - x_0$ (the displacement from the equilibrium position x_0), $\ddot{X} = -\omega^2 X$. This is the equation of simple harmonic motion about $x = x_0$, with period

$$\boxed{\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{mab}{\lambda(a+b)}}}$$

This proves the first result.

Ask yourself Doesn't the motion stop being simple harmonic once one piece is compressed?

It would for an elastic *string*, whose tension formula only holds while it is stretched. A spring obeys the same Hooke's law formula when it is compressed, so the single equation above holds throughout the motion (provided the coils don't close up), and there is no need to restrict l or the amplitude.

Independence of l and of orientation

The length l appears only in x_0: changing l moves the centre of the oscillation but leaves ω, and so the period, unchanged.

If instead B is uppermost, the same working applies with the roles of the two pieces exchanged: the upper piece now has natural length b and the lower one a. That just swaps a and b throughout. The period depends on a and b only through ab and $a+b$, both of which are symmetric, so it is unchanged. (The equilibrium position does change, because gravity now stretches the piece of natural length b rather than a.)

So the period is $2\pi\sqrt{mab/(\lambda(a+b))}$ whatever the value of l and whichever end is uppermost, as required.

Checks

- Two springs on either side of a mass act *in parallel*: the stiffnesses λ/a and λ/b add to give $\lambda(a+b)/(ab)$, as in ω^2 above.
- Limiting case $b \to \infty$: the lower piece becomes very weak and we recover a mass on a single spring of natural length a, with period $2\pi\sqrt{ma/\lambda}$.
- Units: λ is a force, so ma/λ has units of $\mathrm{kg\,m/N} = \mathrm{s}^2$.
- Numerically, with $m = 1$, $\lambda = 50$, $a = 1$, $b = 2$, the formula gives a period of 0.72552. Stepping the full equation of motion numerically for $l = 2$, 3 and 4.5 (including a case where both pieces are compressed), with either end uppermost, gives the same period to five significant figures, and many random choices of the constants agree to within 10^{-5}.

Common slips

- Using modulus $\lambda a/(a+b)$ or similar for the pieces. The modulus of each piece is still λ; it is the stiffness $\lambda/(\text{natural length})$ that changes.
- Getting the sign of the lower tension wrong. A stretched PB pulls the particle *down*, towards B.
- Forgetting to argue the orientation part: you need to say that swapping the ends swaps a and b, and that the period is symmetric in them.

Beyond STEP

Splitting a system into springs and adding stiffnesses is the first step towards coupled oscillators: with several masses along the spring, the equations of motion form a linear system, and the possible frequencies are the eigenvalues of a matrix (the "normal modes"). In the limit of many tiny masses one obtains the wave equation for longitudinal waves in a spring or rod. Kleppner and Kolenkow, *An Introduction to Mechanics*,

treats oscillations and normal modes.

WORKED SOLUTION

STEP I 1999, Question 11

Mechanics • Gravitational Circular Motion

STEP I 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.8, *Three Stars in a Triangle*: https://beyondstep.keithmcnulty.org/ch10/three-stars.html

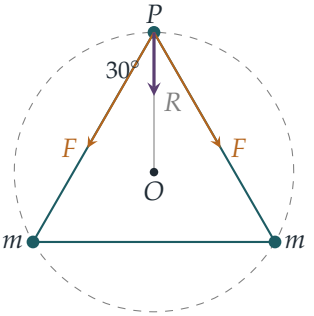
The key idea

Guess the motion and test it. If the triangle turns rigidly about its centre, each star moves in a circle, so it needs a net force of exactly $m\omega^2 R$ pointing to the centre. By symmetry, the gravitational pull on each star from the others points straight at the centre. So all we have to do is match its size to $m\omega^2 R$ and read off ω. A star placed at the centre adds one more inward pull, of a known size.

Solution

We model the stars as particles that move only under their mutual attractions. Let O be the centre of mass. For three equal masses this is the centroid of the triangle, and its distance from each vertex is

$$R = \tfrac{2}{3} \times \tfrac{\sqrt{3}}{2} a = \frac{a}{\sqrt{3}}.$$



Three stars

Take the star P at the top vertex. Each of the other two stars pulls it with a force of size

$$F = \frac{\gamma m^2}{a^2},$$

directed along a side of the triangle. Each side through P makes an angle of 30° with PO, because PO bisects the 60° angle at P. Resolving along and perpendicular to PO:

- perpendicular to PO, the two components are equal and opposite, so they cancel;
- along PO, the resultant is $2F\cos 30^\circ = \sqrt{3}\,\dfrac{\gamma m^2}{a^2}$, towards O.

The other two stars are in the same situation, by symmetry.

Now suppose the triangle rotates rigidly about O with angular velocity ω. Each star then moves in a circle of radius R with constant speed $R\omega$, so it needs an acceleration $R\omega^2$ towards O. Newton's second law towards O gives

$$m\,\frac{a}{\sqrt{3}}\,\omega^2 = \sqrt{3}\,\frac{\gamma m^2}{a^2} \quad\Longrightarrow\quad \omega^2 = \frac{3\gamma m}{a^3}.$$

Ask yourself Why does this show that the motion is *possible*, and not just that the forces balance at one instant?

Start the stars at the vertices, with velocities of size $R\omega$ perpendicular to OP, all turning the same way. While the configuration stays an equilateral triangle of side a centred at O, the force on each star has exactly the size and direction that circular motion with angular velocity ω requires. So the rigidly rotating triangle satisfies the equation of motion of every star at every moment. It is therefore a genuine motion of the system: it is the solution that follows from those initial conditions. The total momentum is zero, so O stays fixed, as it should. Hence each star can revolve about the centre of mass with angular velocity $(3\gamma m/a^3)^{1/2}$.

Adding a fourth star at the centre

Place a star of mass λm at O. The centre of mass is still O. The three outer stars pull the central star with forces of equal size at 120° to one another, so the resultant is zero. The central star therefore stays at rest at O, provided that the outer three keep their symmetric arrangement.

Each outer star now has an extra pull towards O, of size

$$\frac{\gamma(\lambda m)m}{R^2} = \frac{3\gamma\lambda m^2}{a^2},$$

since $R^2 = a^2/3$. This extra force acts along PO, so it simply adds to the inward

resultant. The equation of motion towards O becomes

$$m\,\frac{a}{\sqrt{3}}\,\omega^2 = \sqrt{3}\,\frac{\gamma m^2}{a^2} + 3\lambda\,\frac{\gamma m^2}{a^2}$$

$$\omega^2 = \frac{3\gamma m}{a^3} + \frac{3\sqrt{3}\,\lambda\gamma m}{a^3} = \frac{3\gamma m}{a^3}\left(1+\sqrt{3}\,\lambda\right).$$

The same argument as before shows that this rigid rotation is a genuine motion. So, with the fourth star at rest at the centre, the outer three can revolve about it with angular velocity

$$\omega = \left(\frac{3\gamma m\,(1+\sqrt{3}\,\lambda)}{a^3}\right)^{1/2}.$$

Checks

- Setting $\lambda = 0$ gives back the first answer, as it must.
- Dimensions: $\gamma m/r^2$ is an acceleration, so $\gamma m/a^3$ has units of time^{-2}, which is right for ω^2.
- If λ is very large, the outer stars barely feel one another. They then orbit the central mass alone, with $\omega^2 \approx \gamma\lambda m/R^3 = 3\sqrt{3}\,\gamma\lambda m/a^3$, which matches the dominant term.
- We integrated the full equations of motion numerically (with $\gamma = m = a = 1$). Starting from the predicted circular velocities, the stars returned to their starting positions after three revolutions, with relative error below 10^{-7}, for $\lambda = 0$, 1 and 2.5. For example, $\lambda = 1$ gives $\omega^2 = 3(1+\sqrt{3}) \approx 8.196$.

Common slips

- Using a instead of $R = a/\sqrt{3}$ as the radius of the circle, or as the distance to the central star.
- Adding the two pulls of size F as $2F$, forgetting that they are at 30° to PO.
- Forgetting to say why the central star stays put: the three pulls on it cancel.

Beyond STEP

This is a special case of Lagrange's equilateral solution of the three-body problem, which works for *any* three masses at the vertices of a rotating equilateral triangle. The same idea gives the Lagrange points L_4 and L_5, where the Trojan asteroids share Jupiter's orbit. Whether such configurations are *stable* is a deeper question, answered by linearising the equations of motion about the rotating solution. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and orbits carefully at this level.

WORKED SOLUTION

STEP I 1999, Question 12

Probability and Statistics • Loaded Dice Inequalities

STEP I 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.13, *Loaded Dice*: https://beyondstep.keithmcnulty.org/ch09/loaded-dice.html

The key idea

The inequality in the first line is the whole engine of the question. If every total were equally likely, then the two "corner" totals 2 and 12 would each force a product of probabilities to be $\frac{1}{11}$, and the total 7 would then contain two cross terms whose sum has the form $c\,(x + x^{-1})$. The inequality makes that sum too big. Part (ii) is the same idea in a different guise: a sum of squares is smallest when everything is equal.

Solution

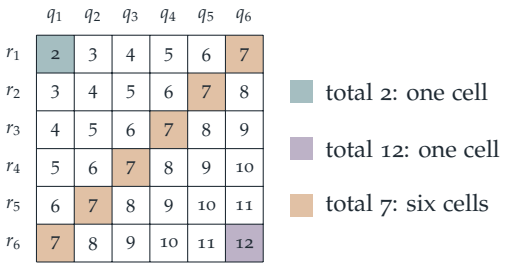
Part (i)

The inequality. For $x > 0$ we can take square roots, and

$$x + x^{-1} - 2 = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0,$$

so $x + x^{-1} \geq 2$, with equality only when $\sqrt{x} = 1/\sqrt{x}$, that is, $x = 1$. (Equivalently, multiply $(x-1)^2 \geq 0$ by the positive number x^{-1}.)

The dice. Let p_s be the probability that the total is s, for $2 \leq s \leq 12$. Since the dice are independent, the probability of the pair (first shows i, second shows j) is $q_i r_j$. The grid below shows which pairs give the totals we need.



The total 2 arises only from $(1,1)$ and the total 12 only from $(6,6)$, so

$$p_2 = q_1 r_1, \qquad p_{12} = q_6 r_6.$$

The total 7 arises from six pairs, including $(1,6)$ and $(6,1)$. All six terms are non-negative, so

$$p_7 = q_1 r_6 + q_2 r_5 + \cdots + q_6 r_1 \;\geq\; q_1 r_6 + q_6 r_1.$$

Now suppose, for contradiction, that all eleven totals are equally likely. They are exhaustive, so each has probability $\frac{1}{11}$. Then $q_1 r_1 = \frac{1}{11}$ and $q_6 r_6 = \frac{1}{11}$. In particular none of q_1, r_1, q_6, r_6 is zero, so we may write $r_1 = \dfrac{1}{11 q_1}$ and $r_6 = \dfrac{1}{11 q_6}$. Substituting,

$$\begin{aligned} p_7 \;\geq\; q_1 r_6 + q_6 r_1 &= \frac{q_1}{11 q_6} + \frac{q_6}{11 q_1} = \frac{1}{11}\left(x + x^{-1}\right), \qquad \text{where } x = \frac{q_1}{q_6} > 0, \\ &\geq \frac{2}{11} \end{aligned}$$

by the inequality we have just proved. But $p_7 = \frac{1}{11} < \frac{2}{11}$, a contradiction. So, however the two dice are loaded, the totals $2, 3, \ldots, 12$ cannot all be equally likely.

Ask yourself Why was it important to note that q_1 and q_6 are non-zero?

We divided by q_1 and q_6, and the inequality $x + x^{-1} \geq 2$ needs $x > 0$. Both are guaranteed because each product $q_1 r_1$ and $q_6 r_6$ equals $\frac{1}{11} \neq 0$. Without that step the argument would have a hole: a die that never shows 1 is perfectly legal in general.

Part (ii)

Let S be the event that the two throws of the first die give the same number. The throws are independent, so the probability that both show j is q_j^2, and these six events are disjoint. Hence

$$\mathrm{P}(S) = q_1^2 + q_2^2 + \cdots + q_6^2.$$

We want the smallest this can be, given $q_j \geq 0$ and $\sum q_j = 1$. Expand a sum of squares that measures how far the die is from fair:

$$0 \leq \sum_{j=1}^{6} \left(q_j - \tfrac{1}{6}\right)^2 = \sum_{j=1}^{6} q_j^2 - \tfrac{1}{3}\sum_{j=1}^{6} q_j + 6 \cdot \tfrac{1}{36}$$
$$= \sum_{j=1}^{6} q_j^2 - \tfrac{1}{3} + \tfrac{1}{6} = \mathrm{P}(S) - \tfrac{1}{6}.$$

So $\mathrm{P}(S) \geq \frac{1}{6}$, with equality exactly when every $q_j = \frac{1}{6}$, the fair die.

Ask yourself Where does $1/36$ come from, and why is it a red herring?

$\frac{1}{36}$ is the probability of one *particular* pair, such as a double six, with a fair die. "The numbers are the same" covers six such pairs, so for a fair die it has probability $6 \times \frac{1}{36} = \frac{1}{6}$, and loading the die can only increase it.

So the answer is **no**: whatever the values of q_j, the probability that the two numbers are the same is at least $\frac{1}{6}$, which is far more than $\frac{1}{36}$.

Checks

- Fair dice: $p_2 = p_{12} = \frac{1}{36}$ but $p_7 = \frac{6}{36}$, clearly unequal, as the result predicts.
- A numerical search over 20,000 random pairs of loaded dice never got all eleven totals within 0.05 of $\frac{1}{11}$, and exact checks with fractions confirmed $p_7 \geq q_1 r_6 + q_6 r_1$ every time.
- For the die with $(q_1, \ldots, q_6) = (0.1, 0.2, 0.3, 0.15, 0.15, 0.1)$ we get $\mathrm{P}(S) = 0.195 \geq \frac{1}{6}$; a simulation of 200,000 double throws gave 0.194.

Common slips

- Assuming the two dice are identical ($q_i = r_i$). The question allows them to be loaded differently, and the argument above doesn't need them to match.
- Writing $p_7 = q_1 r_6 + q_6 r_1$. There are four more terms; we only need an inequality, so throw them away, but say why ($q_i r_j \geq 0$).
- In part (ii), answering with $\sum q_j^2 \geq \frac{1}{36}$ or forgetting that "the same" means any of six outcomes.

Beyond STEP

Part (ii) is the statement that $\sum q_j^2 \geq \frac{1}{n}\left(\sum q_j\right)^2$, a special case of the Cauchy–Schwarz inequality; the quantity $\sum q_j^2$ is called the *collision probability* and appears in information

theory and in the analysis of hashing. Part (i) is a classic result about generating functions: the total of two dice has probability generating function $Q(t)R(t)$, and a neat alternative proof notes that uniform totals would force $1 + t + \cdots + t^{10}$ to be a constant times a product of two real polynomials of degree 5, each of which has a real root, whereas $1 + t + \cdots + t^{10} = (t^{11} - 1)/(t - 1)$ has none. Generating functions of this kind are developed in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP I 1999, Question 13

Probability and Statistics • Indicator Variables

STEP I 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q13

The key idea

Don't count the units directly. Count the *gaps* instead: the number of units is one more than the number of places where neighbouring magnets stand apart. Whether a gap opens depends only on whether the two neighbours point the same way, and these $N-1$ events turn out to be independent fair coin tosses. So the number of gaps is binomial, and both the expectation and the variance follow at once.

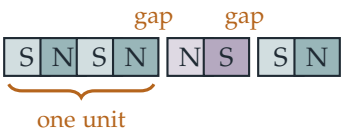
Solution

Setting up the model

We take "placed randomly" to mean that each magnet is equally likely to point either way, independently of the others. Call a magnet *type A* if its north pole is on the right (SN) and *type B* if its north pole is on the left (NS). Let $X_1, \ldots, X_N$ be the types, read from left to right.

Look at two neighbours. The ends facing each other are the right end of the first magnet and the left end of the second.

- Same type (A then A): the facing ends are N and S, which are opposite, so they **join**. (B then B is the same with S and N.)
- Different types (A then B): the facing ends are N and N, so they **stand apart**. (B then A gives S and S.)



So a gap appears between magnets i and $i+1$ exactly when $X_i \neq X_{i+1}$. Define, for

$i = 1, \ldots, N-1$,

$$G_i = \begin{cases} 1 & \text{if } X_i \neq X_{i+1} \text{ (a gap)}, \\ 0 & \text{if } X_i = X_{i+1} \text{ (joined)}. \end{cases}$$

Each gap splits the line into one more piece, so the number of separate units is

$$U = 1 + G_1 + G_2 + \cdots + G_{N-1}.$$

The distribution of the gaps

Each G_i is 1 with probability $\frac{1}{2}$: whatever X_i is, X_{i+1} differs from it with probability $\frac{1}{2}$.

> **Ask yourself** The events "gap at i" and "gap at $i+1$" both involve magnet $i+1$. Are the G_i really independent?

They are, even though neighbouring G's share a magnet. Fix the type of the first magnet, X_1. Then the sequence $G_1, \ldots, G_{N-1}$ determines $X_2, X_3, \ldots, X_N$ one at a time (X_{i+1} equals X_i if $G_i = 0$ and is the other type if $G_i = 1$), and conversely the types determine the G's. So, for either value of X_1, the 2^{N-1} equally likely choices of $(X_2, \ldots, X_N)$ correspond one-to-one with the 2^{N-1} possible sequences $(G_1, \ldots, G_{N-1})$. Every sequence of gaps therefore has probability $1/2^{N-1}$, which says exactly that the G_i are independent, each 0 or 1 with probability $\frac{1}{2}$.

Hence $G_1 + \cdots + G_{N-1} \sim \mathrm{B}\left(N-1, \frac{1}{2}\right)$.

Expectation and variance

A $\mathrm{B}(n, p)$ variable has mean np and variance $np(1-p)$. With $n = N-1$, $p = \frac{1}{2}$, and remembering that adding the constant 1 shifts the mean but not the variance,

$$\mathrm{E}(U) = 1 + \tfrac{1}{2}(N-1) = \frac{N+1}{2},$$
$$\mathrm{Var}(U) = (N-1) \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \frac{N-1}{4}.$$

If you prefer not to quote the binomial results: $\mathrm{E}(G_i) = \frac{1}{2}$ and, since $G_i^2 = G_i$, $\mathrm{Var}(G_i) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. Linearity gives the mean, and independence lets us add the $N-1$ variances.

Checks

- $N = 1$: one magnet is always one unit, and the formulae give mean 1 and variance 0.
- $N = 2$: the two magnets form 1 or 2 units with probability $\frac{1}{2}$ each, so the mean is $\frac{3}{2}$ and the variance is $\frac{5}{2} - \frac{9}{4} = \frac{1}{4}$, as predicted.
- An exact enumeration over all 2^N arrangements for $N \leq 12$ reproduced $\frac{N+1}{2}$ and $\frac{N-1}{4}$ exactly, and a simulation with $N = 20$ gave a mean of about 10.5 and a variance of about 4.73 (predicted 10.5 and 4.75).

Common slips

- Thinking that two magnets of the *same* orientation repel. They face each other with opposite poles (N to S), so they attract and join.
- Adding variances without justifying independence. Here the independence is true but not obvious, because neighbouring gaps share a magnet.
- Writing $\mathrm{Var}(U) = \frac{N}{4}$ by counting N gaps instead of $N - 1$, or forgetting the extra 1 in the mean.

Beyond STEP

Counting units by counting the boundaries between them is the method of *indicator variables*, one of the most useful tools in probability: write a count as a sum of 0–1 variables, then use linearity of expectation, which needs no independence at all. Here U counts the *runs* in a random sequence of two symbols. Run counts are the basis of the Wald–Wolfowitz runs test in statistics, which checks whether a sequence looks random. For more on indicators and their variances, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 1999, Question 14

Probability and Statistics • Conditional Probability

STEP I 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q14

The key idea

The continuous random variable is only there to set up a discrete one. Once we turn the density into the four probabilities of scoring 4, 3, 2 and 1, part (i) is a one-line expectation. Part (ii) is a conditional probability, and since the total must hit 3 exactly, there are very few ways to win, so we can list them all.

Solution

Part (i)

For $0 \leq a \leq b \leq 1$,

$$\mathrm{P}(a \leq X < b) = \int_a^b 2x \,\mathrm{d}x = b^2 - a^2.$$

(Including or excluding an endpoint makes no difference for a continuous random variable.) Let S be the score from one dart. Then

$$\mathrm{P}(S=4) = \left(\tfrac{1}{4}\right)^2 - 0 = \tfrac{1}{16}, \qquad \mathrm{P}(S=3) = \left(\tfrac{1}{2}\right)^2 - \left(\tfrac{1}{4}\right)^2 = \tfrac{3}{16},$$

$$\mathrm{P}(S=2) = \left(\tfrac{3}{4}\right)^2 - \left(\tfrac{1}{2}\right)^2 = \tfrac{5}{16}, \qquad \mathrm{P}(S=1) = 1 - \left(\tfrac{3}{4}\right)^2 = \tfrac{7}{16}.$$

These add up to 1, as they should. Notice that the outer rings are more likely: the density $2x$ grows with x, because there is more area far from the centre.

So

$$\mathrm{E}(S) = 4 \cdot \tfrac{1}{16} + 3 \cdot \tfrac{3}{16} + 2 \cdot \tfrac{5}{16} + 1 \cdot \tfrac{7}{16} = \frac{4+9+10+7}{16} = \frac{30}{16} = \frac{15}{8},$$

as required.

Part (ii)

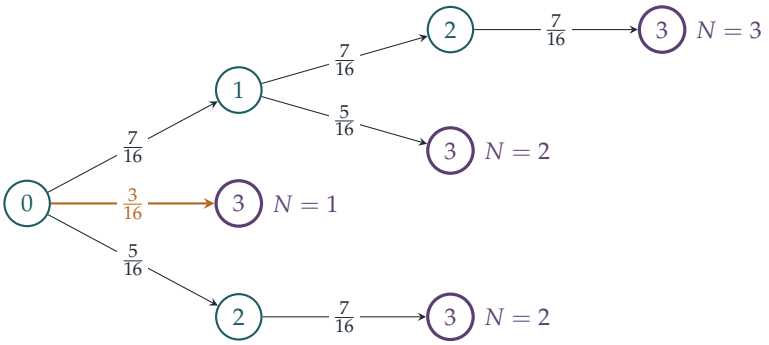
Assume the throws are independent, each with the distribution of S above. Let W be the event that I win, and N the number of throws in the game.

Ask yourself Which sequences of scores lead to a win?

I win exactly when the running total reaches 3 without first going past it. Every score is at least 1, so the total goes up by at least 1 each time and the game lasts at most three throws. A winning game is a sequence of positive scores that adds up to exactly 3. (I only throw again while the total is below 3, so every such sequence really is a complete game.) There are just four:

$$(3), \qquad (1,2), \qquad (2,1), \qquad (1,1,1).$$

A score of 4 never appears, because it takes the total past 3 at once.



The tree shows the running totals along each route that ends in a win (circled in purple), labelled with the probability of each throw.

By independence, the probability of each sequence is the product of the probabilities of its scores:

$$\begin{aligned}
\mathrm{P}(W \cap N = 1) &= \mathrm{P}(3) = \tfrac{3}{16} = \tfrac{768}{4096}, \\
\mathrm{P}(W \cap N = 2) &= \mathrm{P}(1,2) + \mathrm{P}(2,1) = 2 \cdot \tfrac{7}{16} \cdot \tfrac{5}{16} = \tfrac{70}{256} = \tfrac{1120}{4096}, \\
\mathrm{P}(W \cap N = 3) &= \mathrm{P}(1,1,1) = \left(\tfrac{7}{16}\right)^3 = \tfrac{343}{4096}.
\end{aligned}$$

These three events are disjoint and together make up W, so

$$\mathrm{P}(W) = \frac{768 + 1120 + 343}{4096} = \frac{2231}{4096}.$$

Therefore

$$\mathrm{P}(N = 3 \mid W) = \frac{\mathrm{P}(W \cap N = 3)}{\mathrm{P}(W)} = \frac{343/4096}{2231/4096} = \frac{343}{2231},$$

as required.

Checks

- The four score probabilities $\frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}$ add up to 1. They are odd numbers over 16 because $(k+1)^2 - k^2 = 2k + 1$.

- The target $343 = 7^3$ shows straight away that the three-throw win must be $(1,1,1)$, which confirms the setup.
- A computer simulation of 10^6 games gave $\mathrm{P}(W) \approx 0.5452$ (exact value 0.5447) and $\mathrm{P}(N = 3 \mid W) \approx 0.1535$ (exact value $343/2231 \approx 0.1537$).

Common slips

- Treating the score bands as equally likely. The probabilities come from integrating the density, not from the widths of the intervals.
- Counting $(1,2)$ but forgetting $(2,1)$, which gives $\mathrm{P}(W) = \frac{1671}{4096}$.
- Answering $\mathrm{P}(N = 3 \text{ and } W) = \frac{343}{4096}$ instead of dividing by $\mathrm{P}(W)$.

Beyond STEP

Part (ii) is a small Markov chain: the state is the running total, and 3 and "bust" are absorbing states. For bigger targets, listing the routes stops being practical. Instead we write w_t for the probability of winning from total t, which gives a recurrence $w_t = \sum_s \mathrm{P}(S = s)\, w_{t+s}$. This first-step analysis is a key tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers it in their treatment of random walks and Markov chains.

WORKED SOLUTION

STEP II 1999, Question 1

Pure • Comparing Large Numbers

STEP II 1999, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q1

The key idea

These numbers are far too big to compare directly, and even their logarithms are far too big. The way in is to take $\log_{10}$ *twice*. Because x, y and z are each a power of 10, every double logarithm becomes a short sum of 100, x and y, and those are easy to compare. Since $\log_{10}$ is increasing, comparing the double logarithms compares the original numbers.

Solution

Throughout, write log for $\log_{10}$ and

$$L(a) = \log(\log a).$$

The function $t \mapsto \log t$ is strictly increasing, so for numbers $a, b > 10$ (where $\log a, \log b > 1 > 0$) we have $a < b$ exactly when $L(a) < L(b)$. The basic facts we use are

$$\log x = 100, \qquad \log y = x = 10^{100}, \qquad \log z = y = 10^x.$$

Part (i)

Taking $\log_{10}$ of Stirling's approximation with $n = x$,

$$\begin{aligned}\log a_1 &\approx \log\sqrt{2\pi} + \left(x + \tfrac{1}{2}\right)\log x - x\log \mathrm{e} \\ &= 100x - x\log \mathrm{e} + 50 + \log\sqrt{2\pi} \\ &= x\,(100 - \log \mathrm{e}) + 50.4\ldots.\end{aligned}$$

Now $\log \mathrm{e} \approx 0.434$, and the constant $50.4\ldots$ is utterly negligible beside $x = 10^{100}$. So

$$\log a_1 \approx 99.566 \times 10^{100}, \qquad L(a_1) \approx 100 + \log 99.566 \approx 100 + 1.998 = 101.998\ldots \approx 102,$$

as required.

Part (ii)

We compute $\log a_i$ and then $L(a_i)$ for each number.

- $a_2 = x^y$: $\log a_2 = y \log x = 100y$, so $L(a_2) = 2 + \log y = x + 2$.
- $a_3 = y^x$: $\log a_3 = x \log y = x^2 = 10^{200}$, so $L(a_3) = 200$.
- $a_4 = z^x$: $\log a_4 = x \log z = xy$, so $L(a_4) = \log x + \log y = x + 100$.
- $a_5 = \mathrm{e}^{xyz}$: $\log a_5 = xyz \log \mathrm{e}$, so
$$L(a_5) = 100 + x + y + \log(\log \mathrm{e}) \approx y + x + 99.64,$$
since $\log(\log \mathrm{e}) = \log 0.434\ldots \approx -0.36$.
- $a_6 = z^{1/y}$: $\log a_6 = \frac{1}{y} \log z = 1$, so $a_6 = 10$ exactly.
- $a_7 = y^{z/x}$: $\log a_7 = \frac{z}{x} \log y = \frac{z}{x} \cdot x = z$, so $L(a_7) = \log z = y$.

Together with $L(a_1) \approx 102$ from part (i), the double logarithms are

$$\underbrace{0}_{a_6},\ \underbrace{102}_{a_1},\ \underbrace{200}_{a_3},\ \underbrace{x+2}_{a_2},\ \underbrace{x+100}_{a_4},\ \underbrace{y}_{a_7},\ \underbrace{y+x+99.6}_{a_5}.$$

Ask yourself Part (i) only gives an approximation for a_1. Is that good enough to place it?

Yes, and we don't even need Stirling. Each of the x factors of $x!$ is at most x, so $x! \le x^x$ and $\log a_1 \le x \log x = 10^{102}$, giving $L(a_1) \le 102 < 200 = L(a_3)$. And $a_1 = x! > 10 = a_6$ obviously. So $a_6 < a_1 < a_3$ rigorously.

Now we justify the remaining comparisons one at a time.

- $a_3 < a_2$: $200 < 10^{100} + 2$.
- $a_2 < a_4$: $x + 2 < x + 100$.
- $a_4 < a_7$: we need $x + 100 < y = 10^x$. This is true by an enormous margin, because $10^t > t + 100$ for every $t \ge 3$, and here $t = x = 10^{100}$.
- $a_7 < a_5$: $y < y + x + 99.6$, since $x + 99.6 > 0$. (Directly: $\log a_5 = z \cdot xy \log \mathrm{e}$ and $xy \log \mathrm{e} > 1$, so $\log a_5 > z = \log a_7$.)

All the numbers other than a_6 exceed 10, so the monotonicity of L applies, and the order of magnitude is

$$a_6 < a_1 < a_3 < a_2 < a_4 < a_7 < a_5.$$

Checks

For part (i), we compared Stirling's formula with the exact $\log_{10} n!$ (via the log-gamma function) for $n = 10^2, 10^3, 10^5, 10^8$. The double-log formula for a_1 agrees to within 10^{-6},

and for $n = 10^{100}$ it gives 101.998. For part (ii), we repeated the whole calculation with $x = 10^k$ for $k = 2, 3, 4$, where $y = 10^x$ can be held exactly as an integer. The same order $a_6 < a_1 < a_3 < a_2 < a_4 < a_7 < a_5$ comes out each time, so it doesn't hinge on the size of the number 100.

Common slips

- Comparing single logarithms and then getting lost: $\log a_7 = z$ and $\log a_4 = xy$ are still too big to compare by eye. Take logs once more.
- Forgetting the factor $\log \mathrm{e}$ when converting e^{xyz} to base 10. Here it barely matters, but it does shift $L(a_5)$ by about 0.36.
- Missing that a_6 is simply 10, the smallest of all.

Beyond STEP

Iterated logarithms and towers of exponents measure growth in analysis and in computer science. For example, the iterated logarithm $\log^* n$ appears in the running time of some algorithms, and "tower" functions bound results in Ramsey theory. Stirling's formula itself is the first term of an asymptotic series, and proving it carefully is a classic first-year analysis exercise. Abbott's *Understanding Analysis* is a good place to get comfortable with the limits and inequalities involved.

WORKED SOLUTION

STEP II 1999, Question 2

Pure • Discriminants of Quadratics

STEP II 1999, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q2

The key idea

Everything hinges on the discriminant of $(*)$. Because the x-coefficient carries a factor of 2, the quarter-discriminant $\frac{1}{4}(b^2 - 4ac)$ is especially tidy, and it turns out to be a quadratic in n. So each part becomes a question about the sign of a *second* quadratic, this time with n as the variable.

Solution

Throughout we take $pn^2 + q \geq 0$, so that the square root is real and $(*)$ is an ordinary quadratic with real coefficients. (This holds automatically in parts (i) and (iii), where $q > 0$ or $q \geq 0$.) With $a = n$, $b = 2\sqrt{pn^2 + q}$ and $c = rn + s$, the equation has real roots exactly when $b^2 - 4ac \geq 0$, that is, when $\frac{1}{4}b^2 - ac \geq 0$. Now

$$\begin{aligned} \tfrac{1}{4}b^2 - ac &= (pn^2 + q) - n(rn + s) \\ &= (p - r)n^2 - sn + q. \end{aligned}$$

Write $f(n) = (p - r)n^2 - sn + q$. Then

$$(*) \text{ has no real roots} \iff f(n) < 0.$$

Part (i)

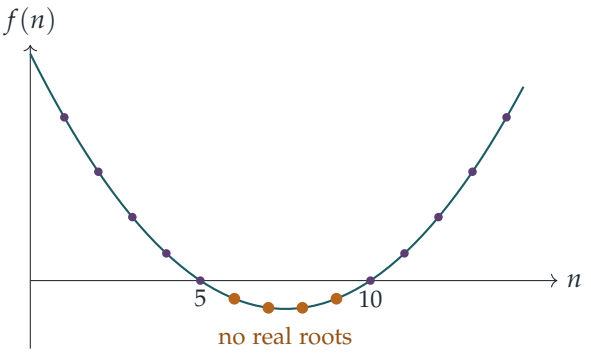
Here $p - r = 1$, so

$$f(n) = n^2 - 15n + 50 = (n - 5)(n - 10).$$

This is negative exactly when $5 < n < 10$. Since n is a positive integer, $(*)$ has no real roots precisely for

$$n \in \{6, 7, 8, 9\}.$$

At $n = 5$ and $n = 10$ we have $f(n) = 0$, so there is a repeated real root, and those values are excluded.



Part (ii)

Now $p < r$, so the leading coefficient $p - r$ of f is negative. Think of f as a quadratic in a real variable t:

$$f(t) = (p-r)t^2 - st + q.$$

Its discriminant is

$$s^2 - 4(p-r)q,$$

which is negative by the hypothesis $4q(p-r) > s^2$.

Ask yourself Why does a negative discriminant for f tell us anything about the sign of $f(n)$?

A quadratic with negative discriminant has no real zeros, so its graph never crosses the axis and it keeps one sign for all real t. With a negative leading coefficient the graph is an upside-down parabola, so that sign is negative. Explicitly, completing the square,

$$f(t) = (p-r)\Big(t - \frac{s}{2(p-r)}\Big)^2 + \frac{4q(p-r) - s^2}{4(p-r)},$$

and both terms are negative or zero: the first because $p - r < 0$, and the second because its numerator is positive and its denominator negative. The second term is strictly negative, so $f(t) < 0$ for every real t.

In particular $f(n) < 0$ for every positive integer n, so $(*)$ has no real roots for any value of n, as required.

Note that the hypotheses force $q < 0$: since $p - r < 0$ and $4q(p-r) > s^2 \geq 0$, q must be negative. So $pn^2 + q$ really can be negative for small n, and our standing assumption that the square root is real matters here. (If an imaginary coefficient were allowed, $x = 0$ could be a root: try $p = 1$, $r = 2$, $s = -2$, $q = -10$, $n = 1$.)

Part (iii)

Now $n = 1$, $p - r = 1$ and $q = s^2/8 \geq 0$, so $p + q > 0$ and the coefficients are real. Here

$$f(1) = 1 - s + \frac{s^2}{8} = \frac{s^2 - 8s + 8}{8}.$$

Real roots occur if, and only if, $f(1) \geq 0$, that is, if, and only if,

$$s^2 - 8s + 8 \geq 0.$$

The roots of $s^2 - 8s + 8 = 0$ are

$$s = \frac{8 \pm \sqrt{64 - 32}}{2} = 4 \pm 2\sqrt{2},$$

so $s^2 - 8s + 8 = \big(s - (4 - 2\sqrt{2})\big)\big(s - (4 + 2\sqrt{2})\big)$. This product is non-negative exactly when s lies outside the open interval between the roots. Hence $(*)$ has real roots if, and only if,

$$s \leq 4 - 2\sqrt{2} \quad \text{or} \quad s \geq 4 + 2\sqrt{2},$$

as required. Every step is reversible, which is what gives us both directions of the "if and only if". At the two endpoints the root is repeated.

Checks

- In (i), take $n = 7$: $\frac{1}{4}b^2 = 3 \cdot 49 + 50 = 197$ and $ac = 7 \cdot 29 = 203$, so $\frac{1}{4}b^2 - ac = -6 < 0$. Take $n = 11$: $413 - 11 \cdot 37 = 6 > 0$.
- In (iii), $4 - 2\sqrt{2} \approx 1.1716$ and $4 + 2\sqrt{2} \approx 6.8284$. With $s = 4$ we get $f(1) = 1 - 4 + 2 = -1 < 0$, which is inside the gap, as expected.
- A computer searched n from 1 to 199 for (i), tested about 5 million random cases of (ii), and tested 10^5 random values of s for (iii). All agreed with the results above.

Common slips

- Including $n = 5$ and $n = 10$ in (i). There the discriminant is zero, so there is a (repeated) real root.
- In (ii), checking only that $f(n) < 0$ for large n. The point of the discriminant condition is that f is negative *everywhere*, including small n.
- Forgetting the factor of 2 in the x-coefficient and getting $4(pn^2 + q)$ in the wrong place in the discriminant.

Beyond STEP

The move here, treating a parameter as a new variable and asking about the sign of a quadratic in it, is the start of studying how the roots of a polynomial depend on its coefficients. At university this leads to discriminants of higher-degree polynomials, and to the idea that a quadratic form $at^2 + bt + c$ is negative definite exactly when $a < 0$ and $b^2 < 4ac$. That same test, for two variables, classifies stationary points of functions of several variables. Siklos's *Advanced Problems in Mathematics* has plenty more STEP questions built on careful discriminant arguments.

WORKED SOLUTION

STEP II 1999, Question 3

Pure • Proof by Induction

STEP II 1999, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q3

The key idea

Don't differentiate e^{-x^3} over and over from scratch. Instead, write the nth derivative as $e^{-x^3}S_n(x)$ and differentiate *that* once more. This gives a recurrence linking S_{n+1} to S_n and S_n', and every part of the question comes out of that one relation: the explicit polynomials, the induction, the degree and leading coefficient, and the final inequality.

Solution

The recurrence

By definition,

$$\frac{d^n}{dx^n}\left(e^{-x^3}\right) = e^{-x^3}S_n(x).$$

Differentiating once more with the product rule,

$$\frac{d^{n+1}}{dx^{n+1}}\left(e^{-x^3}\right) = e^{-x^3}S_n'(x) - 3x^2e^{-x^3}S_n(x),$$

and multiplying by e^{x^3} gives

$$\boxed{S_{n+1}(x) = S_n'(x) - 3x^2S_n(x).}$$

This holds for every $n \geq 0$, where $S_0(x) = 1$ (the zeroth derivative is the function itself).

S_2 and S_3

From $S_0 = 1$ the recurrence gives $S_1 = 0 - 3x^2 = -3x^2$, which you can also see directly. Then

$$S_2 = S_1' - 3x^2S_1 = -6x - 3x^2(-3x^2) = 9x^4 - 6x,$$

as required. Applying the recurrence again,

$$\begin{aligned} S_3 &= S_2' - 3x^2 S_2 = (36x^3 - 6) - 3x^2(9x^4 - 6x) \\ &= 36x^3 - 6 - 27x^6 + 18x^3, \end{aligned}$$

so

$$S_3(x) = -27x^6 + 54x^3 - 6.$$

The induction

We prove a stronger statement, because it costs nothing extra and answers the next question too:

$P(n)$: $S_n(x)$ is a polynomial of degree $2n$ whose leading coefficient is $(-3)^n$.

Base case. $S_1(x) = -3x^2$ has degree 2 and leading coefficient $-3 = (-3)^1$, so $P(1)$ holds. (So does $P(0)$, since $S_0 = 1$.)

Inductive step. Suppose $P(k)$ holds, so

$$S_k(x) = (-3)^k x^{2k} + (\text{terms of degree less than } 2k).$$

The derivative of a polynomial is a polynomial, and so is $3x^2 S_k$, so $S_{k+1} = S_k' - 3x^2 S_k$ is a polynomial. For its degree, look at the two pieces separately:

- S_k' has degree at most $2k - 1$;
- $-3x^2 S_k(x) = -3(-3)^k x^{2k+2} + (\text{lower terms}) = (-3)^{k+1} x^{2k+2} + \cdots$.

> **Ask yourself** Could S_k' cancel the top term of $-3x^2 S_k$?

No: S_k' has degree at most $2k - 1$, which is well below $2k + 2$, so it can't touch the x^{2k+2} term. Hence S_{k+1} has degree exactly $2k + 2 = 2(k + 1)$, with leading coefficient $(-3)^{k+1}$ (non-zero). That is $P(k + 1)$.

By induction, $P(n)$ holds for all $n \geq 1$. So $S_n(x)$ is a polynomial of order (degree) $2n$, and the coefficient of x^{2n} is $(-3)^n$. This agrees with S_2 ($9 = (-3)^2$, degree 4) and S_3 ($-27 = (-3)^3$, degree 6).

The final part

Suppose $S_n'(a) = 0$. Putting $x = a$ in the recurrence, the derivative term vanishes:

$$S_{n+1}(a) = S_n'(a) - 3a^2 S_n(a) = -3a^2 S_n(a).$$

Therefore

$$S_n(a)\, S_{n+1}(a) = -3a^2 \big(S_n(a)\big)^2 \leq 0,$$

since $a^2 \geq 0$ and $\big(S_n(a)\big)^2 \geq 0$. Equality happens exactly when $a = 0$ or $S_n(a) = 0$. That completes the proof.

Checks

- Leading terms: S_3 should start $(-3)^3x^6 = -27x^6$, and it does.
- Constant term of S_3: near $x = 0$, $\mathrm{e}^{-x^3} = 1 - x^3 + \cdots$, so its third derivative at 0 is $-3! = -6$. This matches $S_3(0) = -6$.
- A stationary point: $S_3'(x) = -162x^5 + 162x^2$ vanishes at $x = 1$. There $S_3(1) = 21$ and $S_4(1) = -3 \cdot 21 = -63$, so the product is $-1323 \leq 0$.
- We also checked $S_0, \ldots, S_9$ against the exact Taylor series of $\mathrm{e}^{-(a+h)^3}$ at thirty random rational points a, and tested the inequality at every stationary point of $S_1, \ldots, S_8$ in $[-4, 4]$.

Common slips

- Forgetting to say why the degree is *exactly* $2n + 2$. You need the leading coefficient to be non-zero and S_k' to be unable to cancel it, which is why it pays to carry the leading coefficient in the inductive hypothesis.
- Proving only "S_n is a polynomial" and then guessing the degree from a pattern. The question asks you to get the degree from the induction argument itself.

Beyond STEP

The polynomials here are cousins of the Hermite polynomials, defined by the same trick with e^{-x^2} (a "Rodrigues formula"). They satisfy recurrences like ours and turn up in probability and in the quantum harmonic oscillator. The final inequality is a first glimpse of how such recurrences control where the zeros and turning points of a whole family of polynomials lie. Boyce and DiPrima, *Elementary Differential Equations*, meets Hermite's equation in its chapter on series solutions.

WORKED SOLUTION

STEP II 1999, Question 4

Pure • Binomial Coefficient Identities

STEP II 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q4

The key idea

When two polynomials are identically equal, their coefficients of each power of x agree. So we pick a product of two binomial expansions whose x^n coefficient is the sum we want, and then compute the same coefficient from a simpler form of the product. The only real choice is which two factors to multiply: changing x to $-x$ in one factor brings in the signs for (i), and differentiating one factor brings in the extra t for (ii).

Solution

The first result

By the binomial theorem,

$$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r.$$

Multiply two copies of this together. A term in x^n comes from taking x^s from the first bracket and x^{n-s} from the second, for some s with $0 \le s \le n$. So the coefficient of x^n in $(1+x)^n(1+x)^n$ is

$$\sum_{s=0}^{n} \binom{n}{s}\binom{n}{n-s} = \sum_{s=0}^{n} \binom{n}{s}^2,$$

using the symmetry $\binom{n}{n-s} = \binom{n}{s}$, which is clear from the factorial formula. On the other side of the identity, the coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$. The two polynomials are identical, so their x^n coefficients are equal:

$$\sum_{s=0}^{n} \binom{n}{s}^2 = \binom{2n}{n}.$$

Part (i)

We need alternating signs, so we replace the first factor by $(1-x)^n = \sum_r \binom{n}{r}(-1)^r x^r$ and use the identity

$$(1-x)^n(1+x)^n \equiv (1-x^2)^n.$$

As before, the x^n terms on the left come from x^s in the first bracket and x^{n-s} in the second, so the coefficient of x^n is

$$\sum_{s=0}^{n}(-1)^s\binom{n}{s}\binom{n}{n-s} = \sum_{s=0}^{n}(-1)^s\binom{n}{s}^2.$$

On the right,

$$(1-x^2)^n = \sum_{k=0}^{n}\binom{n}{k}(-1)^k x^{2k}$$

contains only even powers of x. Since n is even, x^n does appear, namely when $2k=n$, that is $k=n/2$, and its coefficient is $(-1)^{n/2}\binom{n}{n/2}$. Equating coefficients,

$$\sum_{s=0}^{n}(-1)^s\binom{n}{s}^2 = (-1)^{n/2}\binom{n}{n/2},$$

as required. (The same argument shows the sum is 0 when n is odd, because then $(1-x^2)^n$ has no x^n term.)

Part (ii)

Ask yourself How can we make a factor of t appear next to $\binom{n}{t}$?

Differentiation does it: $\dfrac{\mathrm{d}}{\mathrm{d}x}x^t = t\,x^{t-1}$. Differentiating the binomial expansion gives

$$n(1+x)^{n-1} = \sum_{t=1}^{n} t\binom{n}{t}x^{t-1}.$$

Now multiply by $(1+x)^n = \sum_u \binom{n}{u}x^u$ and use the identity

$$n(1+x)^{n-1}(1+x)^n \equiv n(1+x)^{2n-1}.$$

On the left, a term in x^{n-1} comes from $t\binom{n}{t}x^{t-1}$ in the first bracket times $\binom{n}{n-t}x^{n-t}$ in the second, so the coefficient of x^{n-1} is

$$\sum_{t=1}^{n} t\binom{n}{t}\binom{n}{n-t} = \sum_{t=1}^{n} t\binom{n}{t}^2.$$

On the right, the coefficient of x^{n-1} in $n(1+x)^{2n-1}$ is $n\binom{2n-1}{n-1}$. Finally,

$$\binom{2n}{n} = \frac{(2n)!}{n!\,n!} = \frac{2n}{n}\cdot\frac{(2n-1)!}{(n-1)!\,n!} = 2\binom{2n-1}{n-1},$$

so equating coefficients and doubling,

$$\sum_{t=1}^{n} 2t\binom{n}{t}^2 = 2n\binom{2n-1}{n-1} = n\binom{2n}{n}.$$

Otherwise: there is a neat symmetry argument. Replacing t by $n-t$ and using $\binom{n}{n-t} = \binom{n}{t}$ shows $\sum_{t=0}^{n} t\binom{n}{t}^2 = \sum_{t=0}^{n}(n-t)\binom{n}{t}^2$. Adding these two expressions for the same sum gives

$$\sum_{t=0}^{n} 2t\binom{n}{t}^2 = n\sum_{t=0}^{n}\binom{n}{t}^2 = n\binom{2n}{n}$$

by the first result, and the $t = 0$ term is zero.

Checks

Take $n = 4$, where the rows of Pascal's triangle are $1, 4, 6, 4, 1$. Then $1 + 16 + 36 + 16 + 1 = 70 = \binom{8}{4}$; the alternating sum is $1 - 16 + 36 - 16 + 1 = 6 = (-1)^2\binom{4}{2}$; and $2(1 \cdot 16 + 2 \cdot 36 + 3 \cdot 16 + 4 \cdot 1) = 280 = 4 \times 70$. We also checked all three identities exactly for every n up to 39.

Common slips

- Forgetting that in part (i) the right-hand side has only even powers, so the evenness of n is exactly what makes x^n appear.
- Comparing the wrong power in part (ii): after differentiating, the exponents drop by one, so we need x^{n-1}, not x^n.

Beyond STEP

The first identity is a special case of Vandermonde's identity, $\sum_s \binom{m}{s}\binom{n}{k-s} = \binom{m+n}{k}$, which also has a counting proof: choose k people from m women and n men by deciding how many women to take. Using polynomials as bookkeeping devices for sequences is the method of *generating functions*, a central tool in combinatorics and probability. Grimmett and Welsh, *Probability: An Introduction*, uses generating functions throughout, and Siklos, *Advanced Problems in Mathematics*, has more questions in this style.

WORKED SOLUTION

STEP II 1999, Question 5

Pure • Trigonometric Equations

STEP II 1999, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q5

The key idea

There are two natural routes. One is to make $\cos\alpha$ the unknown: isolate $\sin\alpha$, square, and use $\sin^2 + \cos^2 = 1$ to get a quadratic in $\cos\alpha$. The other is to write $5\cos x + 12\sin x$ as a single cosine, $13\cos(x-\beta)$, which also shows that both roots of the quadratic really occur. For the final part, $\alpha < \frac{3}{4}\pi$ is a statement about $\cos\alpha$, because cosine is decreasing on $[0, \pi]$. So it comes down to comparing two surds.

Solution

The two values of $\cos\alpha$

Write $c = \cos\alpha$. From the equation, $12\sin\alpha = 7 - 5c$. Squaring, and using $\sin^2\alpha = 1 - c^2$,

$$144(1-c^2) = 49 - 70c + 25c^2,$$
$$169c^2 - 70c - 95 = 0.$$

The quadratic formula gives

$$c = \frac{70 \pm \sqrt{70^2 + 4\cdot 169\cdot 95}}{2\cdot 169} = \frac{70 \pm \sqrt{69120}}{338}.$$

Since $69120 = 576 \times 120 = 24^2 \times 120$, we have $\sqrt{69120} = 24\sqrt{120}$. Hence

$$\cos\alpha = \frac{35 - 12\sqrt{120}}{169} \qquad \text{or} \qquad \cos\alpha = \frac{35 + 12\sqrt{120}}{169},$$

which is the required result, and the other value is $\dfrac{35 + 12\sqrt{120}}{169}$.

Ask yourself Squaring can create false solutions. Do both values really come from solutions of the original equation?

They do, and the single-cosine form shows why. Let β be the acute angle with $\cos\beta = \frac{5}{13}$ and $\sin\beta = \frac{12}{13}$. Then

$$5\cos x + 12\sin x = 13\cos(x-\beta),$$

so the equation says $\cos(x-\beta) = \frac{7}{13}$. Let γ be the acute angle with $\cos\gamma = \frac{7}{13}$, so that $\sin\gamma = \frac{\sqrt{169-49}}{13} = \frac{\sqrt{120}}{13}$. The solutions are $x = \beta \pm \gamma + 2k\pi$, and

$$\cos(\beta\pm\gamma) = \cos\beta\cos\gamma \mp \sin\beta\sin\gamma = \frac{35 \mp 12\sqrt{120}}{169}.$$

So each sign gives a genuine solution: $\alpha = \beta+\gamma$ gives the minus sign and $\alpha = \beta-\gamma$ gives the plus sign.

The second-quadrant solution

Suppose $\frac{1}{2}\pi < \alpha < \pi$. Then $\cos\alpha < 0$. Since $\sqrt{120} > 10$, we have $35 + 12\sqrt{120} > 0$, so this root is ruled out. Also $12\sqrt{120} > 120 > 35$, so the other root is negative. Therefore

$$\cos\alpha = \frac{35 - 12\sqrt{120}}{169}.$$

Cosine is strictly decreasing on $[0,\pi]$. Both α and $\frac{3}{4}\pi$ lie in this interval, so

$$\alpha < \tfrac{3}{4}\pi \iff \cos\alpha > \cos\tfrac{3}{4}\pi = -\tfrac{1}{\sqrt{2}}.$$

So we must show $\dfrac{12\sqrt{120}-35}{169} < \dfrac{1}{\sqrt{2}}$, that is, $12\sqrt{120} - 35 < \dfrac{169}{\sqrt{2}}$.

Ask yourself How do we compare these surds without a calculator, and without squaring something that might be negative?

We first bound $\sqrt{120}$ crudely. Because $120 < 121$, $\sqrt{120} < 11$, so

$$12\sqrt{120} - 35 < 132 - 35 = 97.$$

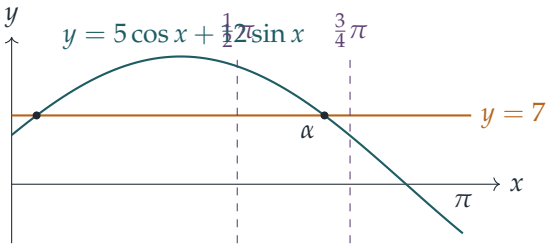
It is now enough to show $97 < \dfrac{169}{\sqrt{2}}$, that is, $97\sqrt{2} < 169$. Both sides are positive, so this is equivalent to the inequality between their squares:

$$2 \times 97^2 = 18\,818 < 28\,561 = 169^2,$$

which is true. Putting it together,

$$\cos\alpha = -\frac{12\sqrt{120}-35}{169} > -\frac{97}{169} > -\frac{1}{\sqrt{2}} = \cos\tfrac{3}{4}\pi,$$

and since cosine is decreasing on $[0,\pi]$, $\alpha < \frac{3}{4}\pi$, as required.



Checks

- Numerically, the roots in $[0, 2\pi)$ are $x \approx 0.1738$ and $x \approx 2.1782$. Their cosines are 0.98493 and -0.57073, which agree with $(35 \pm 12\sqrt{120})/169$.
- The second-quadrant root is 2.1782, which is less than $\frac{3}{4}\pi \approx 2.3562$, as the graph shows. The margin is comfortable: $\cos\alpha \approx -0.571$ against $-1/\sqrt{2} \approx -0.707$. That is why the crude bound $\sqrt{120} < 11$ is enough.
- As a sanity check on the quadratic, the two roots have product $-\frac{95}{169}$ and sum $\frac{70}{169}$.

Common slips

- Squaring an inequality without checking the signs. Here $12\sqrt{120} - 35$ and $169/\sqrt{2}$ are both positive, but the reason should be stated.
- Forgetting that cosine is *decreasing* on $[0, \pi]$, which reverses the inequality: $\alpha < \frac{3}{4}\pi$ corresponds to $\cos\alpha > -\frac{1}{\sqrt{2}}$.
- Not explaining why the positive root is excluded in the second quadrant.

Beyond STEP

The final part is a small exercise in rigorous estimation. You replace an awkward quantity by a simpler bound and check that the margin survives, which is how many inequalities in analysis are proved. The step from cosine values to angles uses the fact that a strictly monotonic function preserves (or reverses) order, which is the idea behind inverse functions in a first analysis course. For more on careful proof with inequalities, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1999, Question 6

Pure • Integration by Substitution

STEP II 1999, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q6

The key idea

The derivative of a fraction $\frac{ax+b}{cx+d}$ is a constant over $(cx+d)^2$. So whenever an integrand is $\frac{1}{(x+3)^2}$ times a function of $\frac{x+1}{x+3}$ or $\frac{x+2}{x+3}$, substituting that fraction as the new variable absorbs the $\frac{1}{(x+3)^2}$ completely, and leaves us with nothing worse than $\int \ln y \, \mathrm{d}y$. The other two integrals then follow from the laws of logarithms.

Solution

The derivative

By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{a(cx+d) - c(ax+b)}{(cx+d)^2} = \frac{ad-bc}{(cx+d)^2}.$$

The x terms in the numerator cancel, leaving a constant. This is what makes the whole question work.

The first integral

Let $y = \dfrac{x+1}{x+3}$. This has the form $(*)$ with $a=1$, $b=1$, $c=1$, $d=3$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3-1}{(x+3)^2} = \frac{2}{(x+3)^2}, \qquad \text{that is,} \qquad \frac{\mathrm{d}x}{(x+3)^2} = \tfrac{1}{2}\,\mathrm{d}y.$$

When $x = 0$, $y = \frac{1}{3}$; when $x = 1$, $y = \frac{1}{2}$. Using $\int \ln y \, dy = y \ln y - y$ (integration by parts with $u = \ln y$):

$$\begin{aligned}\int_0^1 \frac{1}{(x+3)^2} \ln\left(\frac{x+1}{x+3}\right) dx &= \frac{1}{2}\int_{1/3}^{1/2} \ln y \, dy = \frac{1}{2}\Big[y \ln y - y\Big]_{1/3}^{1/2} \\ &= \frac{1}{2}\left(-\tfrac{1}{2}\ln 2 - \tfrac{1}{2} + \tfrac{1}{3}\ln 3 + \tfrac{1}{3}\right) \\ &= \frac{1}{6}\ln 3 - \frac{1}{4}\ln 2 - \frac{1}{12},\end{aligned}$$

using $\ln \frac{1}{2} = -\ln 2$ and $\ln \frac{1}{3} = -\ln 3$. This is the required result. Call this integral I_1.

A companion integral

Ask yourself What has $x^2 + 3x + 2$ got to do with the substitution we have just used?

It factorises as $(x+1)(x+2)$, so

$$\frac{x^2+3x+2}{(x+3)^2} = \frac{x+1}{x+3} \cdot \frac{x+2}{x+3} \qquad \text{and} \qquad \frac{x+1}{x+2} = \frac{x+1}{x+3} \Big/ \frac{x+2}{x+3}.$$

All the logarithms in the last two integrals are therefore sums and differences of $\ln \frac{x+1}{x+3}$ and $\ln \frac{x+2}{x+3}$ (all these fractions are positive for $0 \le x \le 1$). So we need one more integral,

$$J = \int_0^1 \frac{1}{(x+3)^2} \ln\left(\frac{x+2}{x+3}\right) dx.$$

Let $y = \dfrac{x+2}{x+3}$. Now $a = 1$, $b = 2$, $c = 1$, $d = 3$, so $\dfrac{dy}{dx} = \dfrac{1}{(x+3)^2}$, and the limits become $y = \frac{2}{3}$ and $y = \frac{3}{4}$:

$$\begin{aligned}J &= \int_{2/3}^{3/4} \ln y \, dy = \Big[y \ln y - y\Big]_{2/3}^{3/4} \\ &= \tfrac{3}{4}(\ln 3 - 2\ln 2) - \tfrac{3}{4} - \tfrac{2}{3}(\ln 2 - \ln 3) + \tfrac{2}{3} \\ &= \frac{17}{12}\ln 3 - \frac{13}{6}\ln 2 - \frac{1}{12}.\end{aligned}$$

The remaining two integrals

Since $\ln \frac{x^2+3x+2}{(x+3)^2} = \ln \frac{x+1}{x+3} + \ln \frac{x+2}{x+3}$, the second integral is

$$\begin{aligned}I_1 + J &= \left(\tfrac{1}{6} + \tfrac{17}{12}\right)\ln 3 - \left(\tfrac{1}{4} + \tfrac{13}{6}\right)\ln 2 - \tfrac{1}{12} - \tfrac{1}{12} \\ &= \frac{19}{12}\ln 3 - \frac{29}{12}\ln 2 - \frac{1}{6}.\end{aligned}$$

Since $\ln\frac{x+1}{x+2} = \ln\frac{x+1}{x+3} - \ln\frac{x+2}{x+3}$, the third integral is

$$\begin{aligned} I_1 - J &= \left(\tfrac{1}{6} - \tfrac{17}{12}\right)\ln 3 - \left(\tfrac{1}{4} - \tfrac{13}{6}\right)\ln 2 \\ &= \frac{23}{12}\ln 2 - \frac{5}{4}\ln 3, \end{aligned}$$

where the constant terms $-\frac{1}{12}$ have cancelled.

Checks

- Simpson's rule on $[0,1]$ gives $I_1 \approx -0.07352$, $J \approx -0.02878$, and the second and third integrals ≈ -0.10230 and -0.04473, all agreeing with the exact forms to ten decimal places.
- A rough bound: $\int_0^1 (x+3)^{-2}\,\mathrm{d}x = \frac{1}{12}$, and $\ln\frac{x+1}{x+2}$ lies between $\ln\frac{1}{2} \approx -0.69$ and $\ln\frac{2}{3} \approx -0.41$. So the third integral must lie between about -0.058 and -0.034, and -0.0447 does.
- Every integrand is negative on $[0,1]$ (each fraction inside a logarithm is less than 1), so all three answers should be negative, and they are.

Common slips

- Forgetting the factor $\frac{1}{2}$ in the first substitution: $ad - bc = 2$ there, but 1 in the second.
- Not changing the limits, or keeping x-limits 0 and 1 after substituting.
- Trying a fresh substitution for $\frac{x+1}{x+2}$: it works, but leaves $(x+3)^{-2}$ in terms of y as something messy. Splitting the logarithm is far quicker.

Beyond STEP

Maps of the form $x \mapsto \frac{ax+b}{cx+d}$ with $ad - bc \neq 0$ are *Möbius transformations*. Composing two of them corresponds to multiplying the 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and $ad - bc$ is the determinant, which is why it appears in the derivative. In complex analysis they are exactly the conformal bijections of the extended plane, sending circles and lines to circles and lines. Needham's *Visual Complex Analysis* has a beautiful chapter on them.

WORKED SOLUTION

STEP II 1999, Question 7

Pure • Curve Sketching

STEP II 1999, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q7

The key idea

Differentiate and put everything over a common denominator: the numerator collapses almost completely: everything quadratic cancels and a single linear factor is left, which settles the stationary points at once. For the sketches, complete the square under the root. That tells us where the curve exists, how it behaves for large $|x|$, and, when $a = 1$, that the square root becomes a modulus, which turns the curve into two pieces of rectangular hyperbola.

Solution

Write $Q(x) = x^2 - 2x + a = (x-1)^2 + (a-1)$.

Exactly one stationary point when $a > 1$

If $a > 1$ then $Q(x) \geq a - 1 > 0$ for every x, so the curve is defined, and differentiable, for all real x. With $y = xQ^{-1/2}$ and $Q'(x) = 2x - 2$, the product rule gives

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= Q^{-1/2} - \tfrac{1}{2}x(2x-2)Q^{-3/2} = \frac{Q - x(x-1)}{Q^{3/2}} \\ &= \frac{x^2 - 2x + a - x^2 + x}{Q^{3/2}} = \frac{a - x}{(x^2 - 2x + a)^{3/2}}.\end{aligned}$$

The denominator is positive and finite for every x, so $\mathrm{d}y/\mathrm{d}x = 0$ exactly when $x = a$. So C has exactly one stationary point, at

$$x = a, \qquad y = \frac{a}{\sqrt{a^2 - a}} = \sqrt{\frac{a}{a-1}},$$

as required. Since $\mathrm{d}y/\mathrm{d}x$ has the sign of $a - x$, the curve rises for $x < a$ and falls for $x > a$, so this point is a maximum.

Ask yourself Where did $a > 1$ get used?

Twice: it makes $Q > 0$ everywhere, so there are no gaps in the curve and no points where the derivative fails to exist; and it makes $x = a$ a genuine point of the curve, since $Q(a) = a^2 - a > 0$. When $a = 1$ the would-be stationary point $x = a = 1$ is exactly where $Q = 0$, and that is what goes wrong in part (ii).

Two more facts help both sketches. For large $|x|$, $\sqrt{Q} = |x|\sqrt{1 - 2/x + a/x^2} \approx |x|$, so

$$y \to 1 \text{ as } x \to +\infty, \qquad y \to -1 \text{ as } x \to -\infty.$$

Also y has the sign of x, and the curve passes through the origin with gradient $a/a^{3/2} = 1/\sqrt{a}$.

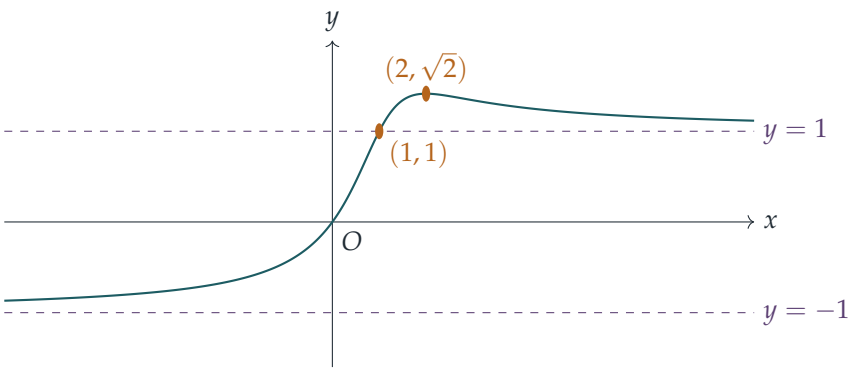
Part (i): $a = 2$

Here $y = \dfrac{x}{\sqrt{(x-1)^2 + 1}}$, defined for all x, with

- a maximum at $(2, \sqrt{2})$, increasing on $x < 2$ and decreasing on $x > 2$;
- the origin on the curve, with gradient $1/\sqrt{2}$ there;
- horizontal asymptotes $y = 1$ (as $x \to \infty$) and $y = -1$ (as $x \to -\infty$).

Which side of each asymptote is the curve on? Setting $y = \pm 1$ and squaring gives $x^2 = x^2 - 2x + 2$, so $x = 1$, and only $y = +1$ fits (since y has the sign of x). So the curve crosses $y = 1$ once, at $(1, 1)$: it is above $y = 1$ for all $x > 1$, coming down towards the asymptote from above after the maximum. It never meets $y = -1$, so it stays above $y = -1$ and rises away from it as x increases from $-\infty$.

(If you want more detail: differentiating again gives $\mathrm{d}^2y/\mathrm{d}x^2 = (2x^2 - 7x + 4)/Q^{5/2}$, so there are points of inflection at $x = \frac{1}{4}(7 \pm \sqrt{17})$, roughly $x \approx 0.72$ and $x \approx 2.78$. The sketch doesn't need these, but they explain its shape.)



Part (ii): $a = 1$

Now $Q = (x-1)^2$, so $x = 1$ is excluded and, because the square root is the positive one,

$$\sqrt{(x-1)^2} = |x-1|.$$

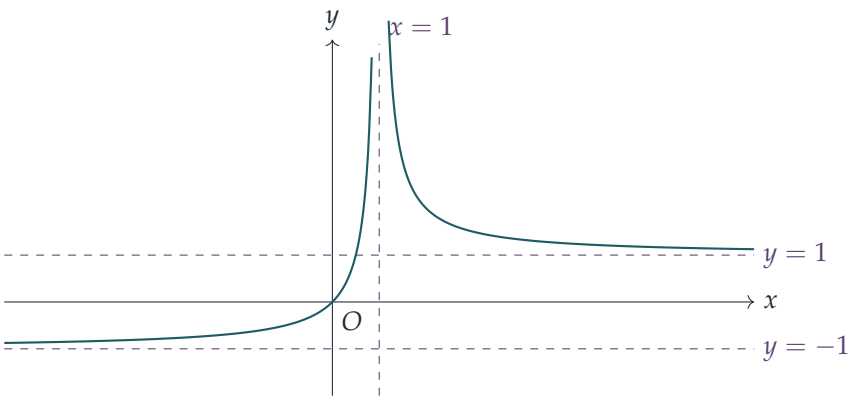
So

$$y = \frac{x}{|x-1|} = \begin{cases} \dfrac{x}{1-x} = -1 + \dfrac{1}{1-x}, & x < 1, \\ \dfrac{x}{x-1} = 1 + \dfrac{1}{x-1}, & x > 1. \end{cases}$$

Each branch is part of a rectangular hyperbola centred at $x = 1$.

- For $x < 1$: y increases (its derivative is $1/(1-x)^2 > 0$), from -1 (approached from above as $x \to -\infty$) through the origin with gradient 1, to $+\infty$ as $x \to 1^-$.
- For $x > 1$: y decreases (derivative $-1/(x-1)^2 < 0$), from $+\infty$ as $x \to 1^+$ down towards 1 (approached from above as $x \to \infty$).

The line $x = 1$ is a vertical asymptote with $y \to +\infty$ on *both* sides (the numerator is near 1, the denominator is small and positive). There is no stationary point: the maximum of the $a > 1$ curves, at height $\sqrt{a/(a-1)}$, has been pushed off to infinity as $a \to 1^+$. The general formula agrees, since $\mathrm{d}y/\mathrm{d}x = (1-x)/|x-1|^3$ is never zero.



Checks

- Numerically, the derivative formula $(a-x)/Q^{3/2}$ agrees with a finite-difference derivative at thousands of random points, and a sign scan finds exactly one sign change for several values of $a > 1$.
- For $a = 2$: $y(2) = 2/\sqrt{2} = \sqrt{2} \approx 1.414$, $y(1) = 1$, and $y(10^6) \approx 1.000001$, $y(-10^6) \approx -0.999999$, confirming the sides from which the asymptotes are approached.
- For $a = 1$: $y(0.999) \approx 999$ and $y(1.001) \approx 1001$, both large and positive, as claimed.

- As $a \to \infty$, the maximum height $\sqrt{a/(a-1)}$ tends to 1, which is sensible: for huge a the curve is nearly $x/\sqrt{a}$ near the origin and the bump flattens.

Common slips

- Writing $\sqrt{(x-1)^2} = x - 1$ in part (ii). That gives $y = x/(x-1)$ everywhere and the wrong left branch, which would then tend to $+1$ as $x \to -\infty$ and go to $-\infty$ as $x \to 1^-$.
- Giving both ends of the curve the asymptote $y = 1$: since $\sqrt{x^2} = |x|$, the limit as $x \to -\infty$ is -1.
- Forgetting to show that the denominator never vanishes when $a > 1$; without that, "the numerator is zero only at $x = a$" does not yet prove there is exactly one stationary point.

Beyond STEP

The family of curves here changes character suddenly at $a = 1$: a smooth curve with a single maximum turns into two separate branches with a vertical asymptote. Studying how the qualitative shape of a family depends on a parameter is the starting point of bifurcation theory, and the care needed with $\sqrt{x^2} = |x|$ and limits as $x \to \pm\infty$ is exactly the care that first-year analysis makes precise. Abbott's *Understanding Analysis* is a friendly place to see limits and continuity done properly.

WORKED SOLUTION

STEP II 1999, Question 8

Pure • Trigonometric Series

STEP II 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q8

The key idea

Multiplying $\sin k\theta$ by $2\sin\frac{1}{2}\theta$ turns it, by a product-to-sum formula, into a *difference* of two cosines, and the sum then telescopes. After that, both parts are about choosing θ well: with $\theta = \pi/n$ the awkward cosine $\cos(n+\frac{1}{2})\theta$ becomes a simple multiple of $\cos\frac{1}{2}\theta$. For (ii), the half-angle formula turns $\sin^2$ into a cosine, and the sum $\sum k\cos k\theta$ is exactly what differentiating $(*)$ produces.

Solution

Proving $(*)$

Assume $\sin\frac{1}{2}\theta \neq 0$, so that the right-hand side makes sense. The identity $2\sin A\sin B = \cos(A-B) - \cos(A+B)$ with $A = k\theta$ and $B = \frac{1}{2}\theta$ gives

$$2\sin\tfrac{1}{2}\theta\,\sin k\theta = \cos\left(k-\tfrac{1}{2}\right)\theta - \cos\left(k+\tfrac{1}{2}\right)\theta.$$

Summing from $k=0$ to n, each $\cos(k+\frac{1}{2})\theta$ subtracted in row k is added back in row $k+1$, so only the first and last terms survive:

$$2\sin\tfrac{1}{2}\theta\sum_{k=0}^{n}\sin k\theta = \cos\left(-\tfrac{1}{2}\theta\right) - \cos\left(n+\tfrac{1}{2}\right)\theta = \cos\tfrac{1}{2}\theta - \cos\left(n+\tfrac{1}{2}\right)\theta,$$

since cosine is even. Dividing by $2\sin\frac{1}{2}\theta$ gives $(*)$.

Part (i)

Put $\theta = \pi/n$ (with $n \geq 1$), so $\sin\frac{1}{2}\theta = \sin\frac{\pi}{2n} \neq 0$. Then

$$\cos\left(n+\tfrac{1}{2}\right)\theta = \cos\left(\pi+\tfrac{\pi}{2n}\right) = -\cos\tfrac{\pi}{2n},$$

so the numerator of $(*)$ is $2\cos\frac{\pi}{2n}$ and

$$\sum_{k=0}^{n}\sin\left(\frac{k\pi}{n}\right)=\frac{2\cos\frac{\pi}{2n}}{2\sin\frac{\pi}{2n}}=\cot\frac{\pi}{2n}.$$

This is exact. When n is large, $\frac{\pi}{2n}$ is small, so $\cos\frac{\pi}{2n}\approx 1$ and $\sin\frac{\pi}{2n}\approx\frac{\pi}{2n}$, giving

$$\sum_{k=0}^{n}\sin\left(\frac{k\pi}{n}\right)\approx\frac{1}{\pi/(2n)}=\frac{2n}{\pi}.$$

Part (ii)

By the double-angle formula, $\sin^2\frac{k\pi}{2n}=\frac{1}{2}\left(1-\cos\frac{k\pi}{n}\right)$, so with $\theta=\pi/n$

$$\sum_{k=0}^{n}k\sin^2\left(\frac{k\pi}{2n}\right)=\frac{1}{2}\sum_{k=0}^{n}k-\frac{1}{2}\sum_{k=0}^{n}k\cos k\theta=\frac{n(n+1)}{4}-\frac{1}{2}\sum_{k=0}^{n}k\cos k\theta. \qquad (1)$$

To find $\sum k\cos k\theta$ we differentiate $(*)$ with respect to θ; the left-hand side becomes exactly this sum. First split the right-hand side as

$$\frac{1}{2}\cot\tfrac{1}{2}\theta-\frac{\cos(n+\frac{1}{2})\theta}{2\sin\frac{1}{2}\theta}.$$

Write $S=\sin\frac{1}{2}\theta$ and $c=\cos\frac{1}{2}\theta$. The first piece differentiates to $-\frac{1}{4}\operatorname{cosec}^2\frac{1}{2}\theta=-\frac{1}{4S^2}$. By the quotient rule the second piece differentiates to

$$\frac{(n+\frac{1}{2})\sin(n+\frac{1}{2})\theta\cdot 2S+\cos(n+\frac{1}{2})\theta\cdot c}{4S^2}.$$

So

$$\sum_{k=0}^{n}k\cos k\theta=\frac{-1+(2n+1)S\sin(n+\frac{1}{2})\theta+c\cos(n+\frac{1}{2})\theta}{4S^2}.$$

Ask yourself Why put $\theta=\pi/n$ *after* differentiating, not before?

The identity $(*)$ holds for all θ (with $\sin\frac{1}{2}\theta\neq 0$), so we may differentiate it. If we fixed $\theta=\pi/n$ first we would have a single number with nothing to differentiate. Once we have the derivative as an identity in θ, we can substitute any value we like.

Now take $\theta=\pi/n$. As in (i), $(n+\frac{1}{2})\theta=\pi+\frac{1}{2}\theta$, so $\sin(n+\frac{1}{2})\theta=-S$ and $\cos(n+\frac{1}{2})\theta=-c$. Hence

$$\sum_{k=0}^{n}k\cos\frac{k\pi}{n}=\frac{-1-(2n+1)S^2-c^2}{4S^2}=\frac{-2-2nS^2}{4S^2}=-\frac{1}{2S^2}-\frac{n}{2},$$

using $c^2=1-S^2$. Substituting into (1):

$$\sum_{k=0}^{n}k\sin^2\left(\frac{k\pi}{2n}\right)=\frac{n(n+1)}{4}+\frac{n}{4}+\frac{1}{4S^2}=\frac{n(n+2)}{4}+\frac{1}{4\sin^2\frac{\pi}{2n}}.$$

This is exact. For large n, $\sin\frac{\pi}{2n} \approx \frac{\pi}{2n}$, so $\frac{1}{4\sin^2(\pi/2n)} \approx \frac{n^2}{\pi^2}$, and $\frac{n(n+2)}{4} = \frac{n^2}{4} + \frac{n}{2}$, whose second term is negligible beside n^2. Therefore

$$\sum_{k=0}^{n} k\sin^2\left(\frac{k\pi}{2n}\right) \approx \left(\frac{1}{4} + \frac{1}{\pi^2}\right)n^2,$$

as required.

Checks

- $n = 1$ in (ii): the sum is $0 + 1\cdot\sin^2\frac{\pi}{2} = 1$, and the exact formula gives $\frac{3}{4} + \frac{1}{4} = 1$.
- $n = 2$ in (i): the sum is $0 + 1 + 0 = 1$, and $\cot\frac{\pi}{4} = 1$.
- Numerically, for $n = 1000$ the sum in (i) is $636.6192\ldots$ against $2n/\pi = 636.6198\ldots$, and the sum in (ii) is 351 821.3, within 0.15% of $(\frac{1}{4} + \frac{1}{\pi^2})n^2$. The relative error in (ii) shrinks like $1/n$ because of the neglected $\frac{n}{2}$ term.

Common slips

- Writing $\cos(\pi + x) = \cos x$: the correct value is $-\cos x$, and that sign is what doubles the numerator in (i).
- Forgetting the $\frac{1}{2}\sum k = \frac{n(n+1)}{4}$ term in (ii), which supplies the $\frac{1}{4}n^2$.

Beyond STEP

Part (i) is a Riemann sum in disguise: $\frac{\pi}{n}\sum\sin\frac{k\pi}{n} \to \int_0^{\pi}\sin x\,\mathrm{d}x = 2$, which is exactly the statement $\sum \approx 2n/\pi$. Similarly, (ii) says $\frac{1}{n^2}\sum k\sin^2\frac{k\pi}{2n} \to \int_0^1 x\sin^2\frac{\pi x}{2}\,\mathrm{d}x = \frac{1}{4} + \frac{1}{\pi^2}$. The same telescoping trick applied to $\sum\cos k\theta$ gives the Dirichlet kernel of Fourier analysis. Abbott's *Understanding Analysis* treats Riemann sums and Fourier series carefully.

WORKED SOLUTION

STEP II 1999, Question 9

Mechanics • Variable Force Motion

STEP II 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q9

The key idea

The acceleration depends only on the distance x, not on time, so the right tool is $\ddot{x} = v\dfrac{\mathrm{d}v}{\mathrm{d}x}$. This turns the equation of motion into one we can integrate directly to get v^2 as a function of x (an energy equation in disguise). Everything else is about asking where v^2 hits zero: that is where a fragment stops and turns back.

Solution

We model each fragment as a particle moving along a straight line through the fixed point G, with x its distance from G and $v = \dot{x}$ its velocity, positive when moving away from G. Initially $x = 0$ and $v = u > 0$. Because the acceleration points towards G, it is negative in this convention.

Part (i)

For $x < R$ we have $\ddot{x} = -k_1 x^3$, and writing $\ddot{x} = v\dfrac{\mathrm{d}v}{\mathrm{d}x}$,

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -k_1 x^3.$$

Separating and integrating from the start ($x = 0$, $v = u$):

$$\tfrac{1}{2}v^2 - \tfrac{1}{2}u^2 = -\tfrac{1}{4}k_1 x^4, \qquad \text{so} \qquad v^2 = u^2 - \tfrac{1}{2}k_1 x^4 \quad (x < R).$$

While the fragment is still moving outwards, $v = \sqrt{u^2 - \tfrac{1}{2}k_1 x^4}$.

Part (ii)

With $2a^2 = k_1R^4$, the formula from (i) reads $v^2 = u^2 - a^2(x/R)^4$. As x increases from 0, v^2 decreases, and it reaches zero at

$$x_0 = \left(\frac{2u^2}{k_1}\right)^{1/4}.$$

If $u < a$ then $2u^2 < 2a^2 = k_1R^4$, so $x_0 < R$.

Ask yourself The speed is zero at x_0, but why does that mean the fragment never gets further?

At x_0 the fragment is momentarily at rest, but its acceleration there is $-k_1x_0^3 \neq 0$, directed towards G. So it is not in equilibrium: it starts to move back towards G. In fact x can never exceed x_0 at all, because the formula from (i) would then give $v^2 < 0$. The furthest the fragment ever gets is $x_0 < R$, so it never reaches a distance R from G, as required.

Part (iii)

First, a fragment with $u \geq b$ certainly reaches R. From the definitions,

$$b^2 = \tfrac{1}{2}k_1R^4 + \frac{2k_2}{3R^3} = a^2 + \frac{2k_2}{3R^3} > a^2,$$

so for $x \leq R$, part (i) gives

$$v^2 = u^2 - \tfrac{1}{2}k_1x^4 \geq u^2 - a^2 \geq b^2 - a^2 = \frac{2k_2}{3R^3} > 0.$$

In particular the fragment arrives at $x = R$ with $v_R^2 = u^2 - \tfrac{1}{2}k_1R^4 = u^2 - a^2$.

For $x \geq R$ the equation of motion is $v\dfrac{\mathrm{d}v}{\mathrm{d}x} = -k_2x^{-4}$. Integrating from $x = R$:

$$\tfrac{1}{2}v^2 - \tfrac{1}{2}v_R^2 = \frac{k_2}{3}\left(\frac{1}{x^3} - \frac{1}{R^3}\right),$$

so

$$\begin{aligned} v^2 &= u^2 - \tfrac{1}{2}k_1R^4 - \frac{2k_2}{3R^3} + \frac{2k_2}{3x^3} \\ &= u^2 - b^2 + \frac{2k_2}{3x^3} \qquad (x \geq R). \end{aligned}$$

That combination is exactly why b was defined as it is: b^2 is the total "cost" in $\frac{1}{2}v^2$ of escaping from G to infinity.

If $u \geq b$ then $u^2 - b^2 \geq 0$ and $\frac{2k_2}{3x^3} > 0$, so $v^2 > 0$ for every $x \geq R$ (this includes the borderline case $u = b$). Together with the bound for $x \leq R$, v^2 is strictly positive at every distance the fragment could reach.

The velocity v is continuous in time and starts at $u > 0$. To become negative it would have to pass through 0, which would need $v^2 = 0$ at some position, and we have just shown that never happens. So $v > 0$ throughout: from the moment of the explosion the fragment is always moving away from G. (Since $v^2 \geq 2k_2/(3x^3)$, it keeps going, with $x \to \infty$.)

Part (iv)

Now $a < u < b$. Since $u > a$, the argument of (iii) shows that $v^2 \geq u^2 - a^2 > 0$ on $[0, R]$, so the fragment reaches R still moving outwards. Beyond R,

$$v^2 = \frac{2k_2}{3x^3} - (b^2 - u^2),$$

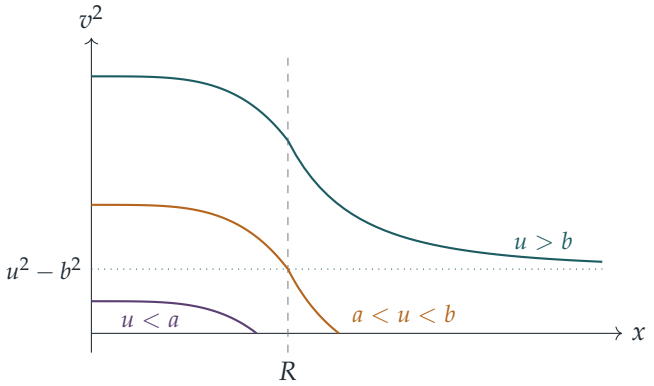
and now $b^2 - u^2 > 0$. This decreases as x increases, and is zero when

$$x^3 = \frac{2k_2}{3(b^2 - u^2)}.$$

This point does lie in the region $x \geq R$: since $u > a$, $b^2 - u^2 < b^2 - a^2 = \frac{2k_2}{3R^3}$, so $x^3 > R^3$. As in (ii), the fragment stops there with a non-zero acceleration $-k_2 x^{-4}$ towards G, and it cannot go further because v^2 would be negative. The maximum distance from G is

$$\boxed{x_{\max} = \left(\frac{2k_2}{3(b^2 - u^2)}\right)^{1/3}}$$

The figure shows v^2 against x for $k_1 = 2$, $k_2 = 3$, $R = 1$ (so $a^2 = 1$, $b^2 = 3$) and the three kinds of initial speed. Each fragment travels until its curve meets the axis.



Checks

- With $k_1 = 2$, $k_2 = 3$, $R = 1$ we get $a = 1$, $b = \sqrt{3}$; for $u = \sqrt{2}$ the formula gives $x_{\max} = (2 \cdot 3/(3 \cdot 1))^{1/3} = 2^{1/3} \approx 1.26$, just beyond R, as it should be.

- As $u \to a^+$, $b^2 - u^2 \to 2k_2/(3R^3)$ and $x_{\max} \to R$; as $u \to b^-$, $x_{\max} \to \infty$. Both match parts (ii) and (iii) at the boundaries.
- Stepping the motion numerically (Runge–Kutta, twelve random choices of k_1, k_2, R) reproduces the formula for v^2, the turning point $(2u^2/k_1)^{1/4}$ when $u < a$, the value of $x_{\max}$ in (iv) to within 0.1%, and no reversal at all when $u \geq b$.

Common slips

- Integrating the second stage from $x = 0$ instead of from $x = R$ with $v_R^2 = u^2 - \frac{1}{2}k_1R^4$: the force law changes at R, so the constant changes too.
- Losing the sign: the acceleration is towards G, so $v\,\mathrm{d}v/\mathrm{d}x$ is *minus* k_1x^3 or k_2x^{-4}.
- In (iii), checking only $x \geq R$ and forgetting that v must also stay positive inside R.

Beyond STEP

The quantity $\frac{1}{2}v^2 + V(x)$, with $V'(x)$ equal to minus the acceleration, is conserved for any force depending only on position. Plotting V (a "potential well") tells you at a glance which motions are bounded and which escape, and b is just the escape speed for this well. The same reasoning gives escape velocity in real gravity, and is the starting point for the phase-plane analysis of nonlinear oscillators. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy diagrams like this in detail.

WORKED SOLUTION

STEP II 1999, Question 10

Mechanics • Successive Collisions

STEP II 1999, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q10

The key idea

Every impact in this chain looks the same: a particle of mass μ moving with speed u hits a particle of mass $q\mu$ at rest. So the first-impact calculation, scaled, describes *every* impact. Once we know each particle's final velocity, the question of whether any extra impacts happen reduces to comparing neighbouring final velocities, and with $q = e$ the velocities become so simple that the energy loss is just a geometric series.

Solution

The first impact

Model the particles as point masses moving along the line, and take the direction from P_1 towards P_2 as positive. Let v_1 and v_2 be the velocities of P_1 and P_2 just after the impact. Conservation of momentum and Newton's law of restitution give

$$\begin{aligned} mV &= mv_1 + qm\,v_2, \\ v_2 - v_1 &= eV. \end{aligned}$$

Substituting $v_2 = v_1 + eV$ into the first equation gives $V = v_1 + qv_1 + qeV$, so

$$v_1 = \left(\frac{1-eq}{1+q}\right)V, \qquad v_2 = v_1 + eV = \frac{1-eq+e+eq}{1+q}\,V = \left(\frac{1+e}{1+q}\right)V,$$

as required.

Every impact has the same shape

Write

$$\alpha = \frac{1-eq}{1+q}, \qquad \beta = \frac{1+e}{1+q}.$$

Nothing in the calculation above used the actual value of m: a particle of mass μ moving at u that hits a particle of mass $q\mu$ at rest leaves with velocity αu and sends the other off

with velocity βu. Since each P_{k+1} has q times the mass of P_k, this applies to an impact of P_k on P_{k+1} whenever P_{k+1} is still at rest.

Exactly $N-1$ impacts when $q \le e$

Suppose $q \le e$. Then $eq \le e^2 < 1$, so $\alpha > 0$; also $\beta > 0$.

Follow the chain. P_1 hits P_2, leaving P_1 with velocity αV and P_2 with βV. Since $\beta V > 0$, P_2 moves towards P_3, which is at rest, and hits it; this leaves P_2 with $\alpha\beta V$ and P_3 with $\beta^2 V$. Continuing, P_k arrives at the stationary P_{k+1} with velocity $u_k = \beta^{k-1}V > 0$, so there is one impact between each neighbouring pair: at least $N-1$ impacts. After them the velocities are

$$w_k = \alpha\beta^{k-1}V \quad (1 \le k \le N-1), \qquad w_N = \beta^{N-1}V.$$

All of these are positive, so everything moves the same way.

Ask yourself How do we know P_k never catches P_{k+1} again?

After their impact, P_k moves at constant velocity w_k from then on, because its only possible later contacts are with P_{k+1} or P_{k-1}. Meanwhile P_{k+1} moves at $u_{k+1} = \beta^k V$ until it hits P_{k+2}, and at w_{k+1} afterwards (or for ever, if $k+1 = N$). The gap between P_k and P_{k+1} can only close if P_k is moving faster, and:

- $u_{k+1} > w_k$, since $\beta^k V > \alpha\beta^{k-1}V$ is equivalent to $\beta > \alpha$, i.e. $1+e > 1-eq$, which is true;
- for $k \le N-2$, $w_{k+1} \ge w_k$, since $w_{k+1}/w_k = \beta = \dfrac{1+e}{1+q} \ge 1$ exactly because $q \le e$;
- for $k = N-1$, $w_N > w_{N-1}$, since $w_N/w_{N-1} = \beta/\alpha > 1$.

So P_{k+1} is never slower than P_k after their impact, and the gap between them never shrinks. (When $q = e$ we get $w_{k+1} = w_k$: the gap stays constant, but it doesn't close.) This holds for every neighbouring pair, so no further impacts occur and there are *exactly* $N-1$ impacts.

This is where $q \le e$ matters. If $q > e$ then $\beta < 1$, each particle ends up slower than the one behind it, and the chain catches up with itself, producing more impacts.

Kinetic energy lost when $q = e$

Now $q = e$, so

$$\beta = \frac{1+e}{1+e} = 1, \qquad \alpha = \frac{1-e^2}{1+e} = 1-e.$$

Every particle is hit at speed V; afterwards $P_1, \dots, P_{N-1}$ move at $(1-e)V$ and P_N moves at V. With P_k of mass $e^{k-1}m$, the final kinetic energy is

$$\begin{aligned} T &= \sum_{k=1}^{N-1} \tfrac{1}{2} e^{k-1} m (1-e)^2 V^2 + \tfrac{1}{2} e^{N-1} m V^2 \\ &= \tfrac{1}{2} m V^2 \left[(1-e)^2 \cdot \frac{1-e^{N-1}}{1-e} + e^{N-1} \right] \\ &= \tfrac{1}{2} m V^2 \left[(1-e)(1-e^{N-1}) + e^{N-1} \right], \end{aligned}$$

summing the geometric series $1 + e + \cdots + e^{N-2} = \dfrac{1-e^{N-1}}{1-e}$. The initial kinetic energy is $\frac{1}{2}mV^2$, so the loss is

$$\begin{aligned} \tfrac{1}{2} m V^2 - T &= \tfrac{1}{2} m V^2 \left[(1-e^{N-1}) - (1-e)(1-e^{N-1}) \right] \\ &= \tfrac{1}{2} m e \left(1 - e^{N-1} \right) V^2, \end{aligned}$$

which is the required result.

Checks

- *Impact by impact.* When $q = e$, the impact of P_k on P_{k+1} takes P_k from V to $(1-e)V$ and P_{k+1} from 0 to V, losing $\frac{1}{2}e^{k-1}mV^2\left[1-(1-e)^2-e\right] = \frac{1}{2}m(1-e)e^kV^2$. Summing over $k = 1, \dots, N-1$ gives $\frac{1}{2}m(1-e)V^2 \cdot e\,\dfrac{1-e^{N-1}}{1-e}$, the same answer.
- *Limits.* As $e \to 0$ the loss tends to 0, and indeed with $q = e \to 0$ the later particles have negligible mass. With $N = 2$ the formula gives $\frac{1}{2}me(1-e)V^2$, which matches a direct calculation for a single impact.
- *Numerically.* We simulated the motion event by event for random $N \le 12$, random gaps and random $q \le e$, and always found exactly $N-1$ impacts. For $N = 5$, $q = e = \frac{1}{2}$, $m = V = 1$ the simulated loss was $0.234375 = \frac{15}{64}$, in agreement with $\frac{1}{2} \cdot \frac{1}{2}\left(1 - \frac{1}{16}\right)$.

Common slips

- Checking only that the first $N-1$ impacts happen. "Exactly" also needs an argument that no particle catches the one in front later on.
- Comparing P_k's final velocity with P_{k+1}'s velocity *just after* their impact (βu_k), rather than with its final velocity after it has hit P_{k+2} and slowed down.
- Getting the sign wrong in the restitution law: it is the speed of separation, $v_2 - v_1$, that equals e times the speed of approach.

Beyond STEP

A chain of masses that pass momentum along one at a time is a one-dimensional model of a shock moving through a granular material. Chains of steadily decreasing masses, like this one when $q < 1$, are studied by physicists as a way to spread out and absorb impacts. A nice way to reorganise the energy calculation is through the *reduced mass* $\mu = m_1 m_2/(m_1 + m_2)$: one impact loses $\frac{1}{2}\mu(1 - e^2)w^2$, where w is the speed of approach. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions in the centre-of-mass frame, where results like this become transparent.

WORKED SOLUTION

STEP II 1999, Question 11

Mechanics • Projectiles and Moving Targets

STEP II 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q11

The key idea

The target stays on the circle, so the shell must land on the horizontal plane at distance exactly R from O; and the gun can always be swung round to face wherever the target will be. So everything comes down to two facts about one projectile: its range and its time of flight. The rotation angle is $\beta = \omega T$, and eliminating α with $\sin^2\alpha + \cos^2\alpha = 1$ produces the quartic. The last part is then Vieta's formulas plus a chord of a circle.

Solution

We model the shell as a particle moving under gravity alone (no air resistance), fired from O on the horizontal plane containing the circle, with $0 < \alpha < \frac{1}{2}\pi$.

Part 1: when $V^2 < gR$

Take axes along the ground (in the plane of firing) and vertically up. At time t,

$$x = Vt\cos\alpha, \qquad y = Vt\sin\alpha - \tfrac{1}{2}gt^2.$$

The shell returns to the plane ($y = 0$, $t > 0$) at

$$T = \frac{2V\sin\alpha}{g}, \qquad \text{at horizontal distance} \qquad x(T) = \frac{2V^2\sin\alpha\cos\alpha}{g} = \frac{V^2\sin 2\alpha}{g}.$$

Since $\sin 2\alpha \le 1$, the shell lands no further than V^2/g from O, whatever the vertical plane. The target is always on the plane at distance R from O, so a hit needs $R \le V^2/g$, that is $V^2 \ge gR$. Hence if $V^2 < gR$ the shell can never hit the target, as required.

Part 2: the equation for β

Now $V^2 > gR$. The shell hits the target at time T and distance R, so

$$T = \frac{2V\sin\alpha}{g}, \qquad R = \frac{2V^2\sin\alpha\cos\alpha}{g}.$$

The target turns at constant rate ω, so $\beta = \omega T$, which gives

$$\sin\alpha = \frac{g\beta}{2V\omega}.$$

Substituting this into the range equation,

$$\cos\alpha = \frac{gR}{2V^2\sin\alpha} = \frac{gR}{2V^2}\cdot\frac{2V\omega}{g\beta} = \frac{R\omega}{V\beta}.$$

Now $\sin^2\alpha + \cos^2\alpha = 1$ gives

$$\frac{g^2\beta^2}{4V^2\omega^2} + \frac{R^2\omega^2}{V^2\beta^2} = 1,$$

and multiplying through by $4V^2\omega^2\beta^2$,

$$g^2\beta^4 - 4\omega^2V^2\beta^2 + 4R^2\omega^4 = 0,$$

which is the required equation.

Ask yourself Why are there exactly two values, and not up to four, or none?

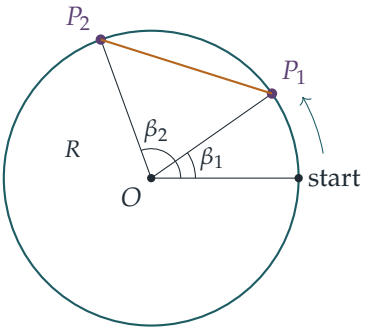
As a quadratic in β^2 its discriminant is

$$16\omega^4V^4 - 16g^2R^2\omega^4 = 16\omega^4\left(V^4 - g^2R^2\right) > 0,$$

since $V^2 > gR$. So there are two distinct real values of β^2. Their sum $4\omega^2V^2/g^2$ and product $4R^2\omega^4/g^2$ are both positive, so both values of β^2 are positive. The angle $\beta = \omega T$ is positive, so each value of β^2 gives exactly one β, and we have exactly two candidates $\beta_1 < \beta_2$.

Each one really does give a hit. For either root, the numbers $g\beta/(2V\omega)$ and $R\omega/(V\beta)$ are positive and (by the equation) their squares add up to 1, so there is an $\alpha \in (0, \frac{1}{2}\pi)$ with these as its sine and cosine. Firing at that elevation, in the vertical plane through the point the target will reach after turning through β, gives a time of flight β/ω and a range R, so the shell meets the target. Hence there are exactly two possible values of β. (They correspond to the two complementary elevations α and $\frac{1}{2}\pi - \alpha$ that give range R.)

Part 3: the distance P_1P_2



Both points of impact are on the circle, measured anticlockwise from the target's position at the moment of firing, at angles β_1 and β_2. So the angle P_1OP_2 is $\beta_2 - \beta_1$ (reduced modulo 2π if necessary, which does not affect the chord), and the isosceles triangle OP_1P_2 gives

$$P_1P_2 = 2R\left|\sin\frac{\beta_2-\beta_1}{2}\right|.$$

From the quadratic in β^2,

$$\beta_1^2+\beta_2^2 = \frac{4\omega^2V^2}{g^2}, \qquad \beta_1^2\beta_2^2 = \frac{4R^2\omega^4}{g^2},$$

and since $\beta_1, \beta_2 > 0$ we can take the positive square root: $\beta_1\beta_2 = 2R\omega^2/g$. Therefore

$$\begin{aligned}(\beta_2-\beta_1)^2 &= \beta_1^2+\beta_2^2-2\beta_1\beta_2\\ &= \frac{4\omega^2V^2}{g^2} - \frac{4R\omega^2}{g} = \frac{4\omega^2}{g^2}\left(V^2-Rg\right),\end{aligned}$$

so $\beta_2-\beta_1 = \dfrac{2\omega}{g}\sqrt{V^2-Rg}$. Hence

$$P_1P_2 = 2R\sin\left(\frac{\omega}{g}\sqrt{V^2-Rg}\right),$$

as required. (Strictly this needs the sine to be non-negative, that is $\frac{\omega}{g}\sqrt{V^2-Rg} \le \pi$; otherwise the distance is the modulus of this expression.)

Checks

- As $V^2 \to gR^+$ the two elevations both tend to 45°, so P_1 and P_2 should merge: indeed the formula tends to 0.
- Check $\beta_1\beta_2 = 2R\omega^2/g$ directly: $\beta_1\beta_2 = (2V\omega/g)^2\sin\alpha_1\sin\alpha_2$, and with $\alpha_2 = \frac{1}{2}\pi - \alpha_1$ this is $(2V\omega/g)^2\sin\alpha_1\cos\alpha_1 = (4V^2\omega^2/g^2)(gR/2V^2)$, as expected.
- Numerically, with $g = 9.8$, $R = 10$, $V = 14$, $\omega = 0.3$ (SI units), stepping both trajectories gives a chord of about 5.97 m, matching $20\sin\left(0.3\sqrt{98}/9.8\right) \approx 5.969$.

Common slips

- Forgetting that the gun can be aimed in any vertical plane: the first part is about range alone, not about where the target happens to be.
- Taking the square root of $\beta_1^2\beta_2^2$ without saying why $\beta_1\beta_2$ is positive.
- Writing $P_1P_2 = R(\beta_2-\beta_1)$, which is the arc length, not the straight-line distance.

Beyond STEP

Hitting a moving target is a small *pursuit* or *interception* problem: you solve for the time of impact first, then for the aim. Eliminating a parameter between two equations to get a polynomial, and then reading off what you need from its coefficients without solving it, is a technique you will use constantly at university. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles and rotating motion with this kind of care.

WORKED SOLUTION

STEP II 1999, Question 12

Probability and Statistics • Bayes' Theorem

STEP II 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q12

The key idea

This is Bayes' theorem with a parameter left in. Once $\mathrm{P}(S)$ is found by the law of total probability, both conditional probabilities are simple rational functions of p, and each is monotonic on the allowed interval. So the "greatest value" in each inspector's procedure sits at an end of the interval $0 \le p \le \frac{1}{3}$, and the interesting part is what that says about estimating from a single observation.

Solution

The conditional probabilities

Let A, B, C be the events that a randomly selected chip was made by A, B, C respectively, and let S be the event that it is sound. These three events partition the sample space, with

$$\begin{aligned} \mathrm{P}(A) = 2p, \quad \mathrm{P}(B) = p, \quad \mathrm{P}(C) = 1 - 3p, \\ \mathrm{P}(S \mid A) = \tfrac{7}{10}, \quad \mathrm{P}(S \mid B) = \tfrac{8}{10}, \quad \mathrm{P}(S \mid C) = \tfrac{9}{10}. \end{aligned}$$

By the law of total probability,

$$\begin{aligned} \mathrm{P}(S) &= 2p \cdot \tfrac{7}{10} + p \cdot \tfrac{8}{10} + (1 - 3p) \cdot \tfrac{9}{10} \\ &= \tfrac{1}{10}(14p + 8p + 9 - 27p) = \tfrac{1}{10}(9 - 5p). \end{aligned}$$

Note that $9 - 5p > 0$ throughout $0 \le p \le \frac{1}{3}$, so we may divide by it. By Bayes' theorem,

$$\mathrm{P}(A \mid S) = \frac{\mathrm{P}(S \mid A)\,\mathrm{P}(A)}{\mathrm{P}(S)} = \frac{\frac{14p}{10}}{\frac{9-5p}{10}} = \frac{14p}{9 - 5p},$$

and similarly

$$\mathrm{P}(C \mid S) = \frac{\frac{9}{10}(1 - 3p)}{\frac{1}{10}(9 - 5p)} = \frac{9(1 - 3p)}{9 - 5p} = \frac{9 - 27p}{9 - 5p}.$$

The first inspector's estimate

As p increases from 0 to $\frac{1}{3}$, the numerator $14p$ increases and the positive denominator $9-5p$ decreases, so $\mathrm{P}(A \mid S)$ is strictly increasing. (Equivalently, its derivative is $126/(9-5p)^2 > 0$.) Its greatest value is therefore at the right-hand end, $p = \frac{1}{3}$, where it equals

$$\frac{14/3}{9-5/3} = \frac{14/3}{22/3} = \frac{7}{11}.$$

So the first inspector's estimate is $\hat{p} = \frac{1}{3}$.

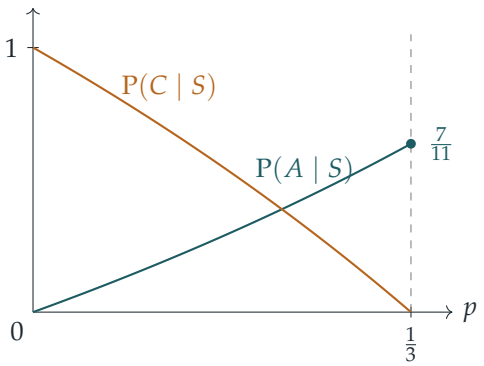
The second inspector's estimate

Ask yourself Is $\mathrm{P}(C \mid S)$ increasing or decreasing? Both its numerator and its denominator decrease as p grows.

When numerator and denominator move in the same direction we can't just read off the answer, so we differentiate with the quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}p}\frac{9-27p}{9-5p} = \frac{-27(9-5p)+5(9-27p)}{(9-5p)^2} = \frac{-243+135p+45-135p}{(9-5p)^2} = \frac{-198}{(9-5p)^2} < 0.$$

So $\mathrm{P}(C \mid S)$ is strictly decreasing on $[0, \frac{1}{3}]$, and its greatest value is at $p = 0$, where it equals 1. The second inspector's estimate is $\hat{p} = 0$.



Comment

The two inspectors, using the same method on equally valid single observations, arrive at the two opposite extremes of the allowed range: $\hat{p} = \frac{1}{3}$ (no chips at all come from C) and $\hat{p} = 0$ (every chip comes from C). Each estimate is simply "the value of p that makes my chip's maker as common as possible", which a sample of one is bound to produce. The second estimate is even contradicted by the first inspector's chip, since $p = 0$ says that A makes no chips. Neither estimate is believable: one observation carries far too little information to estimate p, and a larger sample (with the counts from each manufacturer) is needed.

It is also worth noticing that the soundness of the chip plays almost no part. The likelihood of "sound and from A" is $\frac{7}{10} \cdot 2p$, which is maximised at the same place as $2p$ itself; the factor $\frac{7}{10}$ doesn't depend on p.

Checks

- At $p = 0$ every chip comes from C, so $\mathrm{P}(C \mid S)$ should be 1 and $\mathrm{P}(A \mid S)$ should be 0; both formulae agree.
- Similarly $\mathrm{P}(B \mid S) = 8p/(9 - 5p)$, and the three conditional probabilities add to
$$\frac{14p + 8p + 9 - 27p}{9 - 5p} = 1.$$
- At $p = 0.2$ the formulae give $\mathrm{P}(A \mid S) = 0.35$ and $\mathrm{P}(C \mid S) = 0.45$; a simulation of 400 000 chips gave 0.3505 and 0.4505.

Common slips

- Setting the derivative equal to zero and looking for a turning point. There isn't one: both maxima are at endpoints of $[0, \frac{1}{3}]$.
- Forgetting that $\mathrm{P}(C \mid S)$ needs a careful argument: both numerator and denominator decrease, so "the numerator gets smaller" is not enough.

Beyond STEP

Both inspectors are doing, in effect, maximum likelihood estimation: choosing the parameter value under which what they saw is most probable. With one observation the maximum likelihood estimate often lands on the boundary of the parameter space, which is a warning that it needs plenty of data to behave well. A Bayesian would instead put a prior distribution on p and update it with each chip, which avoids such extreme conclusions from a single observation. Blitzstein and Hwang, *Introduction to Probability*, discusses both points of view.

WORKED SOLUTION

STEP II 1999, Question 13

Probability and Statistics • Continuous Random Variables

STEP II 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q13

The key idea

Although it is filed under mechanics, this is a question about a continuous random variable. Everything becomes easy once you notice that the *shorter* piece has a simple distribution: if the break point is uniform along the stick, the shorter piece is uniform on $[0, \frac{1}{2}]$ of the length. The ratio R is an increasing function of the shorter piece, so "$R < r$" turns into a statement about the shorter piece alone. That gives the distribution function, and differentiating gives the density.

Solution

The distribution of R

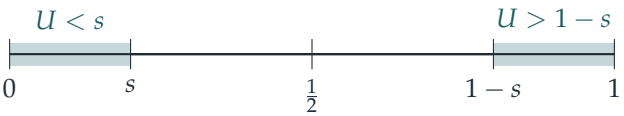
Take the stick to have length 1; ratios don't depend on the units. Let the break be at distance U from one end, where U is uniformly distributed on $[0, 1]$. The two pieces have lengths U and $1 - U$, so the shorter one has length

$$S = \min(U, 1 - U),$$

and the longer one has length $1 - S$. Then $0 \leq S \leq \frac{1}{2}$ and

$$R = \frac{S}{1 - S}.$$

For $0 \leq s \leq \frac{1}{2}$, the event $S < s$ happens exactly when $U < s$ or $U > 1 - s$. These are two disjoint intervals of total length $2s$, so $\mathrm{P}(S < s) = 2s$. In other words, S is uniform on $[0, \frac{1}{2}]$.



Now let $0 \le r \le 1$. The function $s \mapsto \dfrac{s}{1-s}$ is strictly increasing on $[0, \frac{1}{2}]$ (its derivative is $1/(1-s)^2 > 0$), so we may invert the inequality:

$$R < r \iff \frac{S}{1-S} < r \iff S < r(1-S) \iff S < \frac{r}{1+r},$$

where multiplying through by $1-S$ is safe because $1-S \ge \frac{1}{2} > 0$. The value $\dfrac{r}{1+r}$ lies in $[0, \frac{1}{2}]$ when $0 \le r \le 1$, so

$$\boxed{\mathrm{P}(R < r) = \frac{2r}{1+r} \qquad (0 \le r \le 1).}$$

Since the shorter piece is never longer than the longer one, R always lies in $[0,1]$. So $\mathrm{P}(R < r) = 0$ for $r \le 0$ and $\mathrm{P}(R < r) = 1$ for $r \ge 1$, which matches the formula's value 1 at $r = 1$.

Ask yourself Why not work with U directly, as $R = U/(1-U)$?

That's only right when $U \le \frac{1}{2}$. When $U > \frac{1}{2}$ the shorter piece is $1-U$, and $R = (1-U)/U$. You can split into these two cases, and each one contributes $\frac{r}{1+r}$, by symmetry. Passing to S does that bookkeeping once and for all.

The density, mean and variance

Differentiating the distribution function $F(r) = \dfrac{2r}{1+r} = 2 - \dfrac{2}{1+r}$ on $(0,1)$ gives the probability density function

$$f(r) = \frac{2}{(1+r)^2} \quad (0 \le r \le 1), \qquad f(r) = 0 \text{ otherwise.}$$

As a check, $\int_0^1 f(r)\,\mathrm{d}r = F(1) - F(0) = 1$.

For the mean, write $r = (1+r) - 1$:

$$\begin{aligned}\mathrm{E}(R) &= \int_0^1 \frac{2r}{(1+r)^2}\,\mathrm{d}r = 2\int_0^1 \left(\frac{1}{1+r} - \frac{1}{(1+r)^2}\right)\mathrm{d}r \\ &= 2\Big[\ln(1+r) + \frac{1}{1+r}\Big]_0^1 = 2\left(\ln 2 + \tfrac{1}{2} - 1\right) \\ &= 2\ln 2 - 1 \approx 0.386.\end{aligned}$$

For the second moment, write $r^2 = (1+r)^2 - 2(1+r) + 1$, so that

$$\frac{r^2}{(1+r)^2} = 1 - \frac{2}{1+r} + \frac{1}{(1+r)^2}.$$

Then

$$\begin{aligned}\mathrm{E}(R^2) &= 2\int_0^1 \left(1 - \frac{2}{1+r} + \frac{1}{(1+r)^2}\right)\mathrm{d}r = 2\Big[r - 2\ln(1+r) - \frac{1}{1+r}\Big]_0^1 \\ &= 2\left(1 - 2\ln 2 - \tfrac{1}{2} + 1\right) = 3 - 4\ln 2.\end{aligned}$$

Hence

$$\begin{aligned}\mathrm{Var}(R) &= \mathrm{E}(R^2) - \big(\mathrm{E}(R)\big)^2 = 3 - 4\ln 2 - \big(4(\ln 2)^2 - 4\ln 2 + 1\big)\\ &= 2 - 4(\ln 2)^2 \approx 0.0782.\end{aligned}$$

The $\ln 2$ terms cancel, which leaves this tidy form.

Checks

- At $r = 1$ the probability is 1, as it must be. At $r = \frac{1}{2}$ it is $\frac{2}{3}$: the longer piece is more than twice the shorter exactly when the break lands in the outer thirds of the stick, which have total length $\frac{2}{3}$.
- The mean 0.386 lies between 0 and 1, and below $\frac{1}{2}$, which fits a density that decreases from 2 at $r = 0$ to $\frac{1}{2}$ at $r = 1$.
- A simulation of 10^6 random breaks gave a sample mean of 0.3862 and a sample variance of 0.0783, against $2\ln 2 - 1 = 0.38629$ and $2 - 4(\ln 2)^2 = 0.07819$.

Common slips

- Taking $R = U/(1-U)$ for all U. That quantity can exceed 1, so it isn't the ratio of shorter to longer.
- Getting $\mathrm{P}(R < r) = \frac{r}{1+r}$ by forgetting that the break can be near either end.
- Using the mean of the shorter piece, $\frac{1}{4}$, to get $\mathrm{E}(R) = \frac{1/4}{3/4} = \frac{1}{3}$. In general $\mathrm{E}\big(g(S)\big) \neq g\big(\mathrm{E}(S)\big)$.

Beyond STEP

The method here is the standard "distribution function technique" for transforming a random variable. If $Y = g(X)$ with g increasing, then $F_Y(y) = F_X\big(g^{-1}(y)\big)$, and differentiating gives the change-of-variables formula $f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\mathrm{d}g^{-1}/\mathrm{d}y\right|$. The same idea leads on to order statistics and to the classic broken-stick problems, such as the chance that three pieces form a triangle. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read about transforming continuous random variables.

WORKED SOLUTION

STEP II 1999, Question 14

Probability and Statistics • Optimal Stopping

STEP II 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q14

The key idea

The die has no memory. Whenever you decide to throw again, you are back at the very start of the same game, so the best you can hope for from that point on is the same number V, the best expected score of the whole game. That turns the decision after each throw into a simple comparison: keep the number you have if it beats V, otherwise throw again. Writing this down gives one equation for V, and solving it tells you both the value and the strategy.

Solution

Setting up

Call a throw that shows 1 *forced*: the game ends there with score 1. After any other throw $x \in \{2,3,4,5,6\}$ we choose to stop (score x) or continue. Every score lies between 1 and 6, so every strategy has an expected score in $[1,6]$. Let

$$V = \text{the largest expected score that any strategy can achieve}$$

(strictly, the least upper bound over all strategies; we'll see that it is attained).

A strategy that scores 4

Consider the strategy: *stop as soon as you throw* 4, 5 *or* 6 (and, of course, when you throw 1). Let E be its expected score. Condition on the first throw. With probability $\frac{1}{6}$ each, it shows $1,4,5,6$ and the game ends with that score. With probability $\frac{2}{6}$ it shows 2 or 3; we throw again and, because the die has no memory, the expected score from there is E again. So

$$E = \tfrac{1}{6}(1+4+5+6) + \tfrac{2}{6}E \quad\Longrightarrow\quad \tfrac{4}{6}E = \tfrac{16}{6} \quad\Longrightarrow\quad E = 4.$$

(The game ends with probability 1, since the chance of n throws without stopping is $(\frac{1}{3})^n \to 0$.) Hence $V \geq 4$.

No strategy can beat 4

Take any strategy at all; it may even depend on the whole history of throws. Look at its first throw x.

- If $x = 1$ the score is 1.
- If $x \geq 2$ and the strategy stops, the score is x.
- If $x \geq 2$ and the strategy continues, then what it does from now on is just some strategy for a fresh copy of the game (the earlier throws can't affect later ones). So its expected score from here is at most V.

In every case the expected score given $x \geq 2$ is at most $\max(x, V)$. Hence every strategy's expected score is at most

$$g(V) = \tfrac{1}{6}\Big(1 + \sum_{x=2}^{6} \max(x, V)\Big),$$

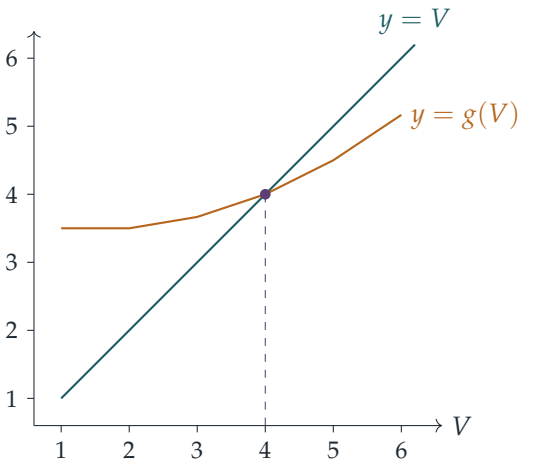
and taking the least upper bound over strategies gives $V \leq g(V)$.

? **Ask yourself** How does the inequality $V \leq g(V)$ pin V down?

Look at $h(V) = g(V) - V$. Each term $\max(x, V)$ rises at rate 0 or 1 as V increases, and there are only five of them, so g rises at rate at most $\frac{5}{6}$. Hence h is strictly decreasing. At $V = 4$,

$$g(4) = \tfrac{1}{6}(1 + 4 + 4 + 4 + 4 + 5 + 6) = \tfrac{24}{6} = 4,$$

so $h(4) = 0$, and $h(V) < 0$ for every $V > 4$. Since $h(V) \geq 0$, we must have $V \leq 4$.



Conclusion

Combining the two parts, $V = 4$: the best possible expected score is 4, and it is achieved by stopping on 4, 5 or 6 and throwing again on 2 or 3. This is exactly the rule "stop if

your throw is at least V": a 2 or 3 is worse than the 4 you expect by carrying on, while a 5 or 6 is better.

A 4 is a tie. Stopping on 4 gives 4, and continuing also gives expected score 4. So "stop only on 5 or 6" is equally good: by the same conditioning,

$$E' = \tfrac{1}{6}(1+5+6) + \tfrac{3}{6}E' \quad \Longrightarrow \quad E' = 4.$$

Any other rule does strictly worse; for instance stopping on 3 or more gives $\frac{1+3+4+5+6}{5} = \frac{19}{5}$, and stopping only on 6 gives $\frac{1+6}{2} = \frac{7}{2}$.

Checks

A neat shortcut for any rule "stop on 1 and on the faces in a set S": each throw ends the game with the same probability, and the face it ends on is equally likely to be any of $\{1\} \cup S$, so the expected score is the mean of those faces. For $S = \{4,5,6\}$ that is $\frac{16}{4} = 4$; for $S = \{5,6\}$ it is $\frac{12}{3} = 4$. We checked all 32 such rules exactly: only these two give 4, and the next best is $\frac{19}{5}$. A simulation of 400 000 games stopping on 4 or more gave an average of 3.998.

Common slips

- Using 3.5, the mean of one throw, as the value of continuing. Continuing is worth more than 3.5 because you keep the option to throw again, so "stop on 4 or more" isn't justified by comparing with 3.5.
- Forgetting that a 1 is forced, and so writing the continuation value as if you could throw forever without risk.
- Checking only threshold strategies without explaining why a cleverer, history-dependent rule can't do better. The memorylessness argument above handles every strategy.

Beyond STEP

This is a small example of *optimal stopping*, and the equation $V = g(V)$ is a *Bellman equation*, the basic tool of dynamic programming and Markov decision processes. The same reasoning prices American options in finance and solves the famous secretary problem. Grimmett and Welsh, *Probability: An Introduction*, covers the conditioning arguments used here, and Blitzstein and Hwang, *Introduction to Probability*, has many problems in the same spirit.

WORKED SOLUTION

STEP III 1999, Question 1

Pure • Roots of Cubic Equations

STEP III 1999, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q1

The key idea

Everything here comes from Vieta's formulas: for the cubic $x^3 - px^2 + qx - r = 0$ with roots α, β, γ,

$$\alpha + \beta + \gamma = p, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = q, \qquad \alpha\beta\gamma = r.$$

A geometric progression is centred on its middle term *multiplicatively*, so the product r and the ratio q/p pick out the middle root. An arithmetic progression is centred *additively*, so the sum p picks out the middle root instead.

Solution

Part (i)

Suppose the roots are a/k, a and ak. For a/k to make sense we need $k \neq 0$, and since the product of the roots is $r \neq 0$, we also have $a \neq 0$. Vieta's formulas give

$$\begin{aligned} p &= \frac{a}{k} + a + ak = a\left(\frac{1}{k} + 1 + k\right), \\ q &= \frac{a}{k} \cdot a + a \cdot ak + \frac{a}{k} \cdot ak = a^2\left(\frac{1}{k} + 1 + k\right), \\ r &= \frac{a}{k} \cdot a \cdot ak = a^3. \end{aligned}$$

The same bracket appears in p and q. It is nonzero because $p \neq 0$, so we can divide:

$$\frac{q}{p} = a.$$

So q/p is a root (the middle one). Then $q = ap$, and so

$$q^3 = a^3p^3 = rp^3, \qquad \text{that is,} \qquad q^3 - rp^3 = 0,$$

as required.

Part (ii)

Now suppose $r = q^3/p^3$, and write $a = q/p$, so that $r = a^3$. Substituting $x = a$ into the cubic, and using $p\,a^2 = q^2/p = q\,a$,

$$a^3 - pa^2 + qa - r = a^3 - qa + qa - a^3 = 0,$$

so q/p is a root.

Since $r \neq 0$, we have $q \neq 0$, so $a \neq 0$. Call the other two roots β and γ. The product of all three roots is r, so

$$a\beta\gamma = r = a^3 \quad \Longrightarrow \quad \beta\gamma = a^2 = \left(\frac{q}{p}\right)^2,$$

dividing by $a \neq 0$.

? **Ask yourself** How do we turn $\beta\gamma = a^2$ into the form a/k, a, ak?

We need to choose k. Since $\beta\gamma = a^2 \neq 0$, neither β nor γ is zero. Define $k = \gamma/a$, so that $\gamma = ak$ and $k \neq 0$. Then

$$\beta = \frac{a^2}{\gamma} = \frac{a^2}{ak} = \frac{a}{k}.$$

So the roots are a/k, a and ak, each obtained from the one before by multiplying by k. They are in geometric progression. (The ratio k may be complex when β and γ are not real. The argument does not need them to be real.)

Together, parts (i) and (ii) show that, when $p \neq 0$ and $r \neq 0$, the roots are in geometric progression if and only if $q^3 = rp^3$.

Part (iii)

Necessity. Suppose the roots are $b - d$, b and $b + d$. Their sum is $3b = p$, so the middle root is $b = p/3$. Substituting $x = p/3$ into the cubic and multiplying by 27:

$$p^3 - 3p^3 + 9pq - 27r = 0 \quad \Longleftrightarrow \quad 2p^3 - 9pq + 27r = 0.$$

Sufficiency. Conversely, suppose $2p^3 - 9pq + 27r = 0$. Reversing the step above, $p/3$ is a root. The other two roots then have sum $p - p/3 = 2p/3$, so we can write them as $p/3 - d$ and $p/3 + d$ for some (possibly complex) d. The three roots $p/3 - d$, $p/3$, $p/3 + d$ are in arithmetic progression.

Hence a necessary and sufficient condition is

$$\boxed{2p^3 - 9pq + 27r = 0}.$$

Checks

- Roots $1, 2, 4$ (a geometric progression) give $p = 7$, $q = 14$, $r = 8$. Then $q/p = 2$ is the middle root, and $q^3 = 2744 = 8 \cdot 343 = rp^3$.

- Roots $1, 2, 3$ (an arithmetic progression) give $p = 6$, $q = 11$, $r = 6$, and $2 \cdot 216 - 9 \cdot 66 + 27 \cdot 6 = 432 - 594 + 162 = 0$. The geometric progression $1, 2, 4$ gives $686 - 882 + 216 = 20 \neq 0$, as it should.
- We also tested all three parts numerically on thousands of random cubics, including ones with complex roots. All agreed to within 10^{-11}.

Common slips

- Dividing by $1/k + 1 + k$ without saying why it is nonzero. It is nonzero because it is p/a and $p \neq 0$.
- In part (ii), showing only that q/p is a root and that $\beta\gamma = (q/p)^2$, and then asserting a geometric progression without exhibiting the common ratio.
- In part (iii), proving only one direction. "Necessary and sufficient" needs both.

Beyond STEP

Vieta's formulas say that the coefficients of a polynomial are the *elementary symmetric functions* of its roots. A classical theorem says that every symmetric polynomial in the roots can be written in terms of these. This is the starting point of Galois theory, which explains when the roots of a polynomial can be found by radicals. Incidentally, the substitution $x = y + p/3$ removes the x^2 term from the cubic, and the constant term of the resulting depressed cubic is a multiple of $2p^3 - 9pq + 27r$. Allenby's *Rings, Fields and Groups* is a good place to see where these ideas lead.

WORKED SOLUTION

STEP III 1999, Question 2

Pure • Counting Real Roots

STEP III 1999, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q2

The key idea

In part (i) the derivative factorises as a perfect square times e^x, so the function never decreases, and it is in fact strictly increasing. In part (ii) the derivative of the new function vanishes exactly when $(1+x^2)e^x = k$, which is the equation from part (i). That tells us the new function has a single turning point, a minimum. After that, everything depends on what happens as $x \to -\infty$, where $\tan^{-1} x \to -\frac{\pi}{2}$. This is where the number $2/\pi$ comes from.

Solution

Part (i)

By the product rule,

$$f'(x) = 2x\,e^x + (1+x^2)e^x = e^x\left(x^2+2x+1\right) = e^x(1+x)^2.$$

Since $e^x > 0$ and $(1+x)^2 \geq 0$, we have $f'(x) \geq 0$ for all x, as required. Equality holds only at $x = -1$.

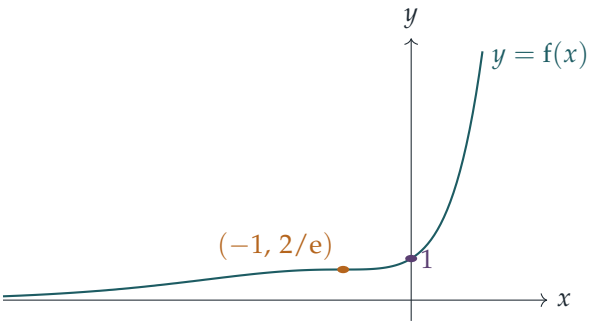
Ask yourself $f'(-1) = 0$, so is f really *strictly* increasing?

Yes. If $a < b$, then $f(b) - f(a) = \int_a^b f'(x)\,dx$, and the integrand is continuous, non-negative and zero at no more than one point of $[a, b]$, so the integral is strictly positive. A single stationary point can't stop a function from increasing. Here $x = -1$ is a stationary point of inflection: f' is positive on both sides of it.

For the sketch, we need:

- $f(x) > 0$ for all x, since both factors are positive;
- $f(0) = 1$, and the point of inflection is $(-1, 2/e) \approx (-1, 0.74)$;
- as $x \to \infty$, $f(x) \to \infty$;

- as $x \to -\infty$, $\mathrm{f}(x) \to 0$, because e^x decays faster than any polynomial grows ($x^2\mathrm{e}^x \to 0$). So the negative x-axis is an asymptote.



Now f is continuous and strictly increasing, with $\mathrm{f}(x) \to 0$ as $x \to -\infty$ and $\mathrm{f}(x) \to \infty$ as $x \to \infty$. So it takes every value in $(0, \infty)$ (intermediate value theorem), and it takes each one exactly once (strictly increasing). It never takes a value ≤ 0, since $\mathrm{f} > 0$. Therefore $(1 + x^2)\mathrm{e}^x = k$ has exactly one real root when $k > 0$ and none when $k \leq 0$. ■

Part (ii)

Let

$$\mathrm{g}(x) = \mathrm{e}^x - 1 - k\tan^{-1} x.$$

Straight away, $\mathrm{g}(0) = 1 - 1 - 0 = 0$, so $x = 0$ is always a root. Next,

$$\mathrm{g}'(x) = \mathrm{e}^x - \frac{k}{1+x^2} = \frac{(1+x^2)\mathrm{e}^x - k}{1+x^2} = \frac{\mathrm{f}(x) - k}{1+x^2}.$$

In both cases $k > 0$, so by part (i) there is exactly one number a with $\mathrm{f}(a) = k$. Because f is strictly increasing, $\mathrm{f}(x) < k$ for $x < a$ and $\mathrm{f}(x) > k$ for $x > a$. Hence

$$\mathrm{g}'(x) < 0 \text{ for } x < a, \qquad \mathrm{g}'(x) > 0 \text{ for } x > a.$$

So g is strictly decreasing on $(-\infty, a]$ and strictly increasing on $[a, \infty)$, with a single minimum at $x = a$. In particular g has at most one root on each side of a, so at most two roots in total.

Where is a? In both cases $k < 1 = \mathrm{f}(0)$, so $\mathrm{f}(a) = k < \mathrm{f}(0)$, which gives $a < 0$. Then $x = 0$ lies in the increasing stretch $[a, \infty)$. It is the only root there, and

$$\mathrm{g}(a) < \mathrm{g}(0) = 0.$$

What remains is to decide whether g also has a root in $(-\infty, a)$.

Behaviour as $x \to -\infty$. Since $\mathrm{e}^x \to 0$ and $\tan^{-1} x \to -\frac{\pi}{2}$,

$$\mathrm{g}(x) \to -1 + \frac{k\pi}{2} = \frac{\pi}{2}\left(k - \frac{2}{\pi}\right) \quad \text{as } x \to -\infty.$$

Call this limit L. Because g is strictly decreasing on $(-\infty, a]$, every value of g there is strictly less than L. To see this, take $x \leq a$ and any $x' < x$. Every $y < x'$ has $g(y) > g(x')$, so letting $y \to -\infty$ gives $L \geq g(x')$. Hence $g(x) < g(x') \leq L$.

(a) $0 < k \leq 2/\pi$**.** Here $L \leq 0$, so $g(x) < L \leq 0$ for every $x \leq a$. There is no root to the left of a, and the equation has **exactly one real root**, namely $x = 0$.

(b) $2/\pi < k < 1$**.** Here $L > 0$, so $g(x) > 0$ for all sufficiently negative x, while $g(a) < 0$. By the intermediate value theorem there is a root in $(-\infty, a)$, and it is the only one there because g is strictly decreasing. Together with $x = 0$, the equation has **exactly two real roots**: one negative root (less than a) and $x = 0$.

? **Ask yourself** Why does the boundary case $k = 2/\pi$ belong with (a)?

When $k = 2/\pi$, the left-hand limit is exactly $L = 0$. The decreasing branch approaches 0 from below but never reaches it, so no second root appears. Strict monotonicity is what settles this.

Checks

- Differentiating f numerically at random points agrees with $e^x(1+x)^2$, and $f(-1) = 2/e \approx 0.736$.
- We counted sign changes of g on a fine grid over $[-200, 10]$. There is one root for $k = 0.05, 0.3, 0.5, 0.6$ and $2/\pi \approx 0.6366$, and two roots for $k = 0.65, 0.7, 0.8, 0.9, 0.99$. For $k = 0.65$ the negative root is far out (near $x \approx -31$), because $L = 0.65 \cdot \frac{\pi}{2} - 1 \approx 0.021$ is so small.
- The minimum point is $a \approx -1.81$ for $k = 0.7$, and $a \to 0^-$ as $k \to 1^-$. This fits: at $k = 1$, $g'(0) = 0$ and $x = 0$ becomes a double root.

Common slips

- Forgetting that $x = 0$ is always a root, and so counting only the roots that come from the sign change at $-\infty$.
- Taking $\tan^{-1} x \to -\infty$ as $x \to -\infty$. It is bounded, tending to $-\frac{\pi}{2}$.
- Concluding from $f'(-1) = 0$ that f has a turning point at $x = -1$.

Beyond STEP

The argument here is a model of a standard real-analysis technique: a continuous function that is strictly monotonic on an interval has at most one zero there, and the intermediate value theorem (applied with limits at infinity) tells us when it has

one. Counting roots by locating the turning points of a function, and then checking signs at the turning points and at the ends, is the idea behind Rolle's theorem and its consequences. Abbott's *Understanding Analysis* develops continuity, the intermediate value theorem and the mean value theorem carefully.

WORKED SOLUTION

STEP III 1999, Question 3

Pure • Riemann Sums and Limits

STEP III 1999, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q3

The key idea

The expression $\frac{1}{n}\sum f(1+m/n)$ is the total area of n thin rectangles of width $\frac{1}{n}$ standing on $[1,2]$, and as the rectangles get thinner they fill out the area under the curve exactly. The other two limits are this same picture in disguise: the job is to pull out a factor $\frac{1}{n}$ and write each term as a function of m/n alone.

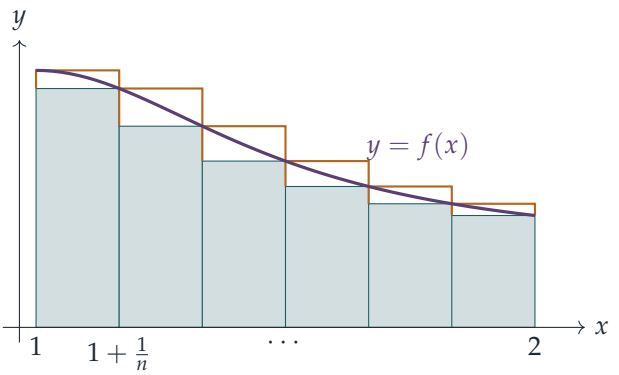
Solution

The limit formula

Split $[1,2]$ into n strips of width $\frac{1}{n}$; the mth strip is $\left[1+\frac{m-1}{n},\, 1+\frac{m}{n}\right]$. On it, stand a rectangle whose height is the value of f at the strip's right-hand end, $f(1+m/n)$. Its area is $\frac{1}{n}f(1+m/n)$, so

$$S_n = \frac{1}{n}\sum_{m=1}^{n} f(1+m/n)$$

is the total area of the rectangles.



The sketch shows a decreasing f. The shaded (teal) rectangles use right-hand heights

and so lie *under* the curve; the amber steps on top show the rectangles we would get using left-hand heights, which lie *over* it. Hence

$$S_n \;\le\; \int_1^2 f(x)\,\mathrm{d}x \;\le\; S_n + (\text{total area of the amber pieces}).$$

Now slide all the amber pieces sideways into a single column of width $\frac{1}{n}$: they stack up to a total height of $f(1) - f(2)$, because the heights telescope. So

$$0 \;\le\; \int_1^2 f(x)\,\mathrm{d}x - S_n \;\le\; \frac{f(1)-f(2)}{n} \;\longrightarrow\; 0 \quad \text{as } n \to \infty,$$

and $S_n \to \int_1^2 f(x)\,\mathrm{d}x$. An increasing f works the same way with the inequalities reversed, and any reasonable (say continuous, piecewise monotonic) function can be cut into finitely many monotonic pieces, each contributing an error of order $\frac{1}{n}$. This justifies

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} f(1 + m/n) = \int_1^2 f(x)\,\mathrm{d}x.$$

The limit $\ln 2$

Ask yourself How do we make $\dfrac{1}{n+m}$ look like $\dfrac{1}{n} f(1+m/n)$?

Take out a factor n from the denominator:

$$\frac{1}{n+m} = \frac{1}{n} \cdot \frac{1}{1+m/n} = \frac{1}{n} f(1+m/n) \qquad \text{with } f(x) = \frac{1}{x}.$$

So

$$\begin{aligned} \lim_{n\to\infty} \left(\frac{1}{n+1} + \cdots + \frac{1}{n+n} \right) &= \lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} \frac{1}{1+m/n} \\ &= \int_1^2 \frac{1}{x}\,\mathrm{d}x = \ln 2 - \ln 1 = \ln 2, \end{aligned}$$

as required. (Here f is decreasing, so the sketch argument applies exactly; it even tells us the sum lies between $\ln 2 - \frac{1}{2n}$ and $\ln 2$.)

The final limit

The mth term is $\dfrac{n}{n^2+m^2}$, for $m = 1, \ldots, n$. Divide top and bottom by n^2:

$$\frac{n}{n^2+m^2} = \frac{1}{n} \cdot \frac{1}{1+(m/n)^2}.$$

To match the formula, which evaluates f at $x = 1 + m/n$, we need $f(x) = \dfrac{1}{1+(x-1)^2}$, since then $f(1+m/n) = \dfrac{1}{1+(m/n)^2}$. This f is decreasing on $[1,2]$. Therefore

$$\lim_{n\to\infty} \sum_{m=1}^{n} \frac{n}{n^2+m^2} = \int_1^2 \frac{\mathrm{d}x}{1+(x-1)^2} = \Big[\tan^{-1}(x-1)\Big]_1^2$$
$$= \tan^{-1} 1 - \tan^{-1} 0 = \frac{\pi}{4}.$$

(Equivalently, substitute $u = x - 1$ to get $\int_0^1 \frac{\mathrm{d}u}{1+u^2}$.)

Checks

The sums can be computed directly. With $n = 1000$, the second sum is 0.69290 against $\ln 2 \approx 0.69315$, and the third is 0.78515 against $\pi/4 \approx 0.78540$. In both cases the shortfall is about $\frac{1}{4000}$, inside the bound $\frac{f(1)-f(2)}{n} = \frac{1}{2000}$ from the sketch. Both sums also come in slightly *below* the limit, as they should for a decreasing f with right-hand heights.

Common slips

- Writing the third limit as $\int_1^2 \frac{\mathrm{d}x}{1+x^2}$: the formula evaluates f at $1 + m/n$, not at m/n, so the shift by 1 must go into f.
- Forgetting that the last term ($m = n$) is $\frac{n}{2n^2}$, which matches $\frac{1}{n}f(2)$: the sum runs over all n right-hand endpoints, not $n + 1$ points.

Beyond STEP

These are Riemann sums, and the sketch argument is the seed of the definition of the Riemann integral: a function is integrable when the upper and lower sums can be squeezed together, and every continuous function on a closed interval has this property. The same idea also runs backwards, turning sums into integrals to estimate them, as in the proof that $1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n$ converges. Abbott's *Understanding Analysis* has a very readable chapter on the Riemann integral.

WORKED SOLUTION

STEP III 1999, Question 4

Pure • Euler's Formula and Polyhedra

STEP III 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 7.9, *Only Five Platonic Solids*: https://beyondstep.keithmcnulty.org/ch07/platonic-solids.html

The key idea

Count the edges in two different ways: once by going round the faces, and once by going round the vertices. Each count expresses E in terms of F (or V), and substituting into Euler's formula leaves a single equation for F. After that the whole classification comes down to one observation: F is positive, so h must be positive, and this leaves only a handful of whole-number pairs (m, n).

Solution

Deriving the formula for F

Each face is an m-gon, so it has m edges. Going round every face and counting its edges gives mF. But each edge is the boundary between exactly two faces, so this counts every edge twice:

$$mF = 2E.$$

In the same way, n edges meet at each vertex, so counting edges vertex by vertex gives nV. Each edge has exactly two endpoints, so this also counts every edge twice:

$$nV = 2E.$$

So $E = \frac{1}{2}mF$ and $V = \frac{2E}{n} = \frac{mF}{n}$. Substituting into $V - E + F = 2$ gives

$$\frac{mF}{n} - \frac{mF}{2} + F = 2$$

$$F\,(2m - mn + 2n) = 4n \qquad \text{(multiplying by } 2n\text{)}.$$

Now expand the given h:

$$4 - (n-2)(m-2) = 4 - (mn - 2m - 2n + 4) = 2m + 2n - mn,$$

which is exactly the bracket. So $Fh = 4n$. Here F and n are positive, so $h \neq 0$, and we can divide:

$$F = \frac{4n}{h}, \qquad h = 4 - (n-2)(m-2),$$

as required.

Only five regular polyhedra

Ask yourself What restrictions on m and n come from geometry rather than from the formula?

A polygon has at least three sides, so $m \geq 3$. At a vertex of a solid at least three faces must meet: two faces meeting along a corner would give a flat flap, not a corner of a solid. So $n \geq 3$ too. Then $m - 2 \geq 1$ and $n - 2 \geq 1$.

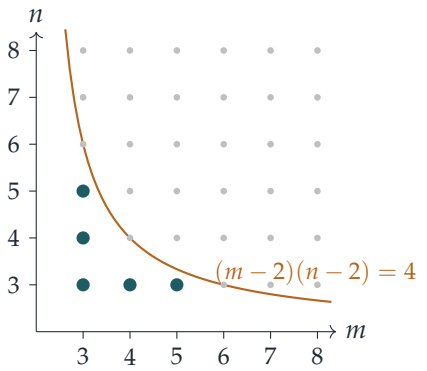
Since $F > 0$ and $4n > 0$, the formula forces $h > 0$, that is,

$$(m-2)(n-2) < 4.$$

The product of the two positive integers $m - 2$ and $n - 2$ is therefore 1, 2 or 3, so $h = 3$, 2 or 1. The only factorisations are

- $(m-2)(n-2) = 1$: $m = n = 3$, so $h = 3$;
- $(m-2)(n-2) = 2$: $(m, n) = (3, 4)$ or $(4, 3)$, so $h = 2$;
- $(m-2)(n-2) = 3$: $(m, n) = (3, 5)$ or $(5, 3)$, so $h = 1$.

The picture shows the same thing: the lattice points with $m, n \geq 3$ lying strictly inside the hyperbola $(m-2)(n-2) = 4$.



For each pair, $F = 4n/h$, $E = \frac{1}{2}mF$ and $V = mF/n$:

	m	n	h	V	E	F
tetrahedron	3	3	3	4	6	4
octahedron	3	4	2	6	12	8
cube	4	3	2	8	12	6
icosahedron	3	5	1	12	30	20
dodecahedron	5	3	1	20	30	12

For example, for the icosahedron $F = 4 \cdot 5/1 = 20$, $E = \frac{1}{2} \cdot 3 \cdot 20 = 30$ and $V = 3 \cdot 20/5 = 12$.

So at most five combinations of (m, n) are possible. Each one really does occur, as the five familiar Platonic solids named in the table show. So there are exactly five regular polyhedra.

Checks

- Every row satisfies $V - E + F = 2$. For the dodecahedron, $20 - 30 + 12 = 2$.
- The cube gives $V = 8$, $E = 12$, $F = 6$, which you can count on a die.
- Swapping m and n swaps V and F and keeps E fixed (cube and octahedron, dodecahedron and icosahedron). This is duality: put a vertex at the centre of each face.
- An independent check: at each vertex the angle deficit $2\pi - n\pi(m-2)/m$ must be positive, and it totals 4π over all V vertices (Descartes' theorem). This holds for all five, and a computer search over $3 \leq m, n < 200$ found no other pairs with $h > 0$.

Common slips

- Forgetting the factor of 2 when counting edges: each edge belongs to two faces and has two ends.
- Not saying why $n \geq 3$. Without it, $n = 2$ would give $h = 4$ for every m, and infinitely many "solutions".
- Concluding "exactly five" from the counting alone. The argument shows there are *at most* five; you need the actual solids to show that each case exists.

Beyond STEP

Euler's formula is the start of topology. The number $V - E + F$ (the *Euler characteristic*) is the same for every way of dividing a surface into faces, so it is 2 for a sphere, 0 for a torus, and so on. Repeating this argument on a torus gives infinitely many "regular" tilings, such as the square grid. The symmetry groups of the five solids are also central examples in a first group theory course. For a lively account, read Richeson, *Euler's Gem*; Armstrong, *Basic Topology*, takes the ideas further.

WORKED SOLUTION

STEP III 1999, Question 5

Pure • Fibonacci Identities and Induction

STEP III 1999, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q5

The key idea

This is the Fibonacci sequence, shifted so that $u_n = F_{n+1}$. For the first identity, write both u_{n+2} and u_{n-1} in terms of the middle pair u_{n+1} and u_n: the cross terms then cancel. For the doubling formulae, prove the two statements *together* by induction, because each one feeds the other through the recurrence. The first identity is exactly the tool needed to finish the odd case.

Solution

The first identity

Fix $n \geq 1$, so that u_{n-1}, u_n, u_{n+1}, u_{n+2} are all defined. The recurrence gives

$$u_{n+2} = u_{n+1} + u_n \qquad \text{and} \qquad u_{n-1} = u_{n+1} - u_n,$$

the second coming from $u_{n+1} = u_n + u_{n-1}$. Squaring and adding,

$$\begin{aligned} u_{n+2}^2 + u_{n-1}^2 &= \left(u_{n+1}^2 + 2u_{n+1}u_n + u_n^2\right) + \left(u_{n+1}^2 - 2u_{n+1}u_n + u_n^2\right) \\ &= 2\left(u_{n+1}^2 + u_n^2\right), \end{aligned}$$

which is the required identity.

The doubling formulae

Write $P(n)$ for the statement that *both*

$$u_{2n} = u_n^2 + u_{n-1}^2 \qquad \text{and} \qquad u_{2n+1} = u_{n+1}^2 - u_{n-1}^2.$$

The first terms are $u_0, u_1, \cdots = 1, 1, 2, 3, 5, 8, \ldots$.

Base case. For $n = 1$: $u_2 = 2 = 1^2 + 1^2 = u_1^2 + u_0^2$, and $u_3 = 3 = 2^2 - 1^2 = u_2^2 - u_0^2$. So $P(1)$ holds.

Ask yourself Why prove the two formulae together rather than one at a time?

To get u_{2n+2} from the recurrence we need u_{2n+1} and u_{2n}, so the even formula at $n+1$ needs *both* formulae at n. The odd formula at $n+1$ then needs the even one at $n+1$ as well. Neither statement can be pushed forward on its own.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$.

Even term. By the recurrence and then $P(n)$,

$$\begin{aligned} u_{2n+2} &= u_{2n+1} + u_{2n} \\ &= (u_{n+1}^2 - u_{n-1}^2) + (u_n^2 + u_{n-1}^2) \\ &= u_{n+1}^2 + u_n^2, \end{aligned}$$

which is the even formula with n replaced by $n+1$.

Odd term. Using the recurrence, the line just proved and $P(n)$,

$$\begin{aligned} u_{2n+3} &= u_{2n+2} + u_{2n+1} \\ &= (u_{n+1}^2 + u_n^2) + (u_{n+1}^2 - u_{n-1}^2) \\ &= 2u_{n+1}^2 + u_n^2 - u_{n-1}^2. \end{aligned}$$

We want this to equal $u_{n+2}^2 - u_n^2$, and here the first identity comes in. Since $n \geq 1$, it gives $u_{n+2}^2 = 2u_{n+1}^2 + 2u_n^2 - u_{n-1}^2$, so

$$u_{n+2}^2 - u_n^2 = 2u_{n+1}^2 + u_n^2 - u_{n-1}^2 = u_{2n+3}.$$

This is the odd formula with n replaced by $n+1$, since $2(n+1)+1 = 2n+3$ and $(n+1) \pm 1 = n+2,\ n$.

So $P(n)$ implies $P(n+1)$. With the base case, $P(n)$ holds for every positive integer n by induction.

Checks

Try $n = 5$: $u_{10} = 89$ and $u_5^2 + u_4^2 = 64 + 25 = 89$; $u_{11} = 144$ and $u_6^2 - u_4^2 = 169 - 25 = 144$. We checked the first identity for all $n \leq 299$, and both doubling formulae for all $n \leq 189$, in exact integer arithmetic. Note also that $u_{n+1}^2 - u_{n-1}^2 = (u_{n+1} - u_{n-1})(u_{n+1} + u_{n-1}) = u_n(u_{n+1} + u_{n-1})$, a handy alternative form for the odd terms.

Common slips

- Inducting on the even formula alone: the step needs u_{2n+1}, so it stalls.
- Using the first identity at $n = 0$, where u_{-1} is not defined. In the inductive step we only use it for $n \geq 1$.
- Forgetting that the indexing is shifted from the usual Fibonacci numbers ($u_0 = u_1 = 1$), and checking the base case with the wrong values.

Beyond STEP

The doubling formulae let you compute u_N in about $\log_2 N$ steps, by halving N repeatedly: this is the "fast doubling" method, a cousin of fast exponentiation. Behind it sits the matrix identity $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, and squaring this matrix gives both formulae at once. Strang's *Introduction to Linear Algebra* treats Fibonacci numbers through powers and eigenvalues of this matrix, and the chapter on continued fractions in Davenport's *The Higher Arithmetic* is where ratios of consecutive Fibonacci numbers reappear, as the convergents of the golden ratio.

WORKED SOLUTION

STEP III 1999, Question 6

Pure • Parametric Curves and Area

STEP III 1999, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q6

The key idea

The equation says that $\left(x^{1/n}\right)^2 + \left(y^{1/n}\right)^2 = \left(a^{1/n}\right)^2$, which is a circle in disguise. So put $x^{1/n} = a^{1/n}\cos t$ and $y^{1/n} = a^{1/n}\sin t$, exactly as we would for a circle, and then raise to the nth power. The fact that n is odd is what lets us take real nth roots of negative numbers, so the curve lives in all four quadrants.

Solution

The parametrisation

Because n is odd, every real number x has exactly one real nth root $x^{1/n}$, with the same sign as x, and $x^{2/n}$ means $\left(x^{1/n}\right)^2$. Writing $X = x^{1/n}$, $Y = y^{1/n}$ and $R = a^{1/n} > 0$, the equation $(*)$ becomes

$$X^2 + Y^2 = R^2,$$

a circle of radius R, which we parametrise by $X = R\cos t$, $Y = R\sin t$. Raising to the nth power (a bijection of $\mathbb{R}$, as n is odd) gives

$$\boxed{x = a\cos^n t, \qquad y = a\sin^n t, \qquad 0 \le t \le 2\pi.}$$

Every point of the curve arises this way, since every point (X, Y) of the circle does.

Ask yourself Why is this anticlockwise, rather than just "some" traversal of the curve?

The map $(X, Y) \mapsto (X^n, Y^n)$ keeps the sign of each coordinate, so it sends each quadrant to itself. As t runs from 0 to 2π, the point $(R\cos t, R\sin t)$ goes once round the circle anticlockwise, visiting the positive x-axis, the first quadrant, the positive y-axis, the second quadrant, and so on, in that order. Its image (x, y) visits the same axes and quadrants in the same order, so it also goes once round the origin anticlockwise. (Within

the first quadrant, for instance, $x = a\cos^n t$ decreases from a to 0 while $y = a\sin^n t$ increases from 0 to a.) When $n = 1$ this is just the familiar circle $x = a\cos t$, $y = a\sin t$.

Sketch for $n = 3$

Now $x = a\cos^3 t$, $y = a\sin^3 t$, and the curve is $x^{2/3} + y^{2/3} = a^{2/3}$.

Symmetry. The equation is unchanged by $x \mapsto -x$, by $y \mapsto -y$ and by swapping x and y. So the curve is symmetric in both axes and in the lines $y = \pm x$, and we need only study the first quadrant, $0 \le t \le \frac{\pi}{2}$.

Intercepts. At $t = 0$ we are at $(a, 0)$ and at $t = \frac{\pi}{2}$ at $(0, a)$. By symmetry the curve meets the axes only at $(\pm a, 0)$ and $(0, \pm a)$. On the line $y = x$ (at $t = \frac{\pi}{4}$) we are at $\left(\frac{a}{2\sqrt{2}}, \frac{a}{2\sqrt{2}}\right)$, only about $0.35a$ along each axis.

Gradient. We have

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -3a\cos^2 t\sin t, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 3a\sin^2 t\cos t,$$

so for $0 < t < \frac{\pi}{2}$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\tan t.$$

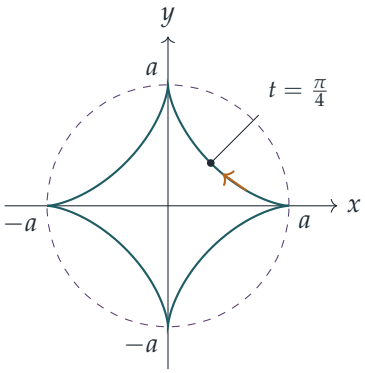
This is 0 as $t \to 0$ and tends to $-\infty$ as $t \to \frac{\pi}{2}$. So at $(a, 0)$ the curve arrives tangent to the x-axis, and at $(0, a)$ it arrives tangent to the y-axis. Both $\mathrm{d}x/\mathrm{d}t$ and $\mathrm{d}y/\mathrm{d}t$ vanish at $t = 0$ and $t = \frac{\pi}{2}$. The arc from the first quadrant meets the reflected arc from the fourth quadrant at $(a, 0)$, and both are tangent to the x-axis there, so the two arcs come together in a sharp point. The curve has *cusps* at the four points $(\pm a, 0)$, $(0, \pm a)$.

Concavity. Differentiating $-\tan t$ with respect to x,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{-\sec^2 t}{-3a\cos^2 t\sin t} = \frac{1}{3a\cos^4 t\sin t} > 0 \quad \text{for } 0 < t < \frac{\pi}{2}.$$

So the first-quadrant arc is convex: it bends in towards the origin, not outwards like a circle. It lies below its chord $x + y = a$. Indeed $\cos^3 t + \sin^3 t \le \cos^2 t + \sin^2 t = 1$, with equality only at the cusps.

Putting this together gives a four-pointed star (the *astroid*). It is shown below with the circle of radius a dashed for comparison.



The area for $n = 3$

Using the derivatives above,

$$\begin{aligned} x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t} &= a\cos^3 t \cdot 3a\sin^2 t\cos t + a\sin^3 t \cdot 3a\cos^2 t\sin t \\ &= 3a^2\sin^2 t\cos^2 t\,(\cos^2 t + \sin^2 t) \\ &= 3a^2\sin^2 t\cos^2 t = \tfrac{3}{4}a^2\sin^2 2t. \end{aligned}$$

Since $\sin^2 2t = \frac{1}{2}(1 - \cos 4t)$, and $\cos 4t$ integrates to zero over $[0, 2\pi]$,

$$\int_0^{2\pi} \sin^2 2t\,\mathrm{d}t = \pi.$$

Therefore

$$A = \frac{1}{2}\cdot\frac{3a^2}{4}\cdot\pi = \boxed{\frac{3\pi a^2}{8}}.$$

The formula gives a positive area because the curve is traversed anticlockwise, which is why the direction mattered in the first part.

Checks

- The astroid lies inside the diamond $|x| + |y| \le a$, which has area $2a^2$, so the answer must be less than $2a^2$. Indeed $\frac{3\pi}{8}a^2 \approx 1.178a^2$.
- As a check that doesn't use the formula, $4\int_0^a \left(a^{2/3} - x^{2/3}\right)^{3/2}\mathrm{d}x$ with $a = 1$, evaluated numerically, gives $1.178097\ldots$, matching $3\pi/8 = 1.1780972\ldots$.
- With $n = 1$ the same method gives $x\dot{y} - y\dot{x} = a^2$ and $A = \pi a^2$, the circle, as it should.

Common slips

- Writing $x = a\cos^n t$ without explaining why negative x is allowed. The oddness of n is exactly what makes $x^{2/n}$ meaningful (and the parametrisation complete) for $x < 0$.
- Drawing the curve bulging outwards like a squashed circle. The second derivative shows it curves inwards, with sharp cusps on the axes.
- Integrating only over $[0, \frac{\pi}{2}]$ and forgetting to multiply by 4, or making a sign error in $-y\,\mathrm{d}x/\mathrm{d}t$ (both terms end up positive).

Beyond STEP

The area formula is a special case of *Green's theorem*, which turns an integral over a region into an integral round its boundary. It is the principle behind the planimeter,

a mechanical instrument that measures area by tracing the boundary. Curves like $|x|^p + |y|^p = 1$ also give the "unit circles" of the ℓ^p norms, which you meet in analysis and linear algebra; the astroid is the case $p = \frac{2}{3}$, which is not convex, so it does not give a norm. The astroid also turns up as the path traced by a point on a circle rolling inside a circle four times its radius. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1999, Question 7

Pure • Groups and Isomorphisms

STEP III 1999, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q7

The key idea

The expression $x + y + axy$ is very nearly a product. Multiplying by a and adding 1 gives

$$1 + a(x * y) = 1 + ax + ay + a^2xy = (1 + ax)(1 + ay),$$

so under the change of variable $X = 1 + ax$ the operation $*$ is just ordinary multiplication in disguise. Every group property then comes from the corresponding property of multiplication, and the one number we must throw out is the one that corresponds to $X = 0$.

Solution

Throughout, write $\phi(x) = 1 + ax$. Since $a \neq 0$, ϕ is a bijection from $\mathbb{R}$ to $\mathbb{R}$, so two real numbers u, v are equal exactly when $\phi(u) = \phi(v)$. The identity above reads

$$\phi(x * y) = \phi(x)\,\phi(y) \qquad \text{for all real } x, y. \tag{†}$$

Associativity

Using (†) twice on each side,

$$\phi\big((x * y) * z\big) = \phi(x * y)\,\phi(z) = \phi(x)\,\phi(y)\,\phi(z),$$
$$\phi\big(x * (y * z)\big) = \phi(x)\,\phi(y * z) = \phi(x)\,\phi(y)\,\phi(z).$$

These are equal because multiplication of real numbers is associative, and since ϕ is one-to-one, $(x * y) * z = x * (y * z)$. So $*$ is associative.

(If you prefer a direct check: both $(x * y) * z$ and $x * (y * z)$ expand to the symmetric expression $x + y + z + a(xy + yz + zx) + a^2xyz$.)

The group

Ask yourself Which number has to be removed, and why can't we keep all of $\mathbb{R}$?

Under ϕ, the number $x = -\frac{1}{a}$ corresponds to $\phi(x) = 0$, and 0 has no multiplicative inverse. Concretely, for any real y,

$$\left(-\tfrac{1}{a}\right) * y = -\tfrac{1}{a} + y - y = -\tfrac{1}{a},$$

so $-\frac{1}{a}$ absorbs everything and can never be combined with anything to give the identity. So we take

$$G = \mathbb{R} \setminus \left\{-\tfrac{1}{a}\right\},$$

which is exactly the set of x with $\phi(x) \neq 0$. We check the four axioms.

Closure. If $x, y \in G$ then $\phi(x) \neq 0$ and $\phi(y) \neq 0$, so by $(\dagger)$, $\phi(x * y) = \phi(x)\phi(y) \neq 0$. Hence $x * y \neq -\frac{1}{a}$, and $x * y$ is certainly real, so $x * y \in G$.

Associativity. This was shown above for all real numbers, so in particular for elements of G.

Identity. We have $x * 0 = x + 0 + 0 = x$ and $0 * x = x$ for every x, and $0 \in G$ since $\phi(0) = 1 \neq 0$. So 0 is the identity. (This matches $\phi(0) = 1$, the multiplicative identity.)

Inverses. Given $x \in G$, we want y with $x * y = 0$, that is $x + y(1 + ax) = 0$. Since $1 + ax \neq 0$ we may divide, giving

$$x^{-1} = \frac{-x}{1 + ax}.$$

This lies in G, because

$$\phi\left(\frac{-x}{1+ax}\right) = 1 - \frac{ax}{1+ax} = \frac{1}{1+ax} \neq 0,$$

and since $*$ is commutative ($x * y = y * x$ is clear from the formula), $x^{-1} * x = 0$ as well.

All four axioms hold, so $(G, *)$ is a group (indeed an abelian one), where the excluded number is $-\frac{1}{a}$.

A subgroup of order 2

The multiplicative group of non-zero reals has the subgroup $\{1, -1\}$. Pulling this back through ϕ: $\phi(x) = 1$ gives $x = 0$, and $\phi(x) = -1$ gives $x = -\frac{2}{a}$. So we try

$$H = \left\{0, -\tfrac{2}{a}\right\}.$$

These are two different elements of G (as $a \neq 0$, $-\frac{2}{a} \neq 0$ and $-\frac{2}{a} \neq -\frac{1}{a}$). Directly,

$$\left(-\tfrac{2}{a}\right) * \left(-\tfrac{2}{a}\right) = -\tfrac{4}{a} + a \cdot \tfrac{4}{a^2} = 0,$$

and $0 * 0 = 0$, $0 * \left(-\frac{2}{a}\right) = -\frac{2}{a}$. So H is closed, contains the identity, and each element is its own inverse. Hence H is a subgroup of $(G, *)$ with exactly two elements.

In fact it is the only one: a group of order 2 is $\{e, h\}$ with $h * h = e$ and $h \neq e$, and $h * h = 0$ means $h(2 + ah) = 0$, so $h = -\frac{2}{a}$.

Checks

Take $a = 1$. Then $x * y = x + y + xy = (1 + x)(1 + y) - 1$, the excluded number is -1, and the order-2 subgroup is $\{0, -2\}$: indeed $(-2) * (-2) = -4 + 4 = 0$. Also $2 * 3 = 11$ and $(1 + 2)(1 + 3) = 12 = 1 + 11$, as $(\dagger)$ predicts. With a computer we verified, in exact rational arithmetic on about 5000 random choices of a, x, y, z, the identity $(\dagger)$, associativity, closure in G, the identity 0, the inverse formula, and closure of $\{0, -\frac{2}{a}\}$.

Common slips

- Forgetting to check that the inverse $\frac{-x}{1+ax}$ is itself in G, or that $x * y$ never equals the excluded number. Closure is the axiom that justifies removing exactly one point.
- Offering $\{0\}$ or $\{0, -\frac{1}{a}\}$ as the subgroup: the first has one element, and $-\frac{1}{a}$ is not in G at all.

Beyond STEP

The map $\phi(x) = 1 + ax$ is an *isomorphism* from $(G, *)$ to the group $(\mathbb{R} \setminus \{0\}, \times)$: a bijection that turns one operation into the other. Once you spot an isomorphism, every structural fact transfers for free, which is why $(G, *)$ has exactly one element of order 2, just as $\times$ has only -1. Isomorphisms, and the idea that two groups can be "the same" while looking different, are central to a first course in abstract algebra; Allenby's *Rings, Fields and Groups* is a good place to read about them.

WORKED SOLUTION

STEP III 1999, Question 8

Pure • Piecewise Differential Equations

STEP III 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q8

The key idea

On each interval the equation is simple harmonic motion with a constant frequency, so the general solution there is a combination of $\cos nx$ and $\sin nx$. The only work is matching: the continuity conditions hand each interval its starting values of y and y' from the end of the previous one. The intervals all have length 2π, so the starting values turn out to be the same every time, and a pattern appears that we can prove by induction.

Solution

Part (i)

For $0 < x < 2\pi$ we have $n = 1$, so $y'' + y = 0$, with general solution

$$y = A\cos x + B\sin x.$$

Since y and y' are continuous at $x = 0$, the conditions $y(0) = 0$ and $y'(0) = 1$ apply to this solution: $y(0) = A = 0$ and $y'(0) = B = 1$. Hence

$$y = \sin x \qquad (0 \le x \le 2\pi).$$

At the right-hand end, $y(2\pi) = \sin 2\pi = 0$ and $y'(2\pi) = \cos 2\pi = 1$.

Part (ii)

For $2\pi < x < 4\pi$ we have $n = 2$, so $y'' + 4y = 0$ and

$$y = A\cos 2x + B\sin 2x.$$

By continuity of y and y' at $x = 2\pi$, this solution must have $y = 0$ and $y' = 1$ there, the values found at the end of part (i). Since $\cos 4\pi = 1$ and $\sin 4\pi = 0$,

$$y(2\pi) = A = 0, \qquad y'(2\pi) = 2B\cos 4\pi = 2B = 1,$$

so $B = \frac{1}{2}$ and $y = \frac{1}{2}\sin 2x$ for $2\pi \le x \le 4\pi$, as required.

Notice that again $y(4\pi) = \frac{1}{2}\sin 8\pi = 0$ and $y'(4\pi) = \cos 8\pi = 1$: the next interval starts from exactly the same values. This suggests the following claim.

Claim. For every positive integer n,

$$y = \frac{1}{n}\sin nx \qquad \text{for } 2(n-1)\pi \le x \le 2n\pi.$$

Proof by induction. The case $n = 1$ is part (i). Suppose the claim holds for $n = k$. At $x = 2k\pi$ the previous piece gives

$$y = \frac{1}{k}\sin(2k^2\pi) = 0, \qquad y' = \cos(2k^2\pi) = 1,$$

because $2k^2\pi$ is a whole multiple of 2π. On $2k\pi < x < 2(k+1)\pi$ the equation is $y'' + (k+1)^2 y = 0$, so

$$y = A\cos(k+1)x + B\sin(k+1)x.$$

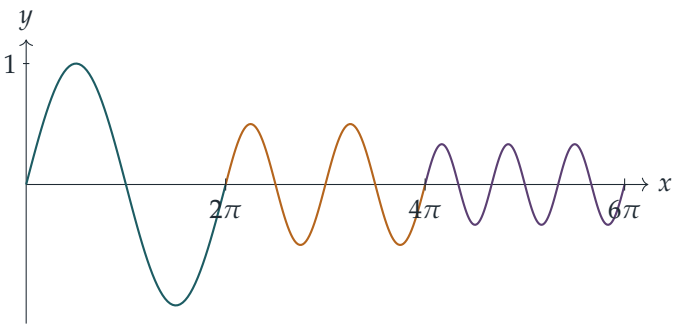
Ask yourself What are $\cos(k+1)x$ and $\sin(k+1)x$ at the left-hand end $x = 2k\pi$?

There $(k+1)x = 2k(k+1)\pi$, again a whole multiple of 2π, so the cosine is 1 and the sine is 0. Continuity of y and y' at $x = 2k\pi$ therefore gives

$$A = 0, \qquad (k+1)B = 1,$$

so $y = \dfrac{1}{k+1}\sin(k+1)x$ on $2k\pi \le x \le 2(k+1)\pi$. This is the claim for $n = k+1$, and by induction it holds for all $n \ge 1$. □

So $y(x) = \dfrac{1}{n}\sin nx$ whenever $2(n-1)\pi \le x \le 2n\pi$. (At each junction $x = 2n\pi$ both neighbouring formulas give $y = 0$, so there is no ambiguity.)



Part (iii)

Using part (ii), on the nth interval

$$\int_{2(n-1)\pi}^{2n\pi} y^2\,\mathrm{d}x = \frac{1}{n^2}\int_{2(n-1)\pi}^{2n\pi} \sin^2 nx\,\mathrm{d}x = \frac{1}{2n^2}\int_{2(n-1)\pi}^{2n\pi} (1-\cos 2nx)\,\mathrm{d}x = \frac{1}{2n^2}\cdot 2\pi = \frac{\pi}{n^2},$$

since $\sin 2nx$ vanishes at both limits (each limit is a multiple of π and $2n$ is an integer), so the cosine term integrates to zero.

Adding the first N intervals,

$$\int_0^{2N\pi} y^2 \,\mathrm{d}x = \pi \sum_{n=1}^{N} \frac{1}{n^2}.$$

The infinite integral is the limit of $\int_0^X y^2 \,\mathrm{d}x$ as $X \to \infty$. Since $y^2 \geq 0$, this is an increasing function of X; and for $2(N-1)\pi \leq X \leq 2N\pi$ it lies between the values at $2(N-1)\pi$ and $2N\pi$. Both of those tend to $\pi \sum_{n=1}^{\infty} \frac{1}{n^2}$ as $N \to \infty$, because the series converges. Hence

$$\int_0^{\infty} y^2 \,\mathrm{d}x = \pi \sum_{n=1}^{\infty} \frac{1}{n^2},$$

as required.

Checks

- Each piece $\frac{1}{n}\sin nx$ satisfies $y'' = -n^2 y$, starts at $y = 0$, $y' = 1$ and ends at $y = 0$, $y' = 1$, so the pieces join smoothly in value and slope.
- Stepping the differential equation numerically (fourth-order Runge–Kutta) through the first six intervals reproduces $\frac{1}{n}\sin nx$ to within 10^{-10}, and the integral of y^2 up to 12π comes out as 4.685336, matching $\pi(1 + \frac{1}{4} + \cdots + \frac{1}{36})$.
- Using Euler's result $\sum 1/n^2 = \pi^2/6$, the integral equals $\pi^3/6 \approx 5.168$.

Common slips

- Applying the initial conditions at $x = 0$ on every interval, rather than at its left-hand end $x = 2(n-1)\pi$. It happens to give the right answer here only because the starting point is a multiple of 2π, and that fact must be stated.
- Forgetting that $y' = nB\cos nx$: the factor n is exactly what produces the amplitude $\frac{1}{n}$.
- Writing the integral as an infinite sum without saying why the limit exists; the positivity of y^2 makes this easy.

Beyond STEP

This is a simple example of a differential equation with piecewise-constant coefficients, the kind that models a spring whose stiffness is switched at fixed times; at each switch the solution is still continuous and differentiable even though y'' jumps. The final answer, $\pi^3/6$, ties a differential equation to the Basel sum $\sum 1/n^2 = \pi^2/6$, which is often proved first in undergraduate courses on Fourier series using Parseval's theorem.

Boyce and DiPrima, *Elementary Differential Equations*, covers both second-order linear equations and Fourier series.

WORKED SOLUTION

STEP III 1999, Question 9

Mechanics • Newton's Shell Theorem

STEP III 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q9

This question has a full guided treatment in *Beyond STEP*, Section 10.7, *Gravity Inside a Shell*: https://beyondstep.keithmcnulty.org/ch10/shell-theorem.html

The key idea

Gravity from an extended body is the vector sum of the pulls of its small pieces. For the ring, symmetry does most of the work: pieces on opposite sides of the ring cancel each other's sideways pulls, and every piece is the same distance from the particle, so only one simple component survives. For the shell, we slice it into rings whose axis passes through the particle, use the ring result for each slice, and add the slices up with an integral. A well-chosen substitution makes that integral collapse to zero.

Solution

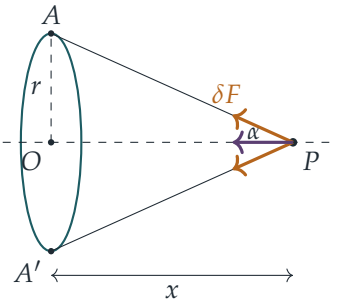
The ring

Model the ring as a thin uniform circle of mass m and the particle as a point of unit mass at P, on the axis at distance x from the centre O. Every point of the ring is at the same distance

$$d = \sqrt{x^2 + r^2}$$

from P, by Pythagoras in the right-angled triangle formed by P, O and that point.

Direction. Pair each small element of the ring at A with the equal element at the diametrically opposite point A'. They are the same distance d from P, so they pull with equal magnitudes along PA and PA'. These two lines are reflections of each other in the axis, so the components perpendicular to the axis cancel, and the components along the axis add, both pointing from P towards O. The whole ring is made of such pairs, so the net force on the particle is along the axis, towards the centre of the ring.



Magnitude. An element of mass δm pulls with force $G\,\delta m/d^2$ along PA. If α is the angle between PA and the axis PO, then $\cos\alpha = x/d$, so its component towards O is

$$\frac{G\,\delta m}{d^2}\cdot\frac{x}{d} = \frac{Gx\,\delta m}{(x^2+r^2)^{3/2}}.$$

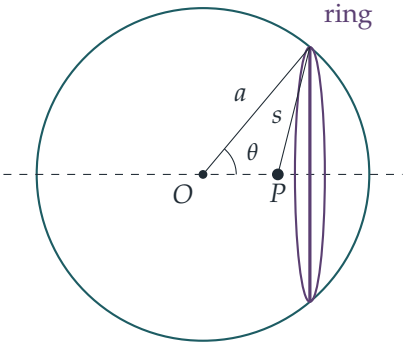
The factor multiplying δm is the same for every element, so summing over the ring just replaces δm by m:

$$F = \frac{Gmx}{(x^2+r^2)^{3/2}},$$

directed towards the centre, as required. If P is on the other side of the ring this holds with x replaced by $|x|$; in the next part we use x as a signed distance, and then the formula with signed x gives the signed component of the force correctly.

The shell

Let the shell have centre O, radius a and mass M, so its surface density is $M/(4\pi a^2)$. Put the particle at P with $OP = b < a$. If $b = 0$, every direction from O is the same and the pulls of opposite points of the shell cancel, so the force is zero. From now on take $0 < b < a$ and use the line OP as an axis, measuring distance along it from O in the direction of P.



Slice the shell into thin rings by planes perpendicular to OP. The ring between polar angles θ and $\theta+\delta\theta$ (measured at O from the direction OP) has radius $a\sin\theta$, centre at

position $a\cos\theta$ on the axis, and width $a\,\delta\theta$. Its mass is

$$\delta M = \frac{M}{4\pi a^2}\cdot 2\pi a\sin\theta\cdot a\,\delta\theta = \tfrac{1}{2}M\sin\theta\,\delta\theta.$$

The centre of this ring is at signed distance $a\cos\theta - b$ from P. By the ring result its pull on P is along the axis, and its component in the positive direction (away from O, through P) is

$$\frac{G\,\delta M\,(a\cos\theta - b)}{\left((a\cos\theta - b)^2 + a^2\sin^2\theta\right)^{3/2}} = \frac{G\,\delta M\,(a\cos\theta - b)}{(a^2 + b^2 - 2ab\cos\theta)^{3/2}}.$$

This sign convention is right in both cases: when $a\cos\theta > b$ the ring is beyond P and pulls in the positive direction; when $a\cos\theta < b$ it pulls the other way. Every ring pulls along the same axis, so the total force is along OP, with component

$$F = \frac{GM}{2}\int_0^{\pi}\frac{(a\cos\theta - b)\sin\theta}{(a^2 + b^2 - 2ab\cos\theta)^{3/2}}\,\mathrm{d}\theta.$$

Ask yourself What substitution tames the denominator?

The quantity $a^2 + b^2 - 2ab\cos\theta$ is, by the cosine rule, the square of the distance s from P to the ring (the line marked s in the figure). So we substitute

$$s^2 = a^2 + b^2 - 2ab\cos\theta, \qquad 2s\,\mathrm{d}s = 2ab\sin\theta\,\mathrm{d}\theta.$$

As θ runs from 0 to π, s increases from $a - b$ to $a + b$ (positive because $b < a$). Also

$$a\cos\theta - b = \frac{a^2 + b^2 - s^2}{2b} - b = \frac{a^2 - b^2 - s^2}{2b}.$$

Using $\sin\theta\,\mathrm{d}\theta = \dfrac{s\,\mathrm{d}s}{ab}$,

$$F = \frac{GM}{2}\int_{a-b}^{a+b}\frac{a^2 - b^2 - s^2}{2b}\cdot\frac{1}{s^3}\cdot\frac{s}{ab}\,\mathrm{d}s = \frac{GM}{4ab^2}\int_{a-b}^{a+b}\left(\frac{a^2 - b^2}{s^2} - 1\right)\mathrm{d}s$$
$$= \frac{GM}{4ab^2}\left[-\frac{a^2 - b^2}{s} - s\right]_{a-b}^{a+b}.$$

At $s = a + b$ the bracket is $-(a - b) - (a + b) = -2a$, and at $s = a - b$ it is $-(a + b) - (a - b) = -2a$. These are equal, so

$$F = \frac{GM}{4ab^2}\big(-2a - (-2a)\big) = 0.$$

The force is along OP and its component along OP is zero, so the particle experiences no gravitational force due to the shell, as required.

Checks

- Ring: at $x = 0$ the force is zero (the particle is at the centre and the pulls balance), and for $x \gg r$ it tends to Gm/x^2, as if all the mass were at O.
- Shell, outside: if instead $b > a$, the lower limit becomes $s = b - a$, where the bracket is $(a + b) - (b - a) = 2a$. Then $F = \frac{GM}{4ab^2}(-4a) = -GM/b^2$: an attraction towards O as if the whole mass were at the centre. That is Newton's other shell theorem, a good sign the integral is right.
- Numerically, summing 2000 point masses round a ring reproduced the formula to ten decimal places. Simpson's rule on the θ-integral gave about 10^{-15} for several random $b < a$, and a Monte Carlo shell of 2×10^5 random points with $a = 2$, $b = 1.3$, $GM = 1$ gave a force below 10^{-3}, against a typical scale $GM/a^2 = 0.25$.

Common slips

- Taking the ring's mass element as $M\,\delta\theta/\pi$. The rings near $\theta = \pi/2$ are bigger, so the mass is proportional to $\sin\theta\,\delta\theta$, not $\delta\theta$.
- Using $|a\cos\theta - b|$ in the ring formula. The rings on the two sides of P pull in opposite directions, and that cancellation is the whole point.
- Getting the limits of s wrong: at $\theta = 0$ the distance is $a - b$, not $b - a$, because P is inside.

Beyond STEP

This is Newton's shell theorem, and it has a slicker proof using Gauss's law for gravity: the flux of the gravitational field through any closed surface is proportional to the mass inside, so by symmetry the field inside an empty spherical shell must vanish. The same mathematics gives the potential, and the fact that the potential is constant inside the shell is an early example of a harmonic function, studied in first-year vector calculus. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the shell theorem and gravitational potential carefully.

WORKED SOLUTION

STEP III 1999, Question 10

Mechanics • Falling Chain Impulse

STEP III 1999, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q10

The key idea

While the chain collapses, the part still hanging is in free fall: nothing pulls on it except gravity. So each link simply drops from wherever it started, and the only thing that varies from link to link is the height of that drop. The table supplies one impulse per link to stop it. Adding these up gives a sum of square roots, and for large n this is a Riemann sum for $\int_0^1 \sqrt{x}\,\mathrm{d}x$.

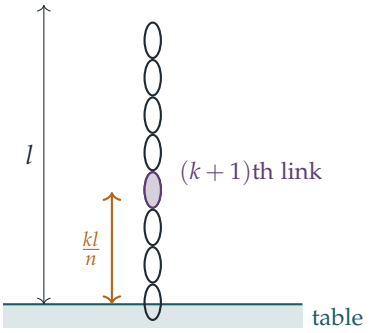
Solution

The model

Each link has mass $\frac{m}{n}$ and length $\frac{l}{n}$. The links are smooth, and the chain is perfectly flexible, so it can't transmit a push. We treat each link as a particle that is brought to rest instantly (the collision is inelastic) when it reaches the table.

? **Ask yourself** Why is the hanging part of the chain in free fall, when the links below it are being stopped by the table?

When the chain is released, every link has acceleration g if there is no tension, and a falling chain with zero tension is consistent: all links move together and stay in contact without stretching. The table stops a link by pushing it up, and a flexible chain can't pass that push on to the links above it. Since the links are smooth, there is no friction between them either. So no force other than gravity acts on any link that is still in the air, and each link falls freely until it lands.



The $(k+1)$th link

Below the $(k+1)$th link there are k links, with total length $\frac{kl}{n}$. As the bottom end just touches the table, the $(k+1)$th link falls a distance $\frac{kl}{n}$ before it reaches the table. Falling freely from rest, its speed on arrival is given by $v^2 = 2gs$:

$$v_k = \sqrt{\frac{2gkl}{n}}.$$

It is then brought to rest, so the magnitude of its change in momentum is

$$\frac{m}{n}\sqrt{\frac{2gkl}{n}} = \frac{m}{n}\sqrt{2gl}\,\sqrt{\frac{k}{n}},$$

directed upwards (the link loses downward momentum $\frac{m}{n}v_k$). For $k = 0$ this is zero: the bottom link is already touching the table.

The total impulse

By Newton's third law, the impulse the table sustains from the $(k+1)$th link is equal in size to that link's change in momentum, and points downwards. The links are numbered $k+1 = 1, 2, \ldots, n$, so the total impulse on the table from the collisions is

$$J = \sum_{k=0}^{n-1} \frac{m}{n}\sqrt{\frac{2gkl}{n}} = m\sqrt{2gl}\ \cdot\ \frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}}.$$

? **Ask yourself** Which integral does $\frac{1}{n}\sum_{k=0}^{n-1}\sqrt{k/n}$ approximate?

It is the sum of the areas of n rectangles of width $\frac{1}{n}$ whose heights are the values of $\sqrt{x}$ at the left-hand ends $x = 0, \frac{1}{n}, \ldots, \frac{n-1}{n}$ of the strips of $[0,1]$. So for large n,

$$\frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}} \approx \int_0^1 \sqrt{x}\,\mathrm{d}x = \left[\tfrac{2}{3}x^{3/2}\right]_0^1 = \frac{2}{3},$$

and therefore

$$J \approx \tfrac{2}{3}\,m\sqrt{2gl},$$

as required.

To see how good the approximation is: $\sqrt{x}$ is increasing, so on the strip $[\frac{k}{n}, \frac{k+1}{n}]$ the left-end height $\sqrt{k/n}$ lies below the curve, and on $[\frac{k-1}{n}, \frac{k}{n}]$ it lies above it. Summing over the strips gives

$$\int_0^{1-\frac{1}{n}} \sqrt{x}\,\mathrm{d}x \;\le\; \frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}} \;\le\; \int_0^1 \sqrt{x}\,\mathrm{d}x, \quad \text{that is} \quad \tfrac{2}{3}\left(1-\tfrac{1}{n}\right)^{3/2} \le \frac{J}{m\sqrt{2gl}} \le \tfrac{2}{3}.$$

Both bounds tend to $\frac{2}{3}$, so the relative error is at most about $\frac{3}{2n}$.

Checks

- Units: $m\sqrt{gl}$ is mass $\times$ speed, which is a momentum, as an impulse must be.
- Numerically, with $m = 2$, $l = 3$, $g = 9.8$: $\frac{2}{3}m\sqrt{2gl} \approx 10.224$, while the exact sum is 9.363 for $n = 10$, 10.144 for $n = 100$ and 10.216 for $n = 1000$. The ratio approaches 1, and always lies inside the bounds above.
- The last link to land falls almost the whole length l, with speed close to $\sqrt{2gl}$. If all the mass arrived at that speed the impulse would be $m\sqrt{2gl}$. Most links fall less far, so it makes sense that the answer is a fraction of this.

Common slips

- Letting the $(k+1)$th link fall a distance $\frac{(k+1)l}{n}$. There are only k links beneath it.
- Including the steady weight of the chain already lying on the table. The question asks only for the impulse from the collisions.

Beyond STEP

Since all the links land in time $\sqrt{2l/g}$, the collisions alone give the table an average force of $\frac{2}{3}m\sqrt{2gl}/\sqrt{2l/g} = \frac{2}{3}mg$. At each moment the impact force is actually twice the weight of chain already on the table, so the reading on a scale under the table climbs to $3mg$ just as the last link lands. Problems like this, where mass is gained or lost continuously (falling chains, rockets, raindrops), are treated with the variable-mass form of Newton's second law. See Kleppner and Kolenkow, *An Introduction to Mechanics*, for a careful treatment.

WORKED SOLUTION

STEP III 1999, Question 11

Mechanics • Rigid Body Rotation

STEP III 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q11

The key idea

Once the hoop is on the edge it is a rigid body pivoting about a fixed point, so two equations do all the work: conservation of energy gives the angular speed at any angle, and Newton's second law for the centre, resolved towards the pivot, gives the normal reaction. Contact is lost when that reaction falls to zero. The moment of inertia from the first part is exactly what the energy equation needs.

Solution

The moment of inertia

Every particle of a thin hoop is at distance a from its centre C, so about the axis through C perpendicular to the plane of the hoop the moment of inertia is

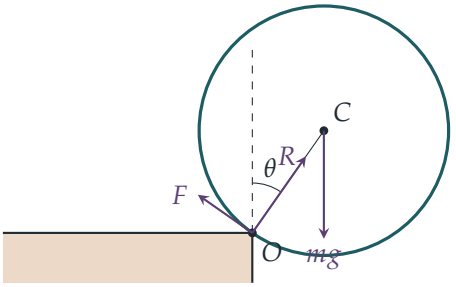
$$I_C = \sum \delta m\, a^2 = ma^2.$$

A point O on the circumference is at distance a from C, and the new axis is parallel to the old one, so by the parallel axis theorem

$$I_O = I_C + ma^2 = 2ma^2.$$

Rolling over the edge

Model the hoop as a uniform rigid ring and the edge of the table as a fixed point O. Both surfaces are rough, and we assume friction is large enough that the hoop never slips: on the table it rolls, and over the edge it turns about O without sliding. Let θ be the angle that OC makes with the upward vertical, so $\theta = 0$ at the moment the hoop reaches the edge, and write $\omega = \dot{\theta}$.



Ask yourself What is the hoop's angular speed about O at the instant it reaches the edge?

A hoop rolling without slipping at speed v has angular speed v/a, and its motion at each instant is a rotation about its point of contact. At the instant it reaches the edge, the point of contact *is* O, and the hoop simply carries on turning about it: no impact occurs, since the hoop does not lose contact. So $\omega = v/a$ when $\theta = 0$.

Energy. While the hoop stays in contact with O, the forces at O do no work (the point of application is at rest), so mechanical energy is conserved. The centre has dropped by $a - a\cos\theta$, so

$$\tfrac{1}{2}(2ma^2)\,\omega^2 = \tfrac{1}{2}(2ma^2)\left(\frac{v}{a}\right)^2 + mga(1-\cos\theta), \qquad \text{that is} \qquad a\omega^2 = \frac{v^2}{a} + g(1-\cos\theta).$$

Towards the pivot. The centre C moves on a circle of radius a about O, so its acceleration has component $a\omega^2$ directed from C towards O. The forces on the hoop are its weight mg, the normal reaction R along OC (pushing away from O), and friction F perpendicular to OC, which has no component along OC. Resolving along CO:

$$mg\cos\theta - R = ma\omega^2.$$

Substituting the energy equation,

$$R = mg\cos\theta - \frac{mv^2}{a} - mg(1-\cos\theta) = mg(2\cos\theta - 1) - \frac{mv^2}{a}.$$

This decreases as θ increases, and the hoop loses contact with the edge when R first reaches zero, which happens when

$$2g\cos\theta = g + \frac{v^2}{a}, \qquad \text{that is} \qquad \cos\theta = \frac{1}{2} + \frac{v^2}{2ag},$$

as required.

Ask yourself Does this make sense for every v?

At $\theta = 0$ we have $R = m(g - v^2/a)$, which is positive only if $v^2 < ag$. That is the condition for the hoop to roll over the edge without immediately losing contact, which

the question assumes, and it is exactly the condition for the right-hand side to be less than 1, so a valid angle θ exists.

For completeness, resolving perpendicular to OC with $I_O\ddot{\theta} = mga\sin\theta$ gives $\ddot{\theta} = g\sin\theta/(2a)$ and $F = \frac{1}{2}mg\sin\theta$. This is the friction the edge must supply to prevent slipping, which is why both surfaces need to be rough.

Checks

- If $v \to 0$ the hoop just topples off the edge and loses contact at $\cos\theta = \frac{1}{2}$, so $\theta = 60°$. As $v^2 \to ag$, $\theta \to 0$: the hoop barely touches the edge. Both limits make sense.
- Numerically, with $a = 0.6\,\text{m}$, $g = 9.8\,\text{m}\,\text{s}^{-2}$ and $v = 1.5\,\text{m}\,\text{s}^{-1}$, stepping $\ddot{\theta} = g\sin\theta/(2a)$ from $\dot{\theta} = v/a$ until $g\cos\theta = a\dot{\theta}^2$ gives $\cos\theta \approx 0.69132$, matching $\frac{1}{2} + v^2/(2ag) = 0.69133$.

Common slips

- Using $I_C = ma^2$ in the energy equation, or adding a separate translational kinetic energy term on top of $\frac{1}{2}I_O\omega^2$. The rotation about O already includes the motion of the centre.
- Forgetting the initial kinetic energy, which gives $\cos\theta = \frac{1}{2}$ for every v.

Beyond STEP

Rolling off the edge is a typical *unilateral constraint*: the edge can push but not pull, so the equations of motion apply only while the reaction they predict is non-negative. The same idea governs a particle sliding off a sphere and a car going over a hump-backed bridge. In a first-year course you meet it alongside rigid-body dynamics and the parallel and perpendicular axis theorems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats all of these carefully.

WORKED SOLUTION

STEP III 1999, Question 12

Probability and Statistics • Discrete Joint Distributions

STEP III 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q12

The key idea

The joint probability depends on j and k mostly through $j+k$, so group the terms by the total $n = j+k$. What is left is $\sum_j \frac{1}{j!(n-j)!}$, which is a binomial sum in disguise: multiply by $n!$ and it becomes $\sum_j \binom{n}{j} = 2^n$. After that, every part is an exponential series.

Solution

Part (i)

Let $n \geq 1$. The event $\{X+Y=n\}$ is the disjoint union of the events $\{X=j,\ Y=n-j\}$ for $j = 0, 1, \ldots, n$. On each of these $j+k=n$, so

$$\begin{aligned} \mathrm{P}(X+Y=n) &= \sum_{j=0}^{n} \mathrm{e}^{-1} \frac{n\lambda^n}{j!\,(n-j)!} = \mathrm{e}^{-1}\frac{n\lambda^n}{n!}\sum_{j=0}^{n}\binom{n}{j} \\ &= \mathrm{e}^{-1}\frac{n\lambda^n 2^n}{n!} = \frac{\mathrm{e}^{-1}(2\lambda)^n}{(n-1)!}, \end{aligned}$$

using $\sum_j \binom{n}{j} = (1+1)^n = 2^n$ by the binomial theorem, and $n/n! = 1/(n-1)!$ for $n \geq 1$. For completeness, $\mathrm{P}(X+Y=0) = \mathrm{P}(X=0,\ Y=0) = 0$, because of the factor $j+k$.

Part (ii)

The probabilities $\mathrm{P}(X+Y=n)$ over all $n \geq 0$ must add up to 1. The $n=0$ term is zero, so

$$1 = \sum_{n=1}^{\infty} \frac{\mathrm{e}^{-1}(2\lambda)^n}{(n-1)!} = \mathrm{e}^{-1}(2\lambda)\sum_{m=0}^{\infty}\frac{(2\lambda)^m}{m!} = \mathrm{e}^{-1}\,2\lambda\,\mathrm{e}^{2\lambda},$$

where we put $m = n-1$ and used the exponential series $\sum_m t^m/m! = \mathrm{e}^t$. Hence $2\lambda\,\mathrm{e}^{2\lambda-1} = 1$, as required.

Part (iii)

Let $g(x) = 2x\,\mathrm{e}^{2x-1}$. By the product rule,

$$g'(x) = 2\mathrm{e}^{2x-1} + 4x\,\mathrm{e}^{2x-1} = 2(1+2x)\,\mathrm{e}^{2x-1}.$$

For $x > 0$ both $1+2x$ and e^{2x-1} are positive, so $g'(x) > 0$ and g is strictly increasing on $x > 0$.

> **Ask yourself** Why does "increasing" give "at most one solution"?

If $0 < x_1 < x_2$ then $g(x_1) < g(x_2)$, so g cannot take the value 1 at two different points. Since $\lambda > 0$, the equation $g(\lambda) = 1$ from (ii) therefore has at most one solution.

Now try the obvious candidate that makes the exponent zero, $x = \frac{1}{2}$: $g(\frac{1}{2}) = 1 \cdot \mathrm{e}^0 = 1$. So $\frac{1}{2}$ is a solution, and by uniqueness it is the only one:

$$\lambda = \tfrac{1}{2}.$$

With this value, part (i) becomes $\mathrm{P}(X+Y=n) = \dfrac{\mathrm{e}^{-1}}{(n-1)!}$ for $n \geq 1$.

Part (iv)

Since $X+Y$ takes only the values $n \geq 1$, with the probabilities just found,

$$\mathrm{E}\big(2^{X+Y}\big) = \sum_{n=1}^{\infty} 2^n \frac{\mathrm{e}^{-1}}{(n-1)!} = 2\mathrm{e}^{-1} \sum_{m=0}^{\infty} \frac{2^m}{m!} = 2\mathrm{e}^{-1}\mathrm{e}^2 = 2\mathrm{e},$$

again with $m = n-1$ and the exponential series (which converges, so the expectation is finite).

Checks

- With $\lambda = \frac{1}{2}$, the formula in (i) says $X+Y-1$ has a Poisson distribution with mean 1. Its probabilities $\mathrm{e}^{-1}/m!$ certainly sum to 1, consistent with (ii).
- Summing the joint probabilities numerically over $j+k < 60$ gives total 1.000000, and summing $2^{j+k}\,\mathrm{P}(X=j, Y=k)$ gives 5.436564, which matches $2\mathrm{e} \approx 5.436564$.
- A quick sense check on (iv): $2^{X+Y} = 2 \cdot 2^N$ with $N \sim \mathrm{Po}(1)$, and $\mathrm{E}(s^N) = \mathrm{e}^{s-1}$ for a Poisson variable, so the answer is $2\mathrm{e}^1$.

Common slips

- Forgetting that $n/n! = 1/(n-1)!$, which leaves an awkward factor of n inside the series in (ii).

- Treating the $n = 0$ term as nonzero: the factor $j + k$ makes it vanish, which is exactly why the sums start at $n = 1$.
- In (iii), showing only that $\lambda = \frac{1}{2}$ works. You also need monotonicity to rule out any other solution.

Beyond STEP

The quantity $\mathrm{E}(s^{X+Y})$ is the *probability generating function* of $X + Y$, here evaluated at $s = 2$. Generating functions turn sums of probabilities into power series, and they are the standard tool in a first university probability course for identifying distributions and finding moments. Grimmett and Welsh, *Probability: An Introduction*, gives a clear account of generating functions and the Poisson distribution.

WORKED SOLUTION

STEP III 1999, Question 13

Probability and Statistics • Bayes with Continuous Prior

STEP III 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q13

The key idea

There are two layers of randomness: first the size X of my portion, then where the currants land. Once we *condition* on $X = x$, each currant is in my portion with probability x, independently of the others, so the number of currants is binomial. After that, part (i) is the law of total expectation and part (ii) is Bayes' theorem with a continuous prior.

Solution

First we find A. A density must integrate to 1:

$$\int_0^1 Ax\,\mathrm{d}x = \frac{A}{2} = 1 \quad \Longrightarrow \quad A = 2,$$

so X has density $\mathrm{f}(x) = 2x$ on $[0, 1]$.

Let N be the number of currants in my portion. We model "randomly placed" in the natural way: each currant's position is uniformly distributed through the cake, independently of the other currants and of how much cake I take. If I take a proportion x of the cake, each currant therefore lies in my portion with probability x, independently of the others, so

$$N \mid (X = x) \sim \mathrm{B}(4, x).$$

Part (i)

Given $X = x$, the binomial mean is $\mathrm{E}(N \mid X = x) = 4x$. Averaging over X (the law of total expectation),

$$\begin{aligned}\mathrm{E}(N) &= \int_0^1 \mathrm{E}(N \mid X = x)\,\mathrm{f}(x)\,\mathrm{d}x = \int_0^1 4x \cdot 2x\,\mathrm{d}x \\ &= 8\int_0^1 x^2\,\mathrm{d}x = \frac{8}{3}.\end{aligned}$$

Equivalently, $\mathrm{E}(N) = 4\,\mathrm{E}(X)$ with $\mathrm{E}(X) = \frac{2}{3}$. The expected number of currants in my portion is $\frac{8}{3}$.

Part (ii)

We want $\mathrm{P}(X > \frac{1}{2} \mid N = 4)$. Given $X = x$, all four currants are in my portion with probability x^4.

> **Ask yourself** How do we use Bayes' theorem when X is continuous and $\mathrm{P}(X = x) = 0$?

We replace the sums in Bayes' theorem by integrals against the density. A slice $[x, x + \mathrm{d}x]$ of possible portion sizes has probability $2x\,\mathrm{d}x$, and given that slice the chance of all four currants is x^4. Adding up over slices,

$$\mathrm{P}(N = 4) = \int_0^1 x^4 \cdot 2x\,\mathrm{d}x = 2\int_0^1 x^5\,\mathrm{d}x = \frac{2}{6} = \frac{1}{3}.$$

In the same way, keeping only the slices with $x > \frac{1}{2}$,

$$\mathrm{P}\big(X > \tfrac{1}{2} \text{ and } N = 4\big) = \int_{1/2}^1 2x^5\,\mathrm{d}x = \frac{1}{3}\Big(1 - \frac{1}{64}\Big) = \frac{1}{3}\cdot\frac{63}{64}.$$

Therefore

$$\mathrm{P}\big(X > \tfrac{1}{2} \mid N = 4\big) = \frac{\frac{1}{3}\cdot\frac{63}{64}}{\frac{1}{3}} = \frac{63}{64}.$$

Another way to see it: given $N = 4$, the conditional density of X is proportional to $x^4 \cdot 2x$, so it is $6x^5$ on $[0,1]$. Its distribution function is x^6, and $1 - (\frac{1}{2})^6 = \frac{63}{64}$.

Checks

- The answer to (ii) is very close to 1, which makes sense: finding every currant is strong evidence that I took a big piece. The prior probability $\mathrm{P}(X > \frac{1}{2}) = 1 - \frac{1}{4} = \frac{3}{4}$ has been pushed up to $\frac{63}{64} \approx 0.984$.
- $\frac{8}{3}$ lies between 0 and 4, and it is more than 2, as it should be: the density $2x$ favours large portions.
- A Monte Carlo simulation with 10^6 trials (taking $X = \sqrt{U}$ with U uniform, which has density $2x$, then placing four currants) gave $\mathrm{E}(N) \approx 2.667$, $\mathrm{P}(N = 4) \approx 0.333$ and a conditional probability ≈ 0.9846, against $\frac{63}{64} = 0.984375$.

Common slips

- Forgetting to find A, or taking $A = 1$. Then f isn't a density, and the expectation in (i) comes out wrong (although A cancels in (ii)).

- Computing $\mathrm{P}(N = 4 \mid X > \frac{1}{2})$ instead of $\mathrm{P}(X > \frac{1}{2} \mid N = 4)$. The conditioning is the other way round.
- Using $(\frac{1}{2})^4$ in place of the integral: the portion size is random, so we must average x^4 against the density rather than plug in a single value.

Beyond STEP

This is Bayesian inference in miniature: the density $2x$ is a *prior* for the unknown proportion, the binomial is the *likelihood*, and $6x^5$ is the *posterior*. Both prior and posterior are beta distributions, an example of a conjugate prior for binomial data, and the same calculation underlies Laplace's rule of succession. Blitzstein and Hwang, *Introduction to Probability*, treats the beta–binomial story and conditioning on continuous random variables very clearly.

WORKED SOLUTION

STEP III 1999, Question 14

Probability and Statistics • Expectation by Conditioning

STEP III 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q14

The key idea

In H1 the winnings depend only on when the first success happens, so the expectation is a finite arithmetico-geometric sum $\sum r\,p\,q^{r-1}$. The standard trick for such sums is to subtract q times the sum from itself. In H2 we condition on the number of turns N and average the H1 answer over the Poisson distribution. Doing this needs only the Poisson sums $\sum \lambda^n x^n/n!$ and $\sum n\lambda^n x^n/n!$, both of which come from the exponential series.

Solution

Game H1

Let W be her winnings in pounds. Turns are independent, each a success with probability p. Her first success is at turn r exactly when the first $r-1$ turns fail and turn r succeeds, so

$$\mathrm{P}(W=r) = q^{r-1}p \qquad (1 \le r \le n),$$

and otherwise $W=0$, which adds nothing to the expectation. Hence

$$\mathrm{E}(W) = p\,S, \qquad \text{where } S = \sum_{r=1}^{n} r\,q^{r-1} = 1 + 2q + 3q^2 + \cdots + nq^{n-1}.$$

Multiply by q and subtract, lining up equal powers of q:

$$\begin{aligned} S - qS &= 1 + q + q^2 + \cdots + q^{n-1} - nq^n \\ &= \frac{1-q^n}{1-q} - nq^n, \end{aligned}$$

by the formula for a geometric series, which is valid because $q \neq 1$. Since $1 - q = p$, this gives $S = \dfrac{1-q^n}{p^2} - \dfrac{nq^n}{p}$, and so

$$\begin{aligned} \mathrm{E}(W) = pS &= \frac{1 - q^n - npq^n}{p} = \frac{1 - q^n - n(1-q)q^n}{p} \\ &= p^{-1}\left[1 + nq^{n+1} - (n+1)q^n\right], \end{aligned}$$

which is the required result.

Game H2

Let $N \sim \mathrm{Po}(\lambda)$ be the number of turns allowed, so $\mathrm{P}(N = n) = \mathrm{e}^{-\lambda}\lambda^n/n!$. Write

$$g(n) = p^{-1}\left[1 + nq^{n+1} - (n+1)q^n\right].$$

Given $N = n \geq 1$, she plays H1 with n turns, so her expected winnings are $g(n)$. Given $N = 0$ she wins nothing. By the law of total expectation (conditioning on N),

$$\mathrm{E}(W) = \sum_{n=1}^{\infty} \mathrm{P}(N = n)\, g(n).$$



Ask yourself The three pieces of $g(n)$ would be easier to handle as full Poisson sums from $n = 0$. Can we start the sum at $n = 0$?

Yes, we can: $g(0) = p^{-1}[1 + 0 - 1] = 0$, which agrees with the winnings of nil when $N = 0$. So

$$\mathrm{E}(W) = \frac{1}{p}\sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\left[1 + nq^{n+1} - nq^n - q^n\right].$$

The first term sums to 1, since the probabilities add to 1. For the others we need two series, both valid for every real x:

$$\begin{aligned} \sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{(\lambda x)^n}{n!} &= \mathrm{e}^{-\lambda}\mathrm{e}^{\lambda x}, \\ \sum_{n=0}^{\infty} n\,\mathrm{e}^{-\lambda}\frac{(\lambda x)^n}{n!} &= \lambda x \sum_{n=1}^{\infty} \mathrm{e}^{-\lambda}\frac{(\lambda x)^{n-1}}{(n-1)!} = \lambda x\,\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda x}. \end{aligned}$$

Put $x = q$, and note that $\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda q} = \mathrm{e}^{-\lambda(1-q)} = \mathrm{e}^{-\lambda p}$. Then

$$\sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\, q^n = \mathrm{e}^{-\lambda p}, \qquad \sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\, nq^n = \lambda q\,\mathrm{e}^{-\lambda p}.$$

The term nq^{n+1} is q times nq^n, so it contributes $\lambda q^2 e^{-\lambda p}$. Altogether,

$$\begin{aligned} \mathrm{E}(W) &= \frac{1}{p}\left[1 + \lambda q^2 e^{-\lambda p} - \lambda q\, e^{-\lambda p} - e^{-\lambda p}\right] \\ &= \frac{1}{p}\left(1 - e^{-\lambda p}\right) + \frac{\lambda q(q-1)}{p}\, e^{-\lambda p} \\ &= \frac{1}{p}\left(1 - e^{-\lambda p}\right) - \lambda q\, e^{-\lambda p}, \end{aligned}$$

using $q - 1 = -p$ in the last step. This is the required expression, in pounds.

Checks

- H1 with $n = 1$: the formula gives $p^{-1}[1 + q^2 - 2q] = p^{-1}(1-q)^2 = p$. This is right, because she wins £1 with probability p.
- As $n \to \infty$ in H1, $q^n \to 0$ and $nq^n \to 0$, so the expectation tends to $1/p$, the mean of a geometric distribution. Similarly, H2 tends to $1/p$ as $\lambda \to \infty$.
- For small λ, expanding the exponentials gives $\lambda - \frac{1}{2}\lambda^2 p - \lambda q + O(\lambda^2) = \lambda p + O(\lambda^2)$. This makes sense: she almost always gets either no turns or one turn, and $\mathrm{P}(N = 1) \approx \lambda$.
- Numerically, the H1 formula matched the direct sum exactly (using fractions) for $n \leq 14$. The H2 formula matched the truncated Poisson sum, and a simulation of 4×10^5 games with $\lambda = 2$, $p = 0.4$ gave 0.836, against the formula's 0.837.

Common slips

- Getting the last term of $S - qS$ wrong: the sum S stops at nq^{n-1}, so qS leaves $-nq^n$ over.
- Forgetting that $e^{-\lambda} e^{\lambda q}$ simplifies to $e^{-\lambda p}$, and then getting lost in the algebra.

Beyond STEP

The sums $\sum \mathrm{P}(N = n)\, x^n$ used in H2 are values of the *probability generating function* $G_N(x) = \mathrm{E}(x^N)$, and $\sum n\, \mathrm{P}(N = n)\, x^n = x G_N'(x)$. For a Poisson variable, $G_N(x) = e^{\lambda(x-1)}$. Generating functions, together with conditioning on a random number of trials, are central tools in first-year probability, for example in the study of random sums and branching processes. Grimmett and Welsh, *Probability: An Introduction*, covers both clearly.

Beyond STEP

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